BRAKES

04
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ABS SYSTEM WIRING DIAGRAM [4W-ABS]

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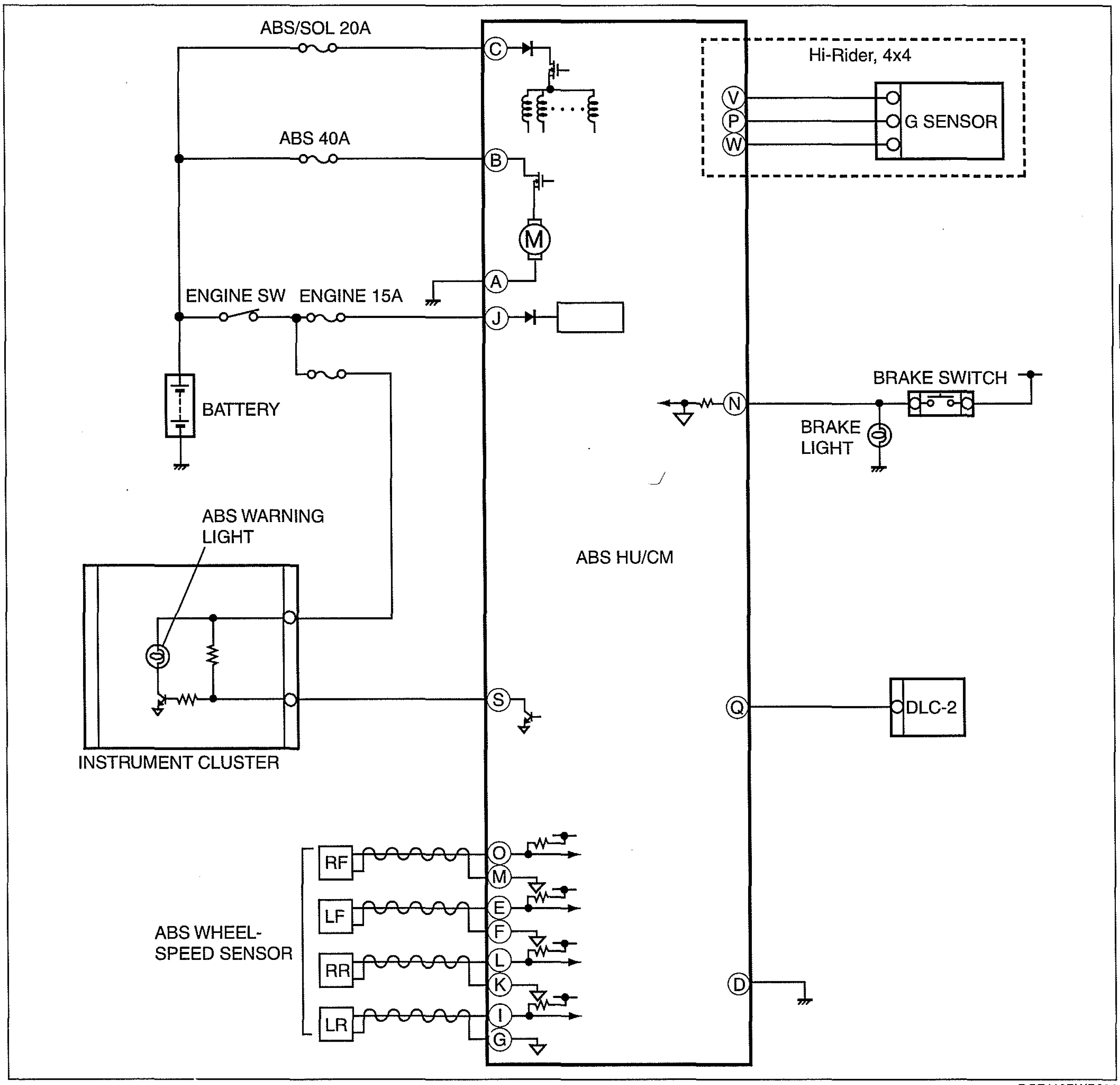

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ON-BOARD DIAGNOSIS [4W-ABS]

dcf040200000w16

On-Board Diagnostic (OBD) Test Description

- The OBD test inspects the integrity and function of the ABS and outputs the results when requested by the specific tests.
- On-board diagnostic test also:
  — Provides a quick inspection of the ABS usually performed at the start of each diagnostic procedure.
  — Provides verification after repairs to ensure that no other faults occurred during service.
- The OBD test is divided into 3 tests:
  — Read/clear diagnostic results, PID monitor and record and active command modes.

Read/clear diagnostic results

- This function allows you to read or clear DTCs in the ABS HU/CM memory.

PID/Data monitor and record

- This function allows you to access certain data values, input signals, calculated values, and system status information.

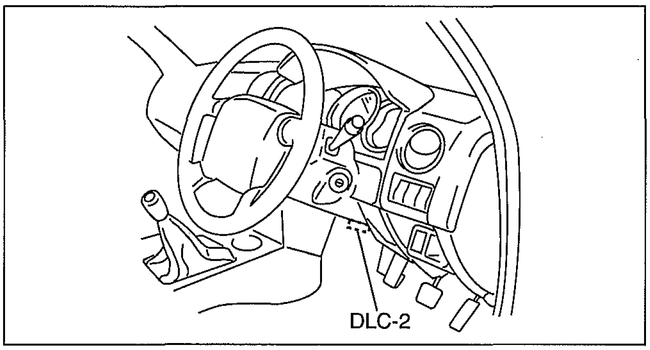
Active command modes

- This function allows you to control devices through the current diagnostic tool.

Reading DTCs Procedure

Using WDS

1. Connect the WDS to the DLC-2 connector.
2. Retrieve DTC using the WDS.



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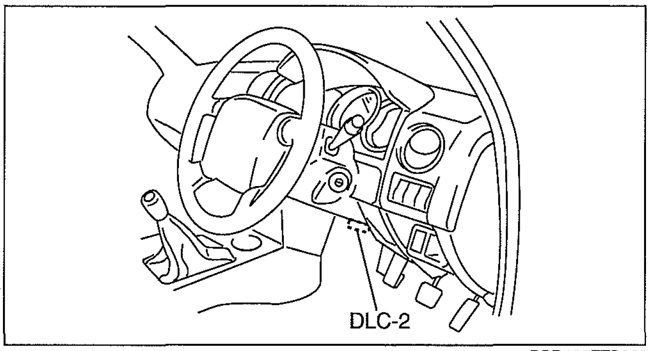
Using IDS/PDS

1. Connect the IDS/PDS to the DLC-2 connector.
2. After the vehicle is identified, select the following items from the initialization screen of the IDS/PDS.
   - When using the IDS (notebook PC)
     1. Select the "Toolbox" tab.
     2. Select "Self Test".
     3. Select "Module".
     4. Select "ABS".
   - When using the PDS (pocket PC)
     1. Select "Module Tests".
     2. Select "ABS".
     3. Select "Self Test".



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3. Verify the DTC according to the directions on the screen.
   - If any DTCs are displayed, perform troubleshooting according to the corresponding DTC inspection.
4. After completion of repairs, clear all DTCs stored in the ABS HU/CM. (See 04-02B-2 Clearing DTCs Procedures.)

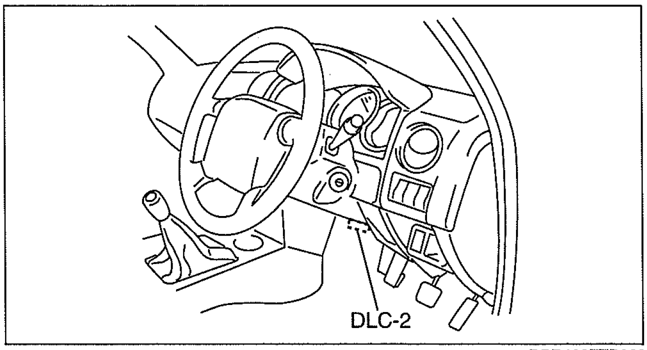
Clearing DTCs Procedures

Using WDS

1. After repairs have been made, perform the **DTCs reading procedure.**
2. Erase DTC using the WDS.
3. Ensure that the customer's concern has been resolved.

Using IDS/PDS

1. Connect the IDS/PDS to the vehicle DLC-2 connector.
2. After the vehicle is identified, select the following items from the initialization screen of the IDS/PDS.
   - When using the IDS (notebook PC)
     1. Select the "Toolbox" tab.
     2. Select "Self Test".
     3. Select "Module".
     4. Select "ABS".
   - When using the PDS (pocket PC)
     1. Select "Module Tests".
     2. Select "ABS".
     3. Select "Self Test".
3. Verify the DTC according to the directions on the screen.
4. Press the clear button on the DTC screen to clear the DTC.
5. Verify that no DTCs are displayed.



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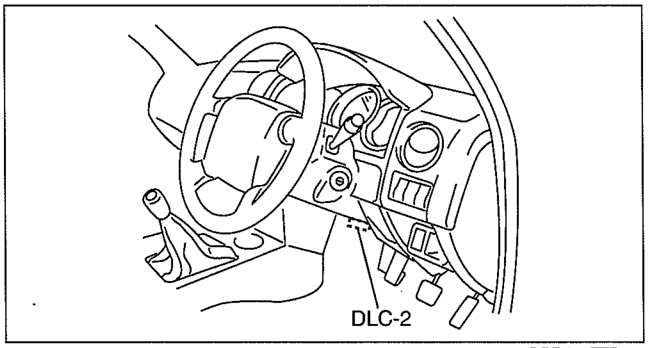
PID/Data Monitor and Record Procedure

Using WDS

1. Connect the WDS to the vehicle DLC-2 connector.
2. Access and monitor PIDs using the WDS.



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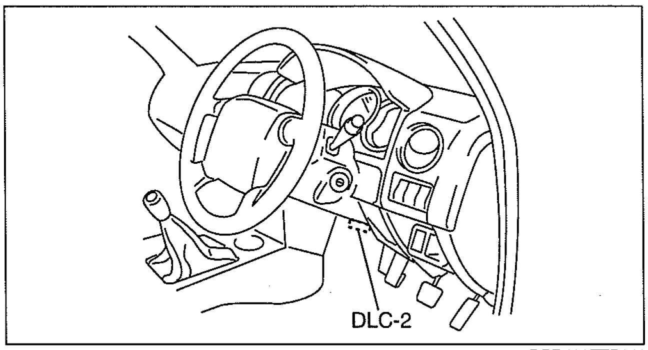
Using IDS/PDS

1. Connect the IDS/PDS to the DLC-2 connector.
2. After the vehicle is identified, select the following items from the initialization screen of the IDS/PDS.
   - When using the IDS (notebook PC)
     1. Select the "Toolbox" tab.
     2. Select "DataLogger".
     3. Select "Module".
     4. Select "ABS".
   - When using the PDS (pocket PC)
     1. Select "Module Tests".
     2. Select "ABS".
     3. Select "DataLogger".
3. Select the applicable PID from the PID table.
4. Verify the PID data according to the directions on the screen.



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Note

- The PID data screen function is used for monitoring the calculated value. Therefore, if the monitored value of the output parts is not within the specification, inspection of the monitored value of input parts corresponding to applicable output part control is necessary. In addition, because the system does not display output part malfunction as abnormality in the monitored value, it is necessary to inspect the output part individually using a active command modes function.

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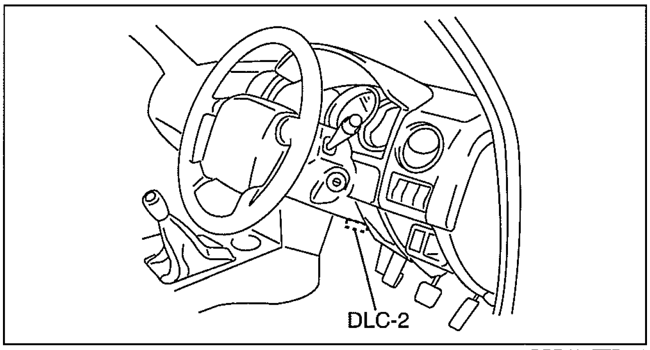
Active Command Modes Procedure

Using WDS

1. Connect the WDS to the vehicle DLC-2 connector.
2. Turn the engine switch to the ON position (engine off) or start the engine.
3. Activate active command modes using the WDS.



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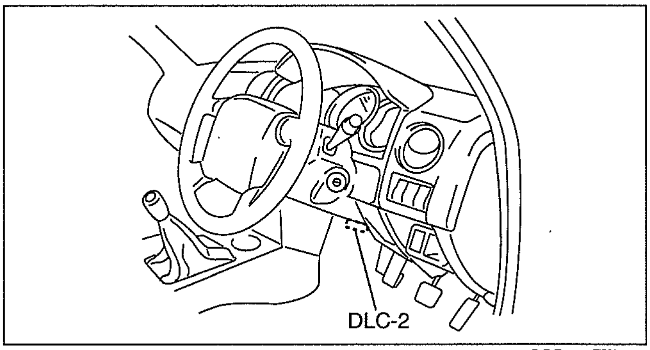
Using IDS/PDS

1. Connect the IDS/PDS to the DLC-2 connector.
2. After the vehicle is identified, select the following items from the initialization screen of the IDS/PDS.
   - When using the IDS (notebook PC)
     1. Select the "Toolbox" tab.
     2. Select "DataLogger".
     3. Select "Module".
     4. Select "ABS".
   - When using the PDS (pocket PC)
     1. Select "Module Tests".
     2. Select "ABS".
     3. Select "DataLogger".
3. Select the active command modes from the PID table.
4. Perform the active commands modes, inspect the operations for each parts.
   - If there is no operation sound from the relay, motor, and solenoid after the active commands mode inspection is performed, it is possible that there ia an open or short circuit in the wiring harness, relay, motor or solenoid, or sticking and operation malfunction.



DBR402ZTB003

DTC Table

DTC	System malfunction location	Page
B1317	Power supply system	(See 04-02B-7 DTC B1317, B1318 [4W-ABS].)
B1318	Power supply system	(See 04-02B-7 DTC B1317, B1318 [4W-ABS].)
B1342	ABS HU/CM system	(See 04-02B-8 DTC B1342 [4W-ABS].)
B1484	Brake switch system	(See 04-02B-8 DTC B1484 [4W-ABS].)
C1095	Pump motor, motor relay system	(See 04-02B-9 DTC C1095, C1096 [4W-ABS].)
C1096	Pump motor, motor relay system	(See 04-02B-9 DTC C1095, C1096 [4W-ABS].)
C1145	RF ABS wheel-speed sensor system	(See 04-02B-11 DTC C1145, C1155, C1165, C1175 [4W-ABS].)
C1155	LF ABS wheel-speed sensor system	(See 04-02B-11 DTC C1145, C1155, C1165, C1175 [4W-ABS].)
C1165	RR ABS wheel-speed sensor system	(See 04-02B-11 DTC C1145, C1155, C1165, C1175 [4W-ABS].)
C1175	LR ABS wheel-speed sensor system	(See 04-02B-11 DTC C1145, C1155, C1165, C1175 [4W-ABS].)
C1186	Fail-safe relay system	(See 04-02B-12 DTC C1186 [4W-ABS].)
C1194	LF outlet solenoid valve system	(See 04-02B-13 DTC C1194, C1198, C1202, C1206, C1210, C1214 [4W-ABS].)
C1198	LF inlet solenoid valve system	(See 04-02B-13 DTC C1194, C1198, C1202, C1206, C1210, C1214 [4W-ABS].)
C1202	Rear outlet solenoid valve system	(See 04-02B-13 DTC C1194, C1198, C1202, C1206, C1210, C1214 [4W-ABS].)
C1206	Rear inlet solenoid valve system	(See 04-02B-13 DTC C1194, C1198, C1202, C1206, C1210, C1214 [4W-ABS].)

DTC	System malfunction location	Page
C1210	RF outlet solenoid valve system	(See 04-02B-13 DTC C1194, C1198, C1202, C1206, C1210, C1214 [4W-ABS].)
C1214	RF inlet solenoid valve system	(See 04-02B-13 DTC C1194, C1198, C1202, C1206, C1210, C1214 [4W-ABS].)
C1222	ABS wheel-speed sensor (slip monitor) system	(See 04-02B-14 DTC C1222 [4W-ABS].)
C1233	LF ABS wheel-speed sensor/ABS sensor rotor system	(See 04-02B-16 DTC C1233, C1234, C1235, C1236 [4W-ABS].)
C1234	RF ABS wheel-speed sensor/ABS sensor rotor system	(See 04-02B-16 DTC C1233, C1234, C1235, C1236 [4W-ABS].)
C1235	RR ABS wheel-speed sensor/ABS sensor rotor system	(See 04-02B-16 DTC C1233, C1234, C1235, C1236 [4W-ABS].)
C1236	LR ABS wheel-speed sensor/ABS sensor rotor system	(See 04-02B-16 DTC C1233, C1234, C1235, C1236 [4W-ABS].)
C1414	Incorrect ABS HU/CM installed	(See 04-02B-17 DTC C1414 [4W-ABS].)
C1730*	G sensor system	(See 04-02B-18 DTC C1730, C1949, C1950 [4W-ABS].)
C1949*	G sensor system	(See 04-02B-18 DTC C1730, C1949, C1950 [4W-ABS].)
C1950*	G sensor system	(See 04-02B-18 DTC C1730, C1949, C1950 [4W-ABS].)

* : Hi-Rider, 4x4

PID/DATA Monitor Table

PID name (definition)	Unit/Condition	Operation condition (reference)	Action	ABS HU/CM terminal
ABS_LAMP (ABS warning light driver output state)	On/Off	• ABS warning light is illuminated: ON • ABS warning light is not illuminated: Off	Inspect the ABS warning light.	S
ABS_VOLT (System battery voltage value)	V	• Engine switch at ON: **Approx. 12.2 V** • Idling: **Approx. 14.1 V**	Inspect the power supply circuit. (See 04-13B-5 ABS HU/CM INSPECTION [4W-ABS].)	J
ACCLMTR* (Forward-G sensor input)	G	• Vehicle is stopped or driving at a constant speed: **0 G** • Cornering to left: Changes between **0 G and 1.5 G** • Cornering to right: Changes between **0 G and -1.5 G**	Inspect the G sensor. (See 04-13B-13 G SENSOR INSPECTION [4W-ABS].)	P
BOO_ABS (Brake pedal switch input)	On/Off	• Brake pedal depressed: On • Brake pedal released: Off	Inspect the brake switch. (See 04-11-6 BRAKE SWITCH INSPECTION.)	N
CCNTABS (Number of continuous codes)	—	• DTCs detected: **1—255** • No DTCs detected: **0**	Perform the DTC inspection.	—
PMP_MOTOR (Pump motor output state)	On/Off	• Pump motor activated: On • Pump motor not activated: Off	Inspect the ABS HU/CM. (See 04-13B-2 ABS HU/CM SYSTEM INSPECTION [4W-ABS].)	—
RLY_PMP (Pump motor relay output state)	On/Off	• Relay activated: On • Relay not activated: Off	Inspect the ABS HU/CM. (See 04-13B-2 ABS HU/CM SYSTEM INSPECTION [4W-ABS].)	—
RLY_VLV (Solenoid valve relay output state)	On/Off	• Solenoid valve relay is activated: On • Solenoid valve relay is deactivated: Off	Inspect ABS HU/CM. (See 04-13B-2 ABS HU/CM SYSTEM INSPECTION [4W-ABS].)	—
V_LF_INL (Left front inlet solenoid valve output state)	On/Off	• Solenoid valve activated: On • Solenoid valve not activated: Off	Inspect the ABS HU/CM. (See 04-13B-2 ABS HU/CM SYSTEM INSPECTION [4W-ABS].)	—

PID name (definition)	Unit/Condition	Operation condition (reference)	Action	ABS HU/CM terminal
V_LF_OTL (Left front outlet solenoid valve output state)	On/Off	• Solenoid valve activated: On • Solenoid valve not activated: Off	Inspect the ABS HU/CM. (See 04-13B-2 ABS HU/CM SYSTEM INSPECTION [4W-ABS].)	—
V_RF_INL (Right front inlet solenoid valve output state)	On/Off	• Solenoid valve activated: On • Solenoid valve not activated: Off	Inspect the ABS HU/CM. (See 04-13B-2 ABS HU/CM SYSTEM INSPECTION [4W-ABS].)	—
V_RF_OTL (Right front outlet solenoid valve output state)	On/Off	• Solenoid valve activated: On • Solenoid valve not activated: Off	Inspect the ABS HU/CM. (See 04-13B-2 ABS HU/CM SYSTEM INSPECTION [4W-ABS].)	—
V_Rear_ INL (Rear inlet solenoid valve output state)	On/Off	• Solenoid valve activated: On • Solenoid valve not activated: Off	Inspect the ABS HU/CM. (See 04-13B-2 ABS HU/CM SYSTEM INSPECTION [4W-ABS].)	—
V_Rear_OTL (Rear outlet solenoid valve output state)	On/Off	• Solenoid valve activated: On • Solenoid valve not activated: Off	Inspect the ABS HU/CM. (See 04-13B-2 ABS HU/CM SYSTEM INSPECTION [4W-ABS].)	—
WSPD_LF (Left front ABS wheel-speed sensor input)	KPH, MPH	• Vehicle stopped: **0 KPH, 0 MPH** • Vehicle running: Vehicle speed	Inspect the ABS wheel-speed sensor.	E, F
WSPD_LR (Left rear ABS wheel-speed sensor input)	KPH, MPH	• Vehicle stopped: **0 KPH, 0 MPH** • Vehicle running: Vehicle speed	Inspect the ABS wheel-speed sensor.	I, G
WSPD_RF (Right front ABS wheel-speed sensor input)	KPH, MPH	• Vehicle stopped: **0 KPH, 0 MPH** • Vehicle running: Vehicle speed	Inspect the ABS wheel-speed sensor.	O, M
WSPD_RR (Right rear ABS wheel-speed sensor input)	KPH, MPH	• Vehicle stopped: **0 KPH, 0 MPH** • Vehicle running: Vehicle speed	Inspect the ABS wheel-speed sensor.	L, K

* : Hi-Rider, 4x4

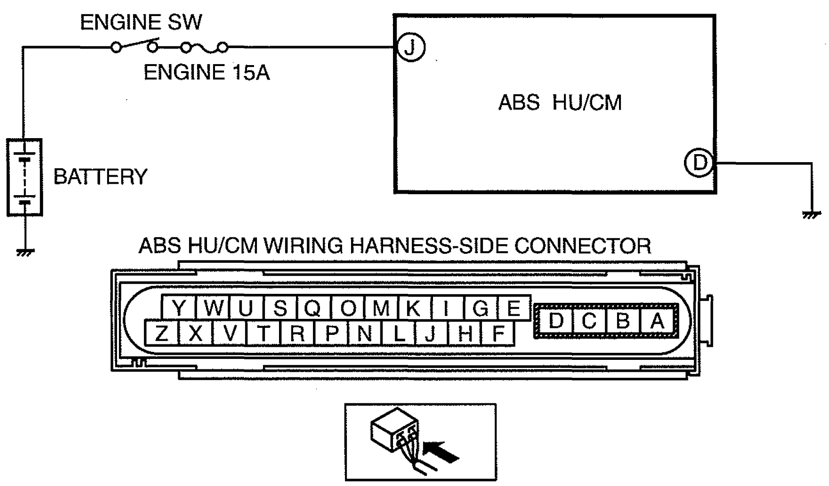
Active Command Modes Table

Command name	Output part	Operation	Operating condition
PMP_MOTOR	Pump motor	On/Off	Engine switch at ON
V_LF_INL	LF inlet solenoid valve		
V_LF_OTL	LF outlet solenoid valve		
V_Rear_INL	Rear inlet solenoid valve		
V_Rear_OTL	Rear outlet solenoid valve		
V_RF_INL	RF inlet solenoid valve		
V_RF_OTL	RF outlet solenoid valve		

DTC B1317, B1318 [4W-ABS]

dcf040200000w17

DTC B1317, B1318	Power supply system
DETECTION CONDITION	• B1317 — The voltage at ABS HU/CM terminal J **16.8 V** or more. • B1318 — The vehicle speed exceeds **6 km/h {3.7 mph}** and the voltage at ABS HU/CM terminal J is less than **9.6 V**.
POSSIBLE CAUSE	• ENGINE 15 A fuse malfunction • Open circuit or short to ground in the wiring harness between the ABS HU/CM terminal J and the battery • Open circuit or faulty ground in the wiring harness between the ABS HU/CM terminal D and the body ground • Battery deterioration • Generator malfunction • Poor connection at connectors (female terminal)



Diagnostic procedure

STEP	INSPECTION		ACTION
1	**INSPECT BATTERY VOLTAGE** • Is the battery terminal voltage normal?	Yes	Make sure that battery terminal connection is normal. Go to the next step.
		No	Charge or replace the battery, then go to Step 6. (See 01-17-4 BATTERY RECHARGING.) (See 01-17-2 BATTERY REMOVAL/INSTALLATION.)
2	**INSPECT BATTERY GRAVITY** • Is battery specific gravity as specified?	Yes	Go to the next step.
		No	Replace the battery, then go to Step 6. (See 01-17-2 BATTERY REMOVAL/INSTALLATION.)
3	**INSPECT CHARGING SYSTEM** • Are the generator and drive belt tensions normal?	Yes	Go to the next step.
		No	Inspect generator and/or drive belt as necessary, then go to Step 6. (See 01-10A-3 DRIVE BELT INSPECTION [WL-3].) (See 01-10B-3 DRIVE BELT INSPECTION [WL-C, WE-C].) (See 01-17-5 GENERATOR INSPECTION.)
4	**INSPECT ABS HU/CM POWER SUPPLY FOR OPEN CIRCUIT** • Start the engine. • Measure the voltage between ABS HU/CM terminal J and ground. • Is the voltage **approx. 10 V**?	Yes	Go to the next step.
		No	Repair or replace the wiring harness for open circuit between the ABS HU/CM and ground, then go to Step 6.
5	**INSPECT ABS HU/CM GROUND FOR POOR GROUND OR OPEN CIRCUIT** • Turn the engine switch off. • Measure the resistance between ground and ABS HU/CM terminal D. • Is the resistance **within 0—1 ohm**?	Yes	Go to the next step.
		No	If there is no continuity: • Repair or replace the wiring harness for open circuit between the ABS HU/CM and ground, then go to the next step. If the resistance is not **within 0—1 ohm**: • Repair or replace harness for poor ground, then go to the next step.

04

STEP	INSPECTION		ACTION
6	**VERIFY TROUBLESHOOTING COMPLETED** • Make sure to reconnect all disconnected connectors. • Clear the DTC from the memory. (See 04-02B-2 Clearing DTCs Procedures.) • Start the engine and drive the vehicle at **6 km/h {3.5 mph} or more**. • Is the same DTC present?	Yes	Replace the ABS HU/CM, then go to the next step. (See 04-13B-3 ABS HU/CM REMOVAL/INSTALLATION [4W-ABS].)
		No	Go to the next step.
7	**VERIFY AFTER REPAIR PROCEDURE** • Are any other DTCs present?	Yes	Go to the applicable DTC inspection. (See 04-02B-4 DTC Table.)
		No	DTC troubleshooting completed.

DTC B1342 [4W-ABS]

dcf040200000w18

DTC B1342	ABS HU/CM system
DETECTION CONDITION	• The ABS HU/CM on-board diagnostic function detects control module malfunction.
POSSIBLE CAUSE	• ABS HU/CM internal malfunction

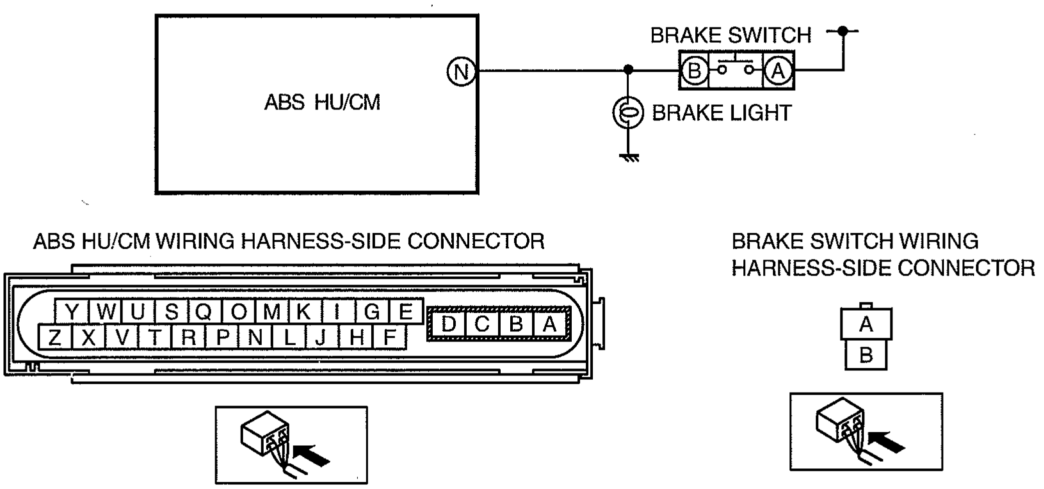
Diagnostic procedure

STEP	INSPECTION		ACTION
1	**VERIFY CURRENT STATUS OF MALFUNCTION** • Clear the DTC from the memory. (See 04-02B-2 Clearing DTCs Procedures.) • Is same DTC present?	Yes	Replace the ABS HU/CM, then go to the next step. (See 04-13B-3 ABS HU/CM REMOVAL/INSTALLATION [4W-ABS].)
		No	Go to the next step.
2	**VERIFY AFTER REPAIR PROCEDURE** • Are any other DTCs present?	Yes	Go to the applicable DTC inspection. (See 04-02B-4 DTC Table.)
		No	DTC troubleshooting completed.

DTC B1484 [4W-ABS]

dcf040200000w19

DTC B1484	Brake switch system
DETECTION CONDITION	• Open circuit in the wiring harness between the ABS HU/CM terminal and the brake switch terminal
POSSIBLE CAUSE	• Brake switch malfunction • Open circuit in the wiring harness between the ABS HU/CM terminal N and the brake switch terminal



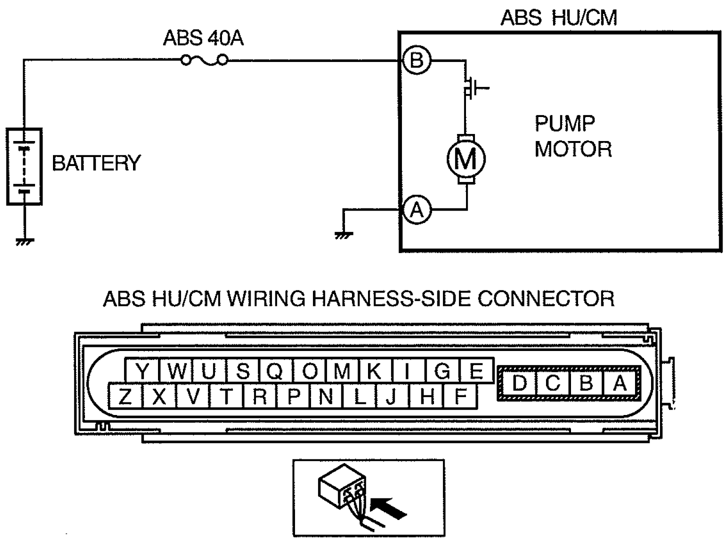
Diagnostic procedure

STEP	INSPECTION		ACTION
1	**INSPECT WIRING HARNESS BETWEEN ABS HU/CM AND BRAKE SWITCH FOR CONTINUITY** • Turn the engine switch off. • Disconnect the ABS HU/CM connector. • Is B+ correctly applied to ABS HU/CM connector terminal N when brake pedal depressed?	Yes	Go to Step 3.
		No	Go to the next step.
2	**INSPECT BRAKE SWITCH** • Inspect the brake switch. (See 04-11-6 BRAKE SWITCH INSPECTION.) • Is the brake switch normal?	Yes	Repair or replace the wiring harness for open circuit between ABS HU/CM and brake switch, then go to the next step.
		No	Replace the brake switch, then go to the next step. (See 04-11-5 BRAKE SWITCH REMOVAL/INSTALLATION.)
3	**VERIFY TROUBLESHOOTING COMPLETED** • Make sure to reconnect all disconnected connectors. • Clear the DTC from the memory. (See 04-02B-2 Clearing DTCs Procedures.) • Are the same DTCs present?	Yes	Replace the ABS HU/CM, then go to the next step. (See 04-13B-3 ABS HU/CM REMOVAL/INSTALLATION [4W-ABS].)
		No	Go to the next step.
4	**VERIFY AFTER REPAIR PROCEDURE** • Are any other DTC present?	Yes	Go to the applicable DTC inspection. (See 04-02B-4 DTC Table.)
		No	DTC troubleshooting completed.

DTC C1095, C1096 [4W-ABS]

dcf040200000w20

DTC C1095, C1096	Pump motor, motor relay system
DETECTION CONDITION	• C1095 — When the pump motor monitor voltage remains at **2.0 V** or more for **1 s**. — ABS motor monitor OFF signal is input within specified time limit when the motor signal is switched from ON to OFF by ABS CM. • C1096 — When the difference between the battery power supply voltage and pump motor power supply voltage remains at **4.0 V** or more for **0.1 s** or more while the pump motor is operating.
POSSIBLE CAUSE	• ABS 40 A fuse malfunction • Open or short to ground circuit in the wiring harness between the battery and the ABS HU/CM terminal B • Open circuit in the wiring harness between the ABS HU/CM terminal A and the body ground • Open or short circuit in the ABS HU/CM internal motor relay, or stuck motor relay • Open or short circuit in the ABS HU/CM internal motor, or frozen motor • Fail-safe relay malfunction • Poor connection at connectors (female terminal)



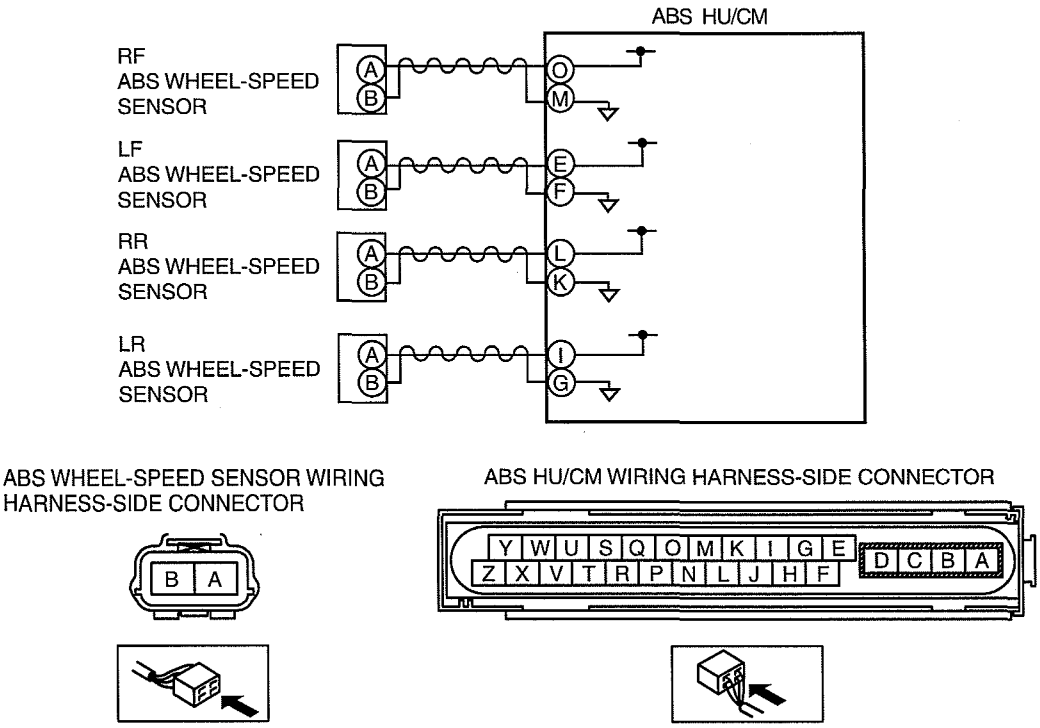
Diagnostic procedure

STEP	INSPECTION		ACTION
1	**INSPECT ABS FUSE CONDITION** • Is the ABS 40A fuse normal?	Yes	Go to the next step.
		No	Replace the fuse, then go to Step 5.
2	**INSPECT MOTOR RELAY POWER SUPPLY FOR OPEN CIRCUIT** • Turn the engine switch off. • Disconnect ABS HU/CM connector. • Turn the engine switch to the ON position (engine off). • Measure voltage between ABS HU/CM terminal B (harness-side) and ground. • Is the voltage **B+**?	Yes	Go to the next step.
		No	Repair or replace the wiring harness for open circuit between battery positive terminal and ABS HU/CM terminal B, then go to Step 5.
3	**INSPECT PUMP MOTOR GROUND FOR OPEN CIRCUIT** • Turn the engine switch off. • Inspect for continuity between ABS HU/CM terminal A (harness-side) and ground. • Is there continuity?	Yes	Go to the next step.
		No	Repair or replace the wiring harness for open circuit between ABS HU/CM terminal A and ground, then go to the next step.
4	**VERIFY PUMP MOTOR OPERATION** • Turn the engine switch off. • Connect the current diagnostic tool to the DLC-2. • Turn the engine switch to the ON position (engine off). • Access PMP_MOTOR active command modes using the current diagnostic tool. • Does the pump motor operate?	Yes	Go to the next step.
		No	Replace the ABS HU/CM, then go to the next step. (See 04-13B-3 ABS HU/CM REMOVAL/INSTALLATION [4W-ABS].)
5	**VERIFY TROUBLESHOOTING COMPLETED** • Make sure to reconnect all disconnected connectors. • Clear the DTC from the memory. (See 04-02B-2 Clearing DTCs Procedures.) • Start the engine and drive the vehicle at **12 km/h {7.5 mph} or more**. • Gradually slow down and stop the vehicle. • Is the same DTC present?	Yes	Replace the ABS HU/CM, then go to the next step. (See 04-13B-3 ABS HU/CM REMOVAL/INSTALLATION [4W-ABS].)
		No	Go to the next step.
6	**VERIFY AFTER REPAIR PROCEDURE** • Are any other DTCs present?	Yes	Go to the applicable DTC inspection. (See 04-02B-4 DTC Table.)
		No	DTC troubleshooting completed.

DTC C1145, C1155, C1165, C1175 [4W-ABS]

dcf040200000w21

DTC	**C1145** **C1155** **C1165** **C1175**	**RF ABS wheel-speed sensor system** **LF ABS wheel-speed sensor system** **RR ABS wheel-speed sensor system** **LR ABS wheel-speed sensor system**
DETECTION CONDITION	• Open circuit or short to ground has been detected in the ABS wheel-speed sensor wiring harness on any of the four vehicle wheels.	
POSSIBLE CAUSE	• Open circuit or short to ground in the wiring harness between the following ABS HU/CM terminal and the ABS wheel-speed sensor terminal: — ABS HU/CM terminal O—RF ABS wheel-speed sensor terminal A — ABS HU/CM terminal M—RF ABS wheel-speed sensor terminal B — ABS HU/CM terminal E—LF ABS wheel-speed sensor terminal A — ABS HU/CM terminal F—LF ABS wheel-speed sensor terminal B — ABS HU/CM terminal L—RR ABS wheel-speed sensor terminal A — ABS HU/CM terminal K—RR ABS wheel-speed sensor terminal B — ABS HU/CM terminal I—LR ABS wheel-speed sensor terminal A — ABS HU/CM terminal G—LR ABS wheel-speed sensor terminal B • ABS wheel-speed sensor malfunction • Poor connection at connectors (female terminal)	



Diagnostic procedure

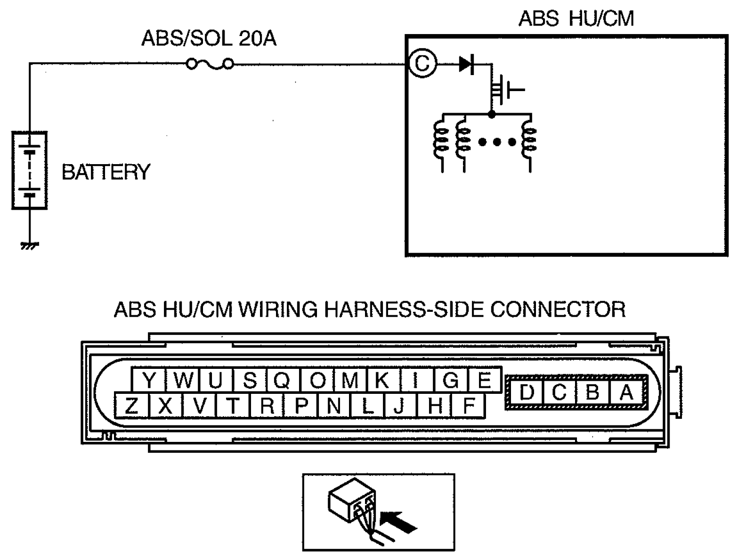
STEP	INSPECTION		ACTION
1	**INSPECT PID TO VERIFY THAT WHEEL SPEED-SIGNALS ARE TRANSMITTED FROM ABS WHEEL- SPEED SENSOR USING CURRENT DIAGNOSTIC TOOL** • Turn the engine switch off. • Connect the current diagnostic tool to the DLC-2. • Select the following PIDs using the current diagnostic tool: WSPD_LF WSPD_LR WSPD_RF WSPD_RR • Drive the vehicle. • Verify that the wheel speed-signals are transmitted from each ABS wheel-speed sensor. • Are the wheel-speed signals transmitted?	Yes	Go to Step 3.
		No	Go to the next step.

STEP	INSPECTION		ACTION
2	**INSPECT FOR OPEN CIRCUIT IN WIRING HARNESS BETWEEN ABS HU/CM AND ABS WHEEL-SPEED SENSOR** • Turn the engine switch off. • Disconnect the ABS HU/CM connector and ABS wheel-speed sensor. • Inspect for continuity in the wiring harness between the following ABS wheel-speed sensor connectors on the vehicle wiring harness-side and ABS HU/CM connectors. — RF ABS wheel-speed sensor (+): O—A — RF ABS wheel-speed sensor (–): M—B — LF ABS wheel-speed sensor (+): E—A — LF ABS wheel-speed sensor (–): F—B — RR ABS wheel-speed sensor (+): L—A — RR ABS wheel-speed sensor (–): K—B — LR ABS wheel-speed sensor (+): I—A — LR ABS wheel-speed sensor (–): G—B • Is there continuity?	Yes	Replace the ABS wheel-speed sensor, then go to the next step. (See 04-13B-7 FRONT ABS WHEEL-SPEED SENSOR REMOVAL/INSTALLATION [4W-ABS (4x2 (EXCEPT Hi-Rider))].) (See 04-13B-8 FRONT ABS WHEEL-SPEED SENSOR REMOVAL/INSTALLATION [4W-ABS (Hi-Rider, 4x4)].) (See 04-13B-10 REAR ABS WHEEL-SPEED SENSOR REMOVAL/INSTALLATION [4W-ABS (4x2 (EXCEPT Hi-Rider))].) (See 04-13B-11 REAR ABS WHEEL-SPEED SENSOR REMOVAL/INSTALLATION [4W-ABS (Hi-Rider, 4x4)].)
		No	Repair or replace the wiring harness, then go to the next step.
3	**VERIFY THAT THE SAME DTC IS NOT PRESENT** • Clear the DTCs from the memory. (See 04-02B-2 Clearing DTCs Procedures.) • Start the engine and drive the vehicle at **12 km/h {7.5 mph} or more**. • Are the same DTCs present?	Yes	Repeat the inspection from Step 1. If the malfunction recurs, replace the ABS HU/CM, then go to the next step. (See 04-13B-3 ABS HU/CM REMOVAL/INSTALLATION [4W-ABS].)
		No	Go to the next step.
4	**VERIFY THAT NO OTHER DTCS ARE PRESENT** • Are any other DTCs output?	Yes	Go to the applicable DTC inspection. (See 04-02B-4 DTC Table.)
		No	DTC troubleshooting completed.

DTC C1186 [4W-ABS]

dcf040200000w23

DTC C1186	Fail-safe relay system
DETECTION CONDITION	• ABS HU/CM internal valve relay remains OFF when valve relay ON is commanded. • ABS HU/CM internal valve relay remains ON (stuck) when valve relay OFF is commanded.
POSSIBLE CAUSE	• ABS/SOL 20 A fuse malfunction • Open circuit or short to ground in the wiring harness between the battery and the ABS HU/CM terminal C • Open or short circuit in the ABS HU/CM internal valve relay, or stuck valve relay • Poor connection at connectors (female terminal)



Diagnostic procedure

STEP	INSPECTION		ACTION
1	**INSPECT ABS FUSE CONDITION** • Is the ABS/SOL 20 A fuse normal?	Yes	Go to the next step.
		No	Replace the fuse, then go to Step 3.
2	**INSPECT VALVE RELAY POWER SUPPLY FOR OPEN CIRCUIT** • Turn the engine switch off. • Disconnect ABS HU/CM connector. • Turn the engine switch to the ON position (engine off). • Measure voltage between ABS HU/CM terminal C (harness-side) and ground. • Is voltage **B+**?	Yes	Go to the next step.
		No	Repair or replace the wiring harness for open circuit between battery positive terminal and ABS HU/CM terminal C, then go to the next step.
3	**VERIFY TROUBLESHOOTING COMPLETED** • Clear the DTC from the memory. (See 04-02B-2 Clearing DTCs Procedures.) • Is the same DTC present?	Yes	Replace the ABS HU/CM, then go to next step. (See 04-13B-3 ABS HU/CM REMOVAL/INSTALLATION [4W-ABS].)
		No	Go to the next step.
4	**VERIFY AFTER REPAIR PROCEDURE** • Are any other DTCs present?	Yes	Go to the applicable DTC inspection. (See 04-02B-4 DTC Table.)
		No	DTC troubleshooting completed.

DTC C1194, C1198, C1202, C1206, C1210, C1214 [4W-ABS]

dcf040200000w24

DTC	**C 1194**	**LF outlet solenoid valve system**
	C 1198	**LF inlet solenoid valve system**
	C 1202	**Rear outlet solenoid valve system**
	C 1206	**Rear inlet solenoid valve system**
	C 1210	**RF outlet solenoid valve system**
	C 1214	**RF inlet solenoid valve system**
DETECTION CONDITION	• Solenoid valve operation does not correspond to solenoid ON/OFF commands from the ABS HU/CM.	
POSSIBLE CAUSE	• Open or short circuit in the ABS HU/CM internal solenoid valves • Solenoid valve malfunction • Poor connection at connectors (female terminal)	

Diagnostic procedure

STEP	INSPECTION		ACTION
1	**VERIFY SOLENOID VALVE OPERATION** • Turn the engine switch off. • Connect the current diagnostic tool to the DLC-2. • Turn the engine switch to the ON position (engine off). • Access the active command mode for the solenoid valve using the current diagnostic tool. • Does the solenoid valve operate?	Yes	Go to the next step.
		No	Replace the ABS HU/CM, then go to the next step. (See 04-13B-3 ABS HU/CM REMOVAL/INSTALLATION [4W-ABS].)
2	**VERIFY DTC TROUBLESHOOTING COMPLETED** • Clear the DTC from the memory. (See 04-02B-2 Clearing DTCs Procedures.) • Start the engine and drive the vehicle at **12 km/h {7.5 mph} or more**. • Gradually slow down and stop vehicle. • Are the same DTCs present?	Yes	Repeat the inspection from Step 1. If the malfunction recurs, replace the ABS HU/CM, then go to the next step. (See 04-13B-3 ABS HU/CM REMOVAL/INSTALLATION [4W-ABS].)
		No	Go to the next step.
3	**VERIFY AFTER REPAIR PROCEDURE** • Are any other DTCs present?	Yes	Go to the applicable DTC inspection. (See 04-02B-4 DTC Table.)
		No	DTC troubleshooting completed.

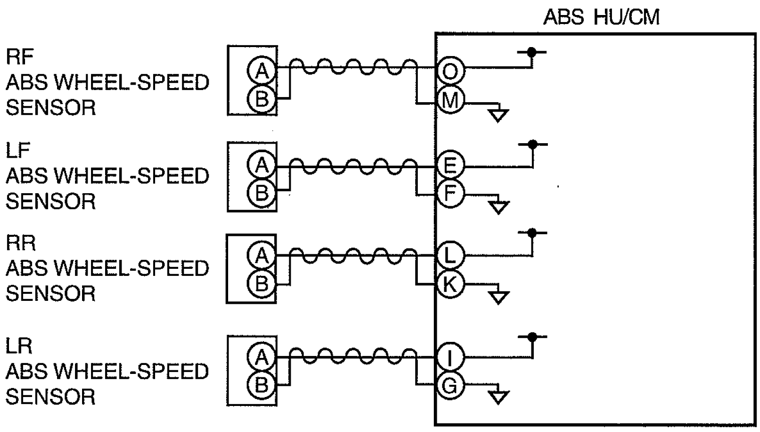
DTC C1222 [4W-ABS]

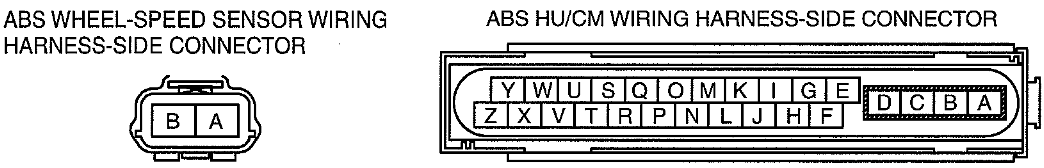
dcf040200000w25

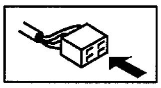
Note

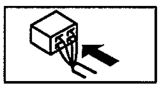
- DTC C1222 will be detected when a malfunctioning ABS wheel-speed sensor cannot be specified.

DTC C1222	ABS wheel-speed sensor (slip monitor) system
DETECTION CONDITION	• Difference between any vehicle wheel speeds exceeds specification when driving at a constant speed. • ABS control operates for **60 s** or more.
POSSIBLE CAUSE	• ABS wheel-speed sensor malfunction (low output, metal shavings on sensor) • ABS sensor rotor malfunction (chipping of sensor rotor teeth) • Poor installation of ABS wheel-speed sensor and/or sensor rotor (If the sensor rotor is installed at an angle, it may cause output of abnormal wave pattern at high speeds.) • Excessive clearance between the ABS wheel-speed sensor and sensor rotor









Diagnostic procedure

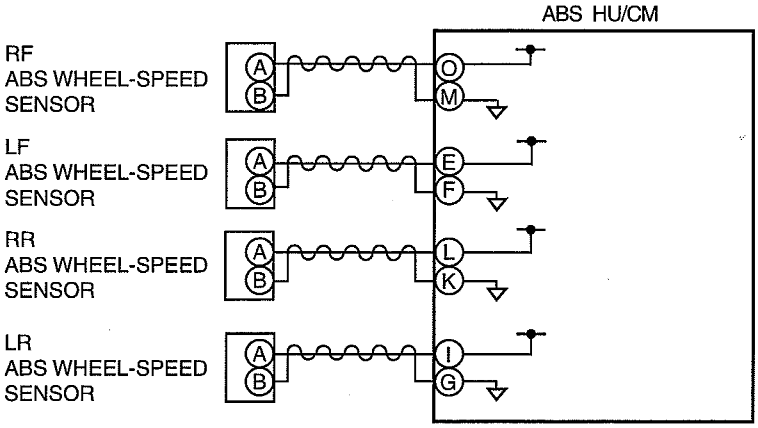
STEP	INSPECTION		ACTION
1	**INSPECT PID FOR ABNORMAL OUTPUT FROM ABS WHEEL-SPEED SENSOR USING CURRENT DIAGNOSTIC TOOL** • Turn the engine switch off. • Connect the current diagnostic tool to the DLC-2. • Select the following PIDs using the current diagnostic tool: WSPD_LF WSPD_LR WSPD_RF WSPD_RR • Start the engine and drive the vehicle. • Verify that the PIDs of the four ABS wheel-speed sensors correspond approximately. • Do the vehicle speeds correspond?	Yes	Go to Step 4.
		No	If there is a difference in speeds of the four wheels, go to the next step.
2	**INSPECT IF MALFUNCTION OCCURRED DUE TO IMPROPER SENSOR CLEARANCE** • Inspect the clearance between the ABS wheel-speed sensor and the ABS sensor rotor. **Clearance** **0.30—1.1 mm {0.012—0.043 in}**	Yes	Go to the next step.
		No	Replace the rear ABS wheel-speed sensor, then go to Step 4. (See 04-13B-7 FRONT ABS WHEEL-SPEED SENSOR REMOVAL/INSTALLATION [4W-ABS (4x2 (EXCEPT Hi-Rider))].) (See 04-13B-8 FRONT ABS WHEEL-SPEED SENSOR REMOVAL/INSTALLATION [4W-ABS (Hi-Rider, 4x4)].) (See 04-13B-10 REAR ABS WHEEL-SPEED SENSOR REMOVAL/INSTALLATION [4W-ABS (4x2 (EXCEPT Hi-Rider))].) (See 04-13B-11 REAR ABS WHEEL-SPEED SENSOR REMOVAL/INSTALLATION [4W-ABS (Hi-Rider, 4x4)].)
3	**VISUALLY INSPECT ABS SENSOR ROTOR FOR FOREIGN MATERIAL ADHERING OR IMPROPER INSTALLATION** • Is the result normal?	Yes	Go to the next step.
		No	Replace the front wheel hub component, front drive shaft or rear axle shaft, then go to the next step. (See 03-11-4 WHEEL HUB, STEERING KNUCKLE REMOVAL/INSTALLATION [4x2 (EXCEPT Hi-Rider)].) (See 03-13-2 FRONT DRIVE SHAFT REMOVAL/ INSTALLATION.) (See 03-12-2 AXLE SHAFT REMOVAL/INSTALLATION.)
4	**VERIFY DTC TROUBLESHOOTING COMPLETED** • Make sure to reconnect all disconnected connectors. • Clear the DTC from the memory. (See 04-02B-2 Clearing DTCs Procedures.) • Start the engine and drive the vehicle at **12 km/h {7.5 mph} or more**. • Are the same DTCs present?	Yes	Repeat the inspection from Step 1. If the malfunction recurs, replace the ABS HU/CM. (See 04-13B-3 ABS HU/CM REMOVAL/INSTALLATION [4W-ABS].)
		No	Go to the next step.
5	**VERIFY AFTER REPAIR PROCEDURE** • Are any other DTCs present?	Yes	Go to the applicable DTC inspection. (See 04-02B-4 DTC Table.)
		No	DTC troubleshooting completed.

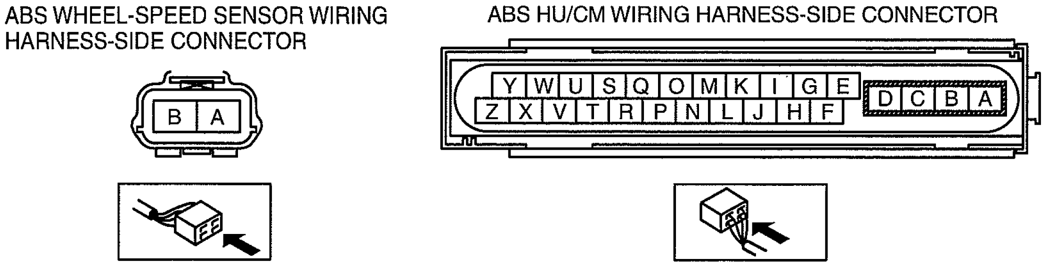
04

DTC C1233, C1234, C1235, C1236 [4W-ABS]

dcf040200000w26

DTC **C1233** **C1234** **C1235** **C1236**	**LF ABS wheel-speed sensor (short to ground) system** **RF ABS wheel-speed sensor (short to ground) system** **RR ABS wheel-speed sensor (short to ground) system** **LR ABS wheel-speed sensor (short to ground) system**
DETECTION CONDITION	• The vehicle wheel speed of any of the four vehicle wheels is **2.75 km/h {1.71 mph} or less** when driving at the specified vehicle speed or more.
POSSIBLE CAUSE	• Short to ground in the wiring harness between the following ABS HU/CM terminal and the ABS wheel-speed sensor terminal: — ABS HU/CM terminal O—RF ABS wheel-speed sensor terminal A — ABS HU/CM terminal M—RF ABS wheel-speed sensor terminal B — ABS HU/CM terminal E—LF ABS wheel-speed sensor terminal A — ABS HU/CM terminal F—LF ABS wheel-speed sensor terminal B — ABS HU/CM terminal L—RR ABS wheel-speed sensor terminal A — ABS HU/CM terminal K—RR ABS wheel-speed sensor terminal B — ABS HU/CM terminal I—LR ABS wheel-speed sensor terminal A — ABS HU/CM terminal G—LR ABS wheel-speed sensor terminal B • ABS wheel-speed sensor malfunction • Poor connection at connectors (female terminal)





Diagnostic procedure

STEP	INSPECTION		ACTION
1	**INSPECT PID TO VERIFY THAT WHEEL SPEED-SIGNALS ARE TRANSMITTED FROM ABS WHEEL- SPEED SENSOR USING CURRENT DIAGNOSTIC TOOL** • Turn the engine switch off. • Connect the current diagnostic tool to the DLC-2. • Select the following PIDs using the current diagnostic tool: WSPD_LF WSPD_LR WSPD_RF WSPD_RR • Drive the vehicle. • Verify that the wheel speed-signals are transmitted from each ABS wheel-speed sensor. • Are the wheel-speed signals transmitted?	Yes	Go to Step 3.
		No	Go to the next step.

STEP	INSPECTION		ACTION
2	**INSPECT A SHORT TO GROUND IN THE WIRING HARNESS BETWEEN THE ABS HU/CM AND THE ABS WHEEL-SPEED SENSOR** • Turn the engine switch off. • Disconnect the ABS HU/CM connector and the ABS wheel-speed sensor connector. • Inspect for a short to ground in the wiring harness between the following ABS wheel-speed sensor connectors on the vehicle wiring harness-side and ABS HU/CM connectors. — RF ABS wheel-speed sensor (+): O—A — RF ABS wheel-speed sensor (–): M—B — LF ABS wheel-speed sensor (+): E—A — LF ABS wheel-speed sensor (–): F—B — RR ABS wheel-speed sensor (+): L—A — RR ABS wheel-speed sensor (–): K—B — LR ABS wheel-speed sensor (+): I—A — LR ABS wheel-speed sensor (–): G—B • Is there continuity?	Yes	Replace the rear ABS wheel-speed sensor, then go to Step 4. (See 04-13B-7 FRONT ABS WHEEL-SPEED SENSOR REMOVAL/INSTALLATION [4W-ABS (4x2 (EXCEPT Hi-Rider))].) (See 04-13B-8 FRONT ABS WHEEL-SPEED SENSOR REMOVAL/INSTALLATION [4W-ABS (Hi-Rider, 4x4)].) (See 04-13B-10 REAR ABS WHEEL-SPEED SENSOR REMOVAL/INSTALLATION [4W-ABS (4x2 (EXCEPT Hi-Rider))].) (See 04-13B-11 REAR ABS WHEEL-SPEED SENSOR REMOVAL/INSTALLATION [4W-ABS (Hi-Rider, 4x4)].)
		No	Repair or replace the wiring harness, then go to the next step.
3	**VERIFY THAT THE SAME DTC IS NOT PRESENT** • Clear the DTCs from the memory. (See 04-02B-2 Clearing DTCs Procedures.) • Start the engine and drive the vehicle at **12 km/h {7.5 mph} or more**. • Are the same DTCs present?	Yes	Repeat the inspection from Step 1. If the malfunction recurs, replace the ABS HU/CM, then go to the next step. (See 04-13B-3 ABS HU/CM REMOVAL/INSTALLATION [4W-ABS].)
		No	Go to the next step.
4	**VERIFY THAT NO OTHER DTCS ARE PRESENT** • Are any other DTCs output?	Yes	Go to the applicable DTC inspection. (See 04-02B-4 DTC Table.)

04

DTC C1414 [4W-ABS]

dcf040200000w27

DTC C1414	ABS HU/CM unit mismatched installation
DETECTION CONDITION	• When G sensor signal input for **more than 1 s** to 4x2 (except Hi-Rider) vehicle-type ABS HU/CM.
POSSIBLE CAUSE	• Mismatched installation of ABS HU/CM — 4x2 (except Hi-Rider) vehicle type ABS HU/CM is mismatched installed on Hi-Rider and 4x4 vehicle.

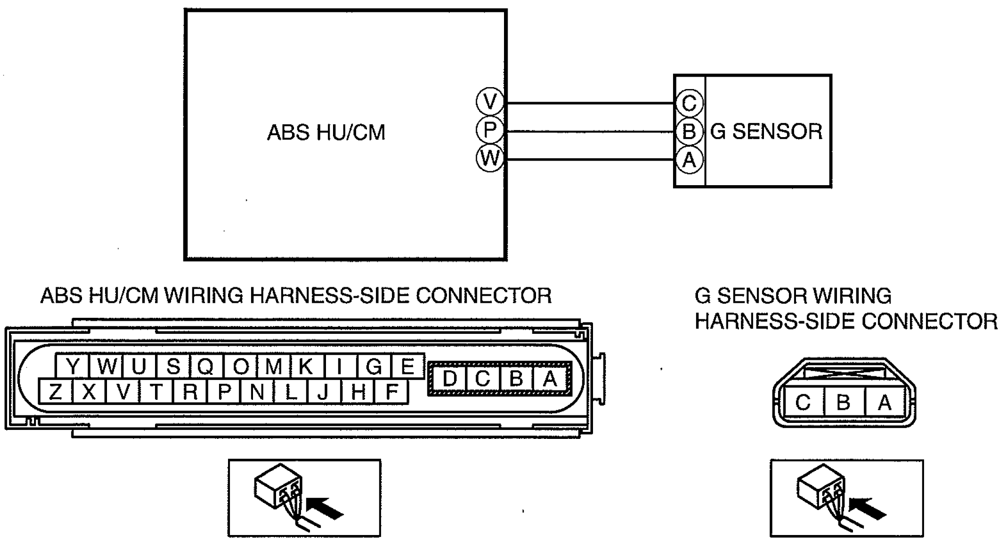
Diagnostic procedure

STEP	INSPECTION		ACTION
1	**INSPECT ABS HU/CM FOR MISMATCHED INSTALLATION** • Verify ABS HU/CM part number • Has a ABS HU/CM with the correct part number been installed?	Yes	Go to next step.
		No	Replace with correct ABS HU/CM part number, then go to next step.
2	**VERIFY CURRENT STATUS OF MALFUNCTION** • Clear DTC from memory. (See 04-02B-2 Clearing DTCs Procedures) • Is same DTC present?	Yes	Replace ABS HU/CM. (See 04-13B-3 ABS HU/CM REMOVAL/INSTALLATION [4W-ABS].)
		No	Go to next step.
3	**VERIFY THAT NO OTHER DTCS ARE PRESENT** • Are any other DTCs output?	Yes	Go to the applicable DTC inspection. (See 04-02B-4 DTC Table.)

DTC C1730, C1949, C1950 [4W-ABS]

dcf040200000w28

DTC C1730, C1949, C1950	G sensor
DETECTION CONDITION	• C1730 — Voltage to G sensor is detected out of range • C1949 — Monitor voltage of G sensor is detected at **4.7 V or more**, or **0.3 V or less** for more than **1 s**. • C1950 — G sensor 0-point correction value is default or more — Output voltage value from G sensor remains absolutely unchanged
POSSIBLE CAUSE	• Open circuit in harness between ABS HU/CM terminal V and G sensor terminal C • Open, short to power, or short to ground circuit in harness between ABS HU/CM terminal P and G sensor terminal B • Open circuit in harness between ABS HU/CM terminal W and G sensor terminal A • Malfunction of G sensor



Diagnostic procedure

STEP	INSPECTION		ACTION
1	**INSPECT G SENSOR POWER SUPPLY CIRCUIT FOR OPEN CIRCUIT** • Turn engine switch to ON (engine OFF). • Measure voltage between G sensor terminal C (harness side) and ground. • Is voltage **B+**?	Yes	Go to the next step.
		No	Repair or replace harness for open circuit between G sensor terminal C and engine switch, then go to Step 7.
2	**INSPECT G SENSOR GROUND CIRCUIT FOR OPEN CIRCUIT** • Turn engine switch to OFF. • Disconnect ABS HU/CM and G sensor connectors. • Inspect continuity between ABS HU/CM terminal W (harness side) and G sensor terminal A (harness side). • Is there continuity?	Yes	Go to the next step.
		No	Repair or replace harness for open circuit between ABS HU/CM terminal W and G sensor terminal A, then go to Step 7.
3	**INSPECT G SENSOR SIGNAL CIRCUIT FOR OPEN CIRCUIT** • Inspect continuity between ABS HU/CM terminal P (harness side) and G sensor terminal B (harness side). • Is there continuity?	Yes	Go to the next step.
		No	Repair or replace harness for open circuit between ABS HU/CM terminal P and G sensor terminal B, then go to Step 7.
4	**INSPECT G SENSOR SIGNAL CIRCUIT FOR SHORT TO POWER** • Turn engine switch to ON (engine OFF). • Measure voltage between ABS HU/CM terminal P (harness side) and ground. • Is voltage **B+**?	Yes	Repair or replace harness for short to power circuit between ABS HU/CM terminal P and G sensor terminal B, then go to Step 7.
		No	Go to the next step.

STEP	INSPECTION		ACTION
5	**INSPECT G SENSOR SIGNAL CIRCUIT FOR SHORT TO GROUND** • Turn engine switch to OFF. • Inspect continuity between ABS HU/CM terminal P (harness side) and ground. • Is there continuity?	Yes	Repair or replace harness for short to ground circuit between ABS HU/CM terminal P and G sensor terminal B, then go to Step 7.
		No	Go to the next step.
6	**INSPECT G SENSOR** • Inspect the G sensor. (See 04-13B-13 G SENSOR INSPECTION [4W-ABS]) • Is it normal?	Yes	Go to the next step.
		No	Replace the G sensor, then go to next step. (See 04-13B-12 G SENSOR REMOVAL/INSTALLATION [4W-ABS].)
7	**VERIFY DTC TROUBLESHOOTING COMPLETED** • Make sure to reconnect all disconnected connectors. • Clear the DTC from the memory. (See 04-02B-2 Clearing DTCs Procedures.) • Are the same DTCs present?	Yes	Replace the ABS HU/CM, then go to next step. (See 04-13B-3 ABS HU/CM REMOVAL/INSTALLATION [4W-ABS].)
		No	Go to the next step.
8	**VERIFY AFTER REPAIR PROCEDURE** • Are any other DTCs present?	Yes	Go to the applicable DTC inspection. (See 04-02B-4 DTC Table.)
		No	DTC troubleshooting completed.

04-03B SYMPTOM TROUBLESHOOTING [4-WHEEL ANTILOCK BRAKE SYSTEM (4W-ABS)]

SYSTEM WIRING DIAGRAM [4W-ABS] 04-03B–1
FOREWORD [4W-ABS] 04-03B–2
PRECAUTION [4W-ABS] 04-03B–2
SYMPTOM TROUBLESHOOTING [4W-ABS] 04-03B–5
NO.1 ABS WARNING LIGHT ILLUMINATE WHEN THE ENGINE SWITCH IS TURNED TO THE ON POSITION [4W-ABS] 04-03B–6
NO.2 ABS WARNING LIGHT DOES NOT ILLUMINATE WHEN THE ENGINE SWITCH IS TURNED TO THE ON POSITION [4W-ABS]..... 04-03B–7
NO.3 ABS WARNING LIGHT STAYS ON 4 S OR MORE WHEN THE ENGINE SWITCH IS TURNED TO THE ON POSITION [4W-ABS] 04-03B–7
NO.4 THERE IS MALFUNCTION IN THE SYSTEM EVEN THOUGH ABS WARNING LIGHT, DO NOT ILLUMINATE [4W-ABS]............. 04-03B–8

SYSTEM WIRING DIAGRAM [4W-ABS]

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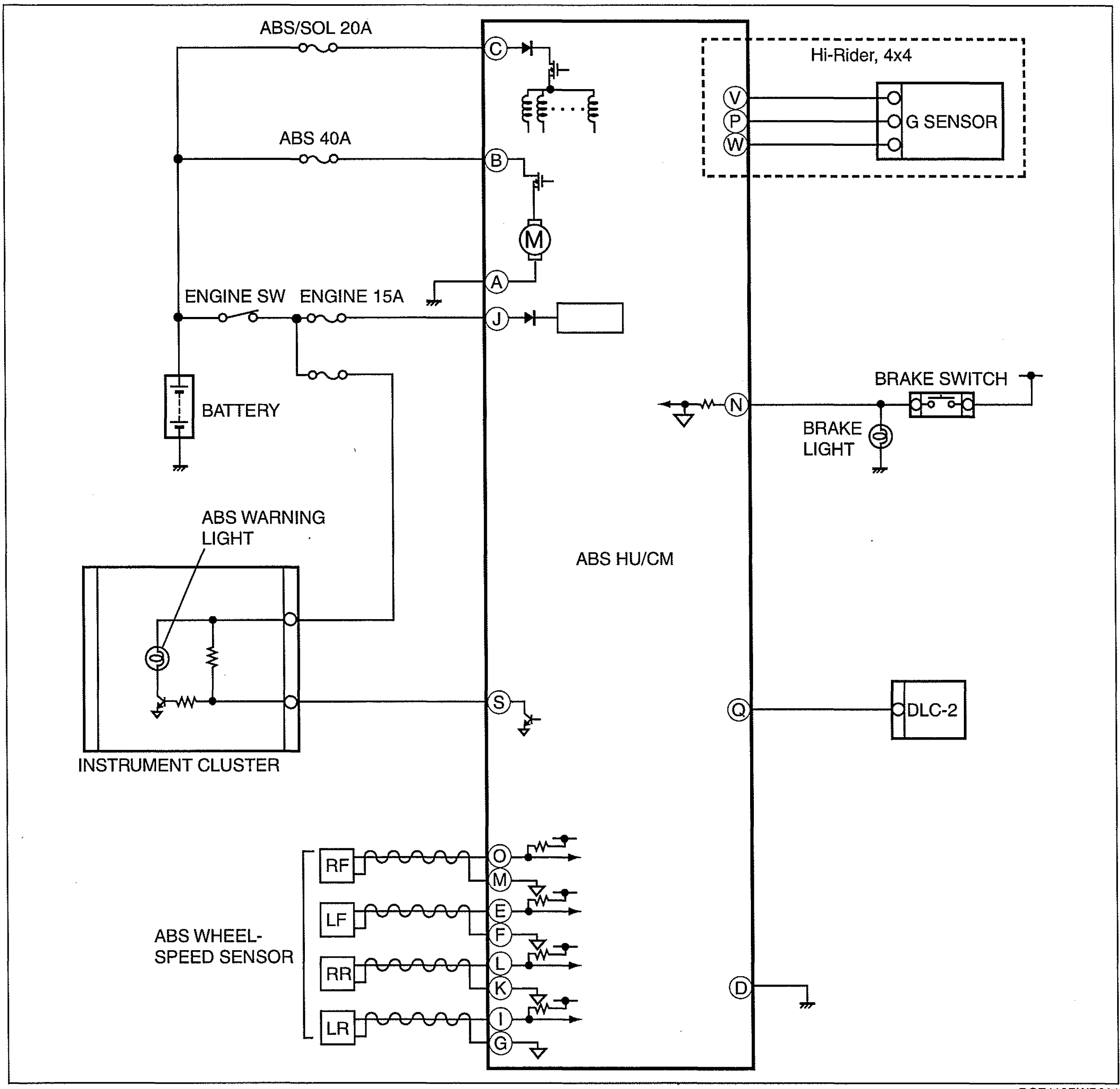

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SYMPTOM TROUBLESHOOTING [4-WHEEL ANTILOCK BRAKE SYSTEM (4W-ABS)]

FOREWORD [4W-ABS]

dcf040300000w23

- Before performing the steps in Symptom Troubleshooting, perform the On-board Diagnostic inspection. To inspect the DTC, follow the DTC Inspection steps. (See 04-02B-4 DTC Table.)

PRECAUTION [4W-ABS]

dcf040300000w24

1. Any one or a combination of the ABS warning illuminates even when the system is normal.

Warning lights that may illuminate and/or flash	Cases in which the light may illuminate	Conditions in which the light will go out	ABS control
• ABS warning light	When the front wheels are jacked up, stuck, or placed on a chassis roller, and only the front wheel ABS wheel-speed sensors are spun.	After turning engine switch off, vehicle is driven at speed **greater than 12 km/h {7.5 mph}** and normal operation is confirmed.	• ABS: Cuts control.
	Parking brake is not fully released while driving.		
	Brake drag.		
	Sudden acceleration/ deceleration.		
	Left/right or front/rear tires are different. (Size, radius, tire pressure, or wear is other than that listed on tire label.)		
	Battery voltage at ABS HU/CM ignition terminal J drops **below approx. 10 V.**(*1)	Battery voltage rises **above approx. 10 V.**	ABS: Cuts control.
	Battery voltage at ABS HU/CM ignition terminal J rises **above approx. 16 V.**(*2)	Battery voltage drops **below approx. 16 V.**	ABS: Cuts control.

*1 : If battery voltage drops **below 10 V** while vehicle speed is **greater than 6 km/h {3.7 mph}**, ABS HU/CM records DTC B1318.

*2 : If battery voltage rises **above 16 V**, ABS HU/CM records DTC B1317.

2. Precautions during servicing of ABS
   The ABS is composed of electrical and mechanical parts. It is necessary to categorize malfunctions as being either electrical or hydraulic when performing troubleshooting.
   (1) Malfunctions in electrical system
   - The ABS hydraulic unit and control module (ABS HU/CM) has an on-board diagnostic function. With this function, the ABS warning light and/or BRAKE system warning light will illuminate when there is a problem in the electrical system. Also, past and present malfunctions are recorded in the ABS HU/CM. This function can find malfunctions that do not occur during periodic inspections. Connect the current diagnostic tool to the DLC-2. The stored malfunctions will be displayed in the order of occurrence. To find out the causes of ABS malfunctions, use these on-board diagnostic results.
   - If a malfunction occurred in the past but is now normal, the cause is likely a temporary poor connection of the wiring harness. The ABS HU/CM usually operates normally. Be careful when searching for the cause of malfunction.
   - After repair, it is necessary to clear the DTC from the ABS HU/CM memory. Also, if the ABS related parts have been replaced, verify that the no DTC has been displayed after repairs.
   - After repairing the ABS wheel-speed sensor or ABS sensor rotor, or after replacing the ABS CM (ABS motor or ABS motor relay or solenoid valve), the ABS warning light may not go out even when the engine switch is turned to the ON position. In this case, drive the vehicle at a speed of **12 km/h {7.5 mph} or more**, make sure that ABS warning light goes out, and then clear the DTC.
   - When repairing, if the ABS related connectors are disconnected and the engine switch is turned to the ON position, the ABS CM will mistakenly detect a fault and record it as a malfunction.
   - To protect the ABS HU/CM, make sure the engine switch is turned off before connecting or disconnecting the ABS CM connector.

   (2) Malfunctions in hydraulic system
   - Symptoms in a hydraulic system malfunction are similar to those in a conventional brake malfunction. However, it is necessary to determine if the malfunction is in an ABS component or the conventional brake system.
   - The ABS hydraulic unit contains delicate mechanical parts. If foreign material enters into the component, the ABS may fail to operate. Also, it will likely become extremely difficult to find the location of the malfunction in the event that the brakes operate but the ABS does not. Make sure foreign material does not enter when servicing the ABS (e.g. brake fluid replacement, pipe removal).

Intermittent Concern Troubleshooting

Vibration method

- If a malfunction occurs or becomes worse while driving on a rough road or when engine is vibrating, perform the following steps.

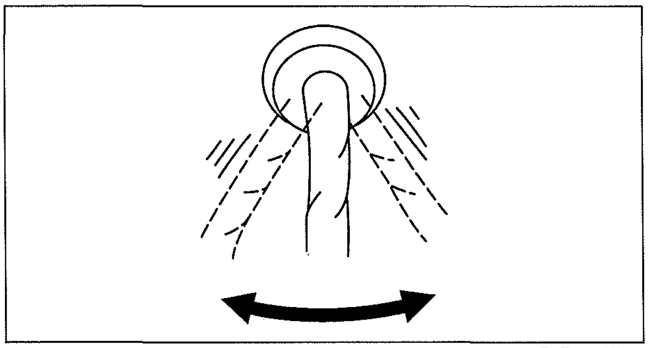
Note
- There are several reasons why vehicle or engine vibration could cause an electrical malfunction. Inspect the following:
  - — Connectors not fully seated.
  - — Wiring harness not having full play.
  - — Wiring harness laying across brackets or moving parts.
  - — Wiring harness routed too close to hot parts.
- An improperly routed, improperly clamped, or loose wiring harness can cause pinching between parts.
- The connector joints, points of vibration, and places where wiring harness pass through the firewall, body panels, etc. are the major areas to be inspected.

Inspection method for switch connectors or wiring harnesses

1. Connect the current diagnostic tool to the DLC-2.
2. Turn the engine switch to the ON position (Engine OFF).

Note
- If the engine starts and runs, perform the following steps at idle.

3. Access PIDs for the switch you are inspecting.
4. Turn the switch on manually.
5. Slightly shake each connector or wiring harness vertically and horizontally while monitoring the PID.
   - If the PID value is unstable, inspect poor connection.



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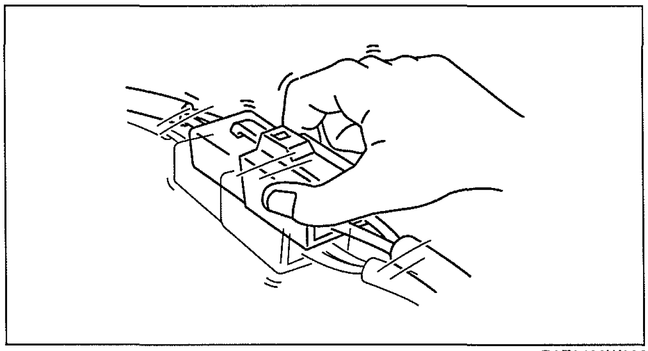
Inspection method for sensor connectors or wiring harnesses

1. Connect the current diagnostic tool to the DLC-2.
2. Turn the engine switch to the ON position (Engine OFF).

Note
- If the engine starts and runs, perform the following steps at idle.

3. Access PIDs for the switch you are inspecting.
4. Slightly shake each connector or wiring harness vertically and horizontally while monitoring the PID.
   - If the PID value is unstable, inspect poor connection.



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Inspection method for sensors

1. Connect the current diagnostic tool to the DLC-2.
2. Turn the engine switch to the ON position (Engine OFF).

Note

- If the engine starts and runs, perform the following steps at idle.

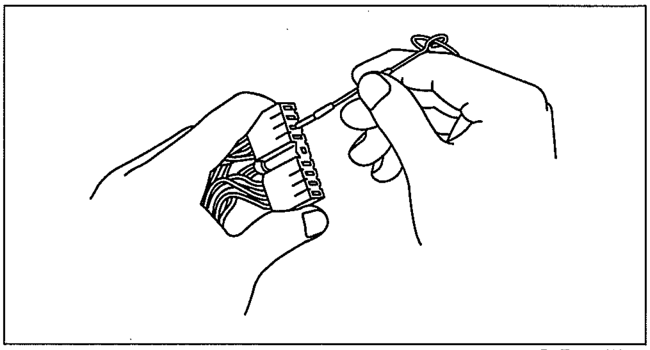
3. Access PIDs for the switch you are inspecting.
4. Vibrate the sensor slightly with your finger.
   - If the PID value is unstable or a malfunction occurs, inspect the sensor for poor connection and/or poor mounting.

Malfunction data monitor method

1. Perform malfunction reappearance test according to malfunction reappearance mode and malfunction data monitor. The malfunction cause is found in the malfunction data.

Connector terminal inspection method

1. Inspect the connection of each female terminal.
2. Insert the male terminal to the female terminal and Inspect the female terminal for looseness.



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SYMPTOM TROUBLESHOOTING [4-WHEEL ANTILOCK BRAKE SYSTEM (4W-ABS)]

SYMPTOM TROUBLESHOOTING [4W-ABS]

dcf040300000w25

- Verify the symptoms, and perform troubleshooting according to the appropriate number.

No.	Symptom
1	ABS warning light illuminate when the engine switch is turned to the ON position.
2	ABS warning light does not illuminate when the engine switch is turned to the ON position.
3	ABS warning light stays on **4 s or more** when the engine switch is turned to the ON position.
4	There is a malfunction in the system even through ABS warning light, do not illuminate.

x: Applicable

Troubleshooting item \ Possible factor		ABS HU/CM	Instrument cluster	Charging system	Instrument cluster power supply (terminal 2A)	Instrument cluster GND	Conventional brakes	Brake pipe routing	Open or short circuit in ABS/brake warning light control circuit
1	ABS warning light illuminate when the engine switch is turned to the ON position.	X	X		X	X			
2	ABS warning light does not illuminate when the engine switch is turned to the ON position.		X						X
3	ABS warning light stays on 4 s or more when the engine switch is turned to the ON position.	X	X	X					X
4	There is a malfunction in the system even though ABS warning light, do not illuminate.	X					X	X	

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SYMPTOM TROUBLESHOOTING [4-WHEEL ANTILOCK BRAKE SYSTEM (4W-ABS)]

NO.1 ABS WARNING LIGHT ILLUMINATE WHEN THE ENGINE SWITCH IS TURNED TO THE ON POSITION [4W-ABS]

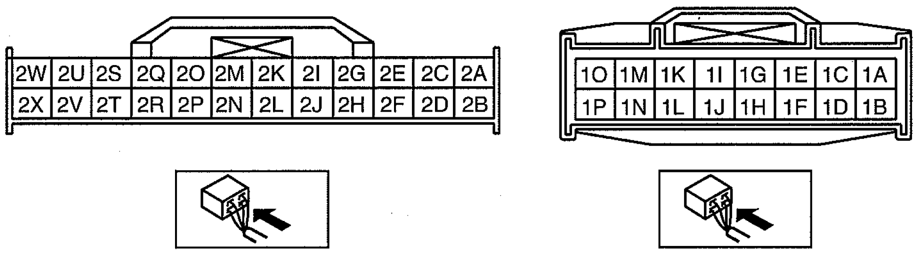
dcf040300000w26

1	ABS warning light illuminate when the engine switch is turned to the ON position.
[TROUBLESHOOTING HINTS] • Instrument cluster or ABS HU/CM malfunction	

Diagnostic procedure

STEP	INSPECTION		ACTION
1	**INSPECT FOR DTCs IN ABS HU/CM** • Have DTCs been stored in memory?	Yes	Perform the applicable DTC inspection. (See 04-02B-4 DTC Table.)
		No	Inspect the instrument cluster. If the instrument cluster is normal, inspect CAN communication. If instrument cluster has a malfunction, go to the next step.
2	**VERIFY WHETHER MALFUNCTION IS IN WARNING LIGHTS AND INDICATOR LIGHT'S COMMON POWER SUPPLY, OR IN OTHER WARNING LIGHTS AND INDICATOR LIGHTS** • Do other warning and indicator lights illuminate when the engine switch is turned to the ON position?	Yes	Replace the instrument cluster. (open circuit in instrument cluster) (See 09-22-2 INSTRUMENT CLUSTER REMOVAL/INSTALLATION.)
		No	Go to the next step.
3	**INSPECT INSTRUMENT CLUSTER POWER SUPPLY FUSE** • Is the instrument cluster ignition power supply fuse normal?	Yes	Go to the next step.
		No	Inspect for a short to ground on circuit of blown fuse. Repair or replace if necessary. Install appropriate amperage fuse.
*4	**VERIFY WHETHER MALFUNCTION IS IN WIRING HARNESS (BETWEEN INSTRUMENT CLUSTER POWER SUPPLY AND INSTRUMENT CLUSTER FOR CONTINUITY) OR INSTRUMENT CLUSTER** • Turn the engine switch to the ON position. • Measure voltage at instrument cluster connector terminal 2A. • Is the voltage **approx. 12 V?**	Yes	Replace the instrument cluster (open circuit in instrument cluster). (See 09-22-2 INSTRUMENT CLUSTER REMOVAL/INSTALLATION.)
		No	Inspect for open circuit in wiring harness between the instrument cluster and ground. Repair or replace if necessary. Replace the ABS HU/CM. (See 04-13B-3 ABS HU/CM REMOVAL/INSTALLATION [4W-ABS].)

INSTRUMENT CLUSTER WIRING HARNESS-SIDE CONNECTOR



- When performing an asterisked (*) troubleshooting inspection, slightly shake the wiring harness and connectors while performing the inspection to discover whether poor contact points are the cause of any intermittent malfunction. If there is a problem, verify that the connectors, terminals and wiring harness are connected correctly and undamaged.

SYMPTOM TROUBLESHOOTING [4-WHEEL ANTILOCK BRAKE SYSTEM (4W-ABS)]

NO.2 ABS WARNING LIGHT DOES NOT ILLUMINATE WHEN THE ENGINE SWITCH IS TURNED TO THE ON POSITION [4W-ABS]

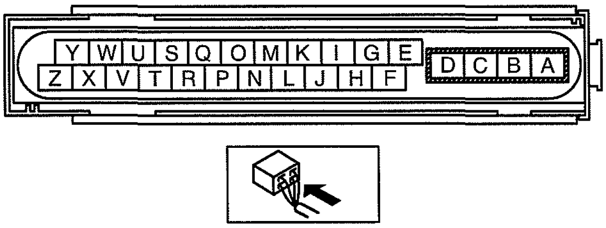
dcf040300000w27

2	ABS warning light does not illuminate when the engine switch is turned to the ON position.
[TROUBLESHOOTING HINTS] • **Instrument cluster malfunction** • **Open circuit between ABS HU/CM and instrument cluster**	

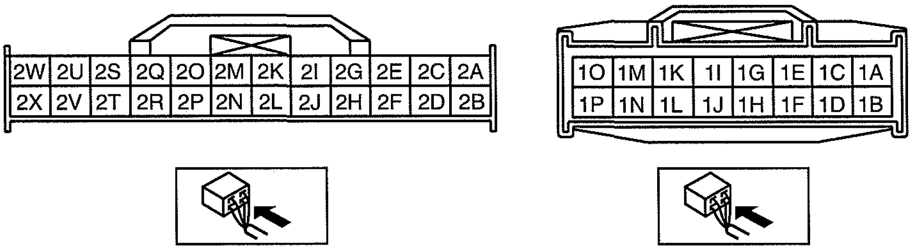
Diagnostic procedure

STEP	INSPECTION		ACTION
1	**VERIFY WHETHER MALFUNCTION IS IN WIRING LIGHT OR CONTROL CIRCUIT** • Turn the engine switch OFF. • Disconnect ABS HU/CM connector. • Short to ground at ABS HU/CM harness-side connector terminal S using a jumper wire. • Turn the engine switch ON. • Does the ABS warning light illuminate?	Yes	Inspect for open circuit between ABS HU/CM terminal S and instrument cluster terminal 2L.
		No	Replace instrument cluster. (See 09-22-2 INSTRUMENT CLUSTER REMOVAL/ INSTALLATION.)

ABS HU/CM WIRING HARNESS-SIDE CONNECTOR



INSTRUMENT CLUSTER WIRING HARNESS-SIDE CONNECTOR



NO.3 ABS WARNING LIGHT STAYS ON 4 S OR MORE WHEN THE ENGINE SWITCH IS TURNED TO THE ON POSITION [4W-ABS]

dcf040300000w29

3	ABS warning light stays on 4 s or more when the engine switch is turned to the ON position.
[TROUBLESHOOTING HINTS] • **ABS CM detects ABS system malfunction.** • **Short to ground in ABS warning light control circuit**	

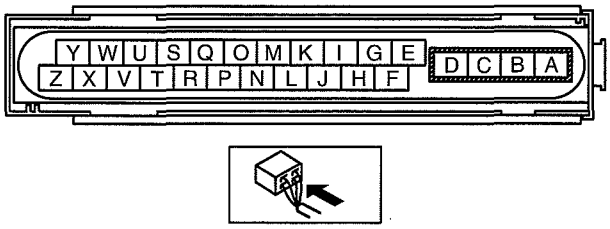
Diagnostic procedure

STEP	INSPECTION		ACTION
1	**INSPECT FOR DTCs IN ABS HU/CM** • Have DTCs been stored in memory?	Yes	Perform the applicable DTC inspection. (See 04-02B-4 DTC Table.)
		No	Inspect the instrument cluster If the instrument cluster is normal, go to the next step. If the instrument cluster has a malfunction, repair the instrument cluster, go to the next step.
2	**VERIFY WHETHER MALFUNCTION IS IN WARNING LIGHT OR CONTROL CIRCUIT** • Turn the engine switch OFF. • Disconnect the ABS HU/CM connector. • Turn the engine switch to ON. • Does the ABS warning light stay on **4 s or more**?	Yes	Go to the next step.
		No	Replace the ABS HU/CM. (See 04-13B-3 ABS HU/CM REMOVAL/INSTALLATION [4W-ABS].)

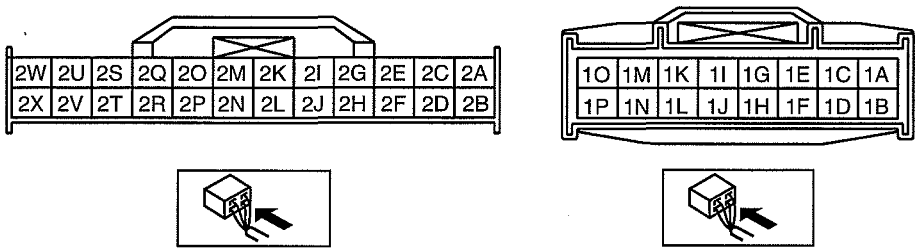
SYMPTOM TROUBLESHOOTING [4-WHEEL ANTILOCK BRAKE SYSTEM (4W-ABS)]

STEP	INSPECTION		ACTION
3	**INSPECT ABS WARNING LIGHT CONTROL CIRCUIT FOR SHORT TO GROUND** • Turn the engine switch OFF. • ABS HU/CM connector disconnected. • Disconnect the instrument cluster connector. • Verify continuity between ABS HU/CM harness-side connector terminal S and body ground. • Is there any continuity?	Yes	Repair or replace for short to ground.
		No	Replace instrument cluster. (See 09-22-2 INSTRUMENT CLUSTER REMOVAL/ INSTALLATION.)

ABS HU/CM WIRING HARNESS-SIDE CONNECTOR



INSTRUMENT CLUSTER WIRING HARNESS-SIDE CONNECTOR



NO.4 THERE IS MALFUNCTION IN THE SYSTEM EVEN THOUGH ABS WARNING LIGHT, DO NOT ILLUMINATE [4W-ABS]

dcf040300000w31

4	There is a malfunction in system even though ABS warning light, do not illuminate.
[TROUBLESHOOTING HINTS] • **There is a difference in size or air pressure between the front and rear tires.**	

Diagnostic procedure

STEP	INSPECTION		ACTION
1	**INSPECT FOR DTCs IN ABS HU/CM** Have DTCs been stored in memory?	Yes	Perform the applicable DTC inspection. (See 04-02B-4 DTC Table.)
		No	Go to the next step.
2	**INSPECT ABS HYDRAULIC UNIT** Perform the ABS hydraulic unit system inspection. Is the system normal?	Yes	Inspect the conventional brake system.
		No	**If the wheels do not rotate:** Replace the ABS HU/CM. (See 04-13B-3 ABS HU/CM REMOVAL/INSTALLATION [4W-ABS].) **If the wheels rotate but order in which wheels rotate is incorrect:** Inspect the brake pipe passage to the ABS HU/CM.

04-10 GENERAL PROCEDURES

GENERAL PROCEDURES (BRAKE). . . 04-10–1

GENERAL PROCEDURES (BRAKE)

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Wheel and Tire Installation

1. When installing the wheels and tires, tighten the wheel nuts in a criss-cross pattern to the following tightening torque.

Tightening torque
88.2—117.6 N·m {9.00—11.99 Kgf·m, 65.06—86.73 ft·lbf}

Brake Lines Disconnection

1. If any brake line has been disconnected during the procedures, add brake fluid, bleed the brakes, and inspect for leakage after the procedure has been completed.

Caution
- **Brake fluid will damage painted surfaces. Be careful not to spill any on painted surfaces. If it is spilled, wipe it off immediately.**

Brake Pipe Flare Nut Tightening

1. Tighten the brake pipe flare nut using the **SST** (49 0259 770B).

Connector Disconnection

1. Disconnect the negative battery cable before performing any work that requires handling of connectors. (See 01-17-2 BATTERY REMOVAL/INSTALLATION.)

ABS Related Parts

1. Make sure that there are no DTCs in the ABS memory after working on ABS related parts. If there are any DTCs in the memory, clear them.

04-11 CONVENTIONAL BRAKE SYSTEM

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CONVENTIONAL BRAKE SYSTEM LOCATION INDEX

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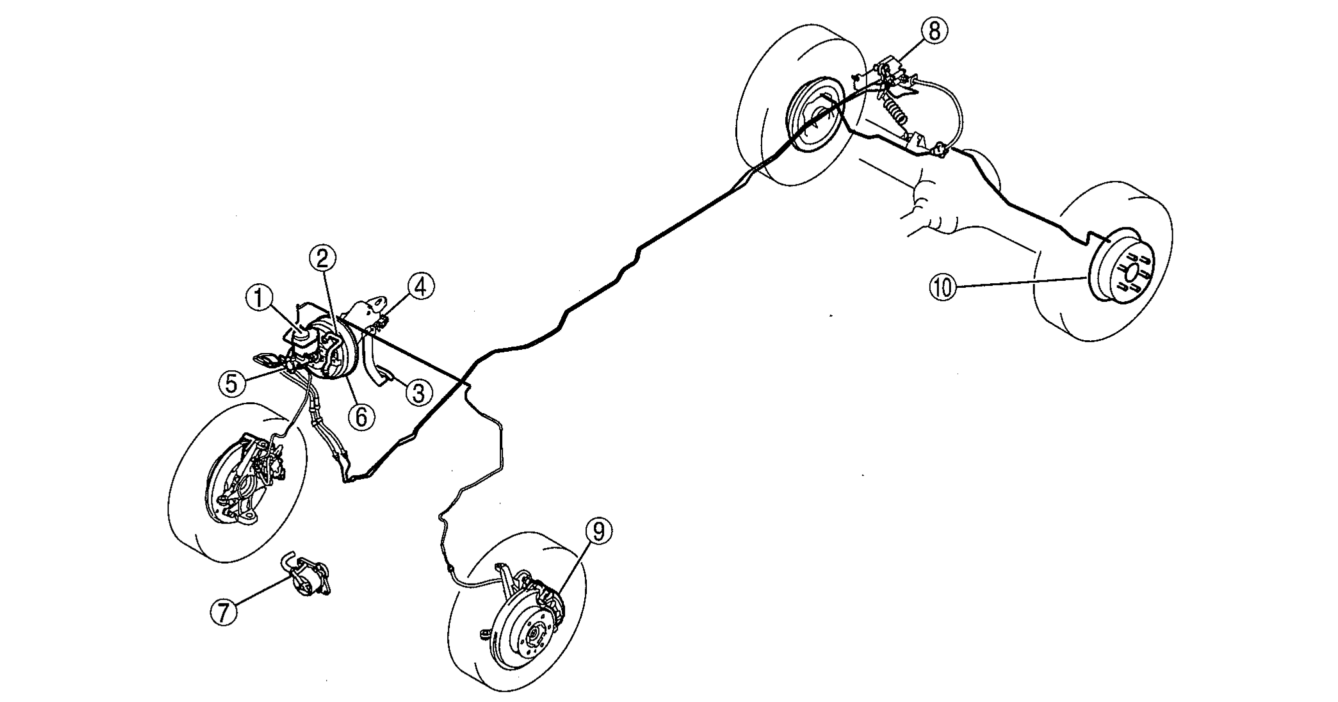

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1	Brake fluid (See 04-11-3 AIR BLEEDING.)
2	Vacuum line (See 04-11-3 VACUUM LINE INSPECTION.)
3	Brake pedal (See 04-11-4 BRAKE PEDAL INSPECTION.) (See 04-11-5 BRAKE PEDAL REMOVAL/ INSTALLATION.)
4	Brake switch (See 04-11-5 BRAKE SWITCH REMOVAL/ INSTALLATION.) (See 04-11-6 BRAKE SWITCH INSPECTION.)
5	Master cylinder (See 04-11-6 MASTER CYLINDER REMOVAL/ INSTALLATION.) (See 04-11-7 BRAKE FLUID LEVEL SENSOR INSPECTION.) (See 04-11-8 MASTER CYLINDER DISASSEMBLY/ ASSEMBLY.)
6	Power brake unit (See 04-11-9 POWER BRAKE UNIT INSPECTION.) (See 04-11-10 POWER BRAKE UNIT REMOVAL/ INSTALLATION.)
7	Vacuum pump (See 04-11-11 VACUUM PUMP INSPECTION.) (See 04-11-11 VACUUM PUMP REMOVAL/ INSTALLATION.)
8	Load sensing proportioning valve (LSPV) (See 04-11-12 LOAD SENSING PROPORTIONING VALVE (LSPV) INSPECTION.) (See 04-11-13 LOAD SENSING PROPORTIONING VALVE (LSPV) ADJUSTMENT.) (See 04-11-14 LOAD SENSING PROPORTIONING VALVE (LSPV) REMOVAL/INSTALLATION.)
9	Front brake (disc) (See 04-11-15 FRONT BRAKE (DISC) INSPECTION.) (See 04-11-16 FRONT BRAKE (DISC) REMOVAL/ INSTALLATION [4x2 (EXCEPT Hi-Rider)].) (See 04-11-17 FRONT BRAKE (DISC) REMOVAL/ INSTALLATION [Hi-Rider, 4x4].) (See 04-11-19 DISC PAD (FRONT) REPLACEMENT [4x2 (EXCEPT Hi-Rider)].) (See 04-11-20 DISC PAD (FRONT) REPLACEMENT [Hi-Rider, 4x4].) (See 04-11-20 CALIPER (FRONT) DISASSEMBLY/ ASSEMBLY [4x2 (EXCEPT Hi-Rider)].) (See 04-11-21 CALIPER (FRONT) DISASSEMBLY/ ASSEMBLY [Hi-Rider, 4x4].)
10	Rear brake (drum) (See 04-11-22 REAR BRAKE (DRUM) INSPECTION.) (See 04-11-23 REAR BRAKE (DRUM) REMOVAL/ INSTALLATION.) (See 04-11-25 WHEEL CYLINDER DISASSEMBLY/ ASSEMBLY.)

AIR BLEEDING

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Caution

- **Brake fluid will damage painted surfaces. Be careful not to spill any on painted surfaces. If it is spilled, wipe it off immediately.**

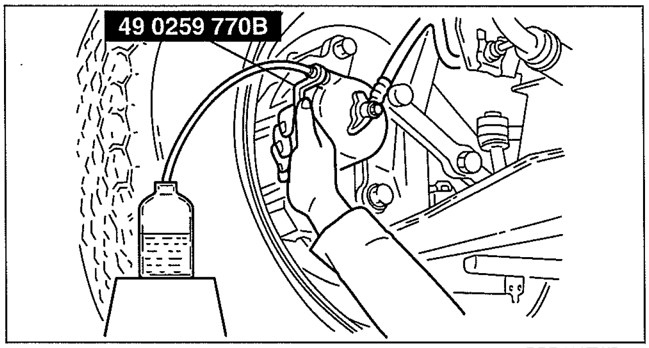
Note

- Keep the fluid level in the reserve tank at 3/4 full or more during the air bleeding.
- Begin air bleeding with the brake caliper that is furthest from the master cylinder.

Brake fluid type
SAE J1703, FMVSS 116 DOT-3

1. Remove the bleeder cap on the brake caliper, and attach a vinyl tube to the bleeder screw.
2. Place the other end of the vinyl tube in a clear container and fill the container with fluid during air bleeding.
3. Working with two people, one should pump the brake pedal several times and depress and hold the pedal down.
4. While the brake pedal is depressed, the other should loosen the bleeder screw using the **SST**, drain out any fluid containing air bubbles, and tighten the bleeder screw.

   Tightening torque
   Front (4x2 (except Hi-Rider)): 9—14 N·m {92—142 kgf·cm, 80—123 in·lbf}
   Front (Hi-Rider, 4x4): 5.9—8.8 N·m {61—89 kgf·cm, 53—77 in·lbf}
   Rear: 5.9—8.8 N·m {61—89 kgf·cm, 53—77 in·lbf}



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5. Repeat Steps 3 and 4 until no air bubbles are seen.
6. Perform air bleeding as described in the above procedures for all brake calipers.
7. After air bleeding, inspect the following:
   - Brake operation
   - Fluid leakage
   - Fluid level

VACUUM LINE INSPECTION

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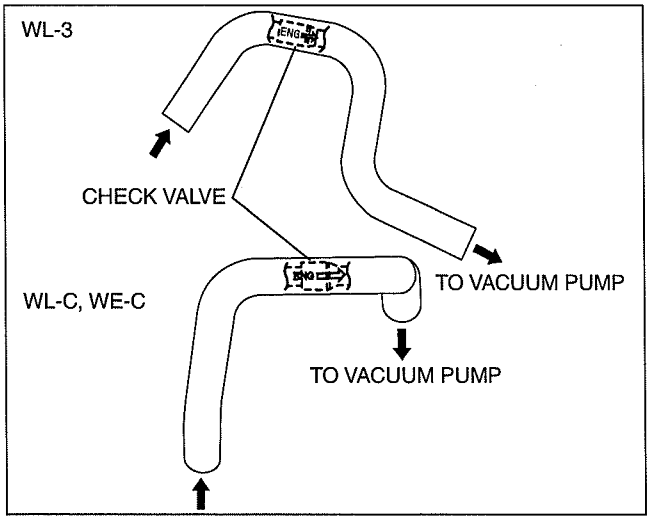
1. Remove the vacuum hose.
2. Verify that air can be blown from the power brake unit side of the vacuum hose towards the vacuum pump side, and that air cannot be blown in the opposite direction.
   - If there is any malfunction of the inner check valve, replace it together with the vacuum hose as a single unit.



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04

BRAKE PEDAL INSPECTION

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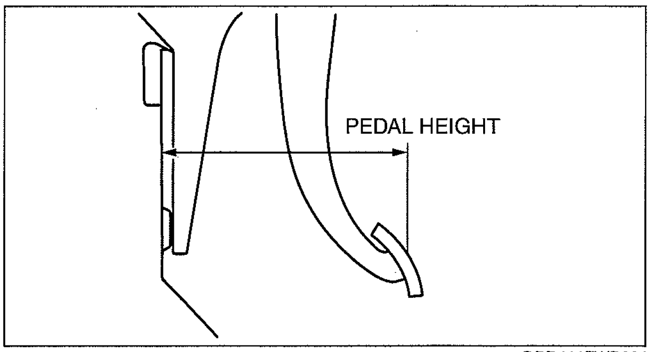
Brake pedal Height Inspection

1. Measure the distance from the center of the upper surface of the pedal pad to the floor panel and verify that it is as specified.
   - If not within the specification, replace the brake pedal.

Brake pedal height
214—219 mm {8.43—8.62 in}



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Brake Pedal Height Adjustment

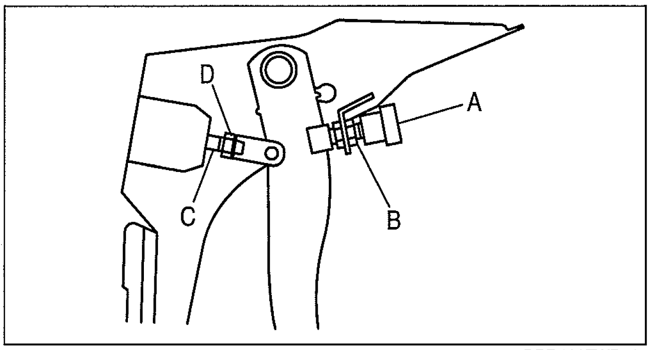
1. Disconnect the brake switch connector.
2. Loosen locknut B and turn switch A until it does not contact the pedal.
3. Loosen locknut D and turn rod C to adjust the height.
4. Tighten the turn rod C with locknut D.

Tightening torque
24—34 N·m {2.5—3.4 kgf·m, 18—25 ft·lbf}

5. Tighten the bolt with locknut B so that clearance between the bolt for brake switch A and pedal stopper is within the specification.

Specification
0.1—1.0 mm {0.004—0.039}
Tightening torque
13.7—17.6 N·m {140—179 kgf·cm, 122—155 in·lbf}

6. Connect the brake switch connector.
7. After adjustment, inspect the pedal play and the brake light operation.



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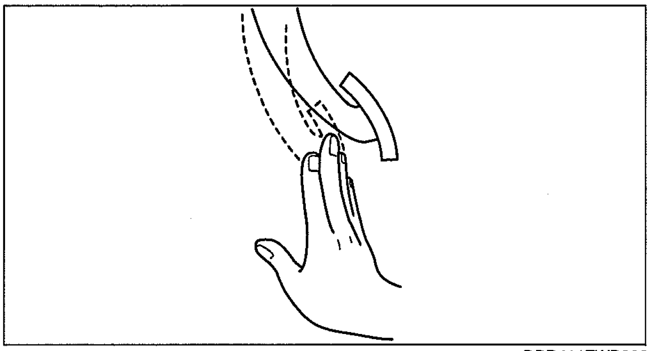
Brake Pedal Play Inspection

1. Pump the pedal several times to release the vacuum in the power brake unit.
2. Remove the snap pin, verify that the holes in the fork and in the pedal are aligned, and reinstall the pin.
3. Gently depress the pedal by hand until resistance is felt (**approx. 20 N {2.0 kgf, 4.5 lbf}**) and measure the pedal play.

Standard pedal play
3.0—8.0 mm {0.12—0.31 in}



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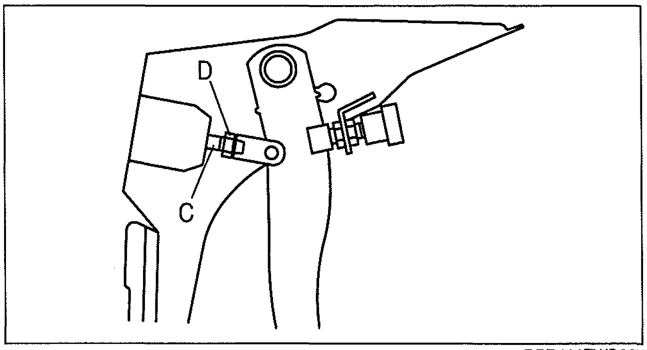
Brake Pedal Play Adjustment

1. Remove the snap pin and clevis pin.
2. Loosen locknut D and turn rod C to align the holes in the fork and in the pedal.
3. Install the clevis pin ant the snap pin.
4. Tighten locknut D.

   Tightening torque
   24—34 N·m {2.5—3.4 kgf·m, 18—25 ft·lbf}

5. After adjustment, inspect the pedal height and the brake light operation.



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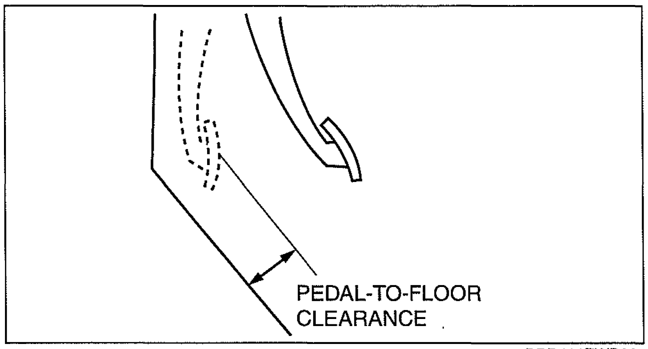
Pedal-to-floor Clearance Inspection

1. Start the engine and depress the brake pedal with a force of **589 N {60 kgf, 132 lbf}**.
2. Measure the distance from the center of the upper surface of the pedal pad to the floor covering and verify that it is as specified.
   - If it is less than the specification, inspect the following:
     — Air in brake system
     — Malfunction of the automatic adjuster
     — Excessive shoe clearance

   Standard pedal-to-floor clearance
   105 mm {4.14 in} or more



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04

BRAKE PEDAL REMOVAL/INSTALLATION

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1. The brake pedal is mounted on the bracket integrated with the clutch pedal and accelerator pedal. Refer to section 01, INTAKE AIR SYSTEM, ACCELERATOR PEDAL REMOVAL/INSTALLATION for the brake pedal removal/installation. (See 01-13A-9 ACCELERATOR PEDAL COMPONENT REMOVAL/INSTALLATION [WL-3].) (See 01-13B-11 ACCELERATOR PEDAL COMPONENT REMOVAL/INSTALLATION [WL-C, WE-C].)

BRAKE SWITCH REMOVAL/INSTALLATION

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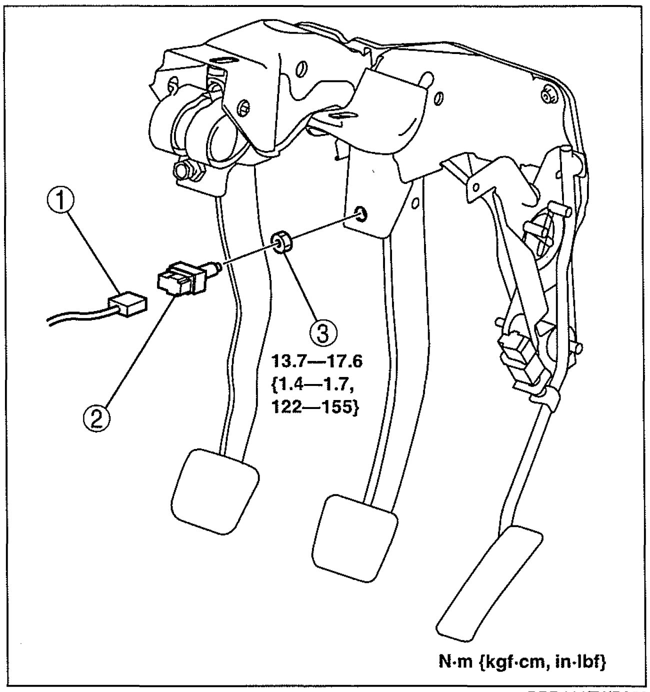
1. Remove in the order indicated in the table.

1	Brake switch connector
2	Brake switch
3	Nut

2. Install in the reverse order of removal.
3. After installation, inspect the brake pedal. (See 04-11-4 BRAKE PEDAL INSPECTION.)



DBR411ZWB053

BRAKE SWITCH INSPECTION

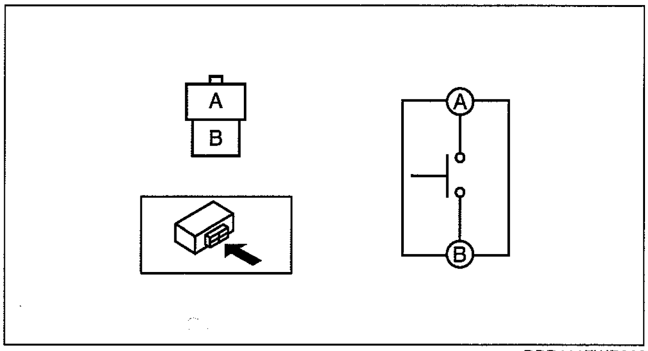
dcf041166490w02

1. Remove the lower panel.
2. Disconnect the brake switch connector.
3. Verify that the continuity is as indicated in the table.
    - If not as indicated in the table, replace the brake switch. (See 04-11-5 BRAKE SWITCH REMOVAL/INSTALLATION.)

O—O : Continuity

Condition	Terminal A	Terminal B
When the brake pedal is depressed	O—	—O
When the brake pedal is not depressed		

DBR411ZWB052

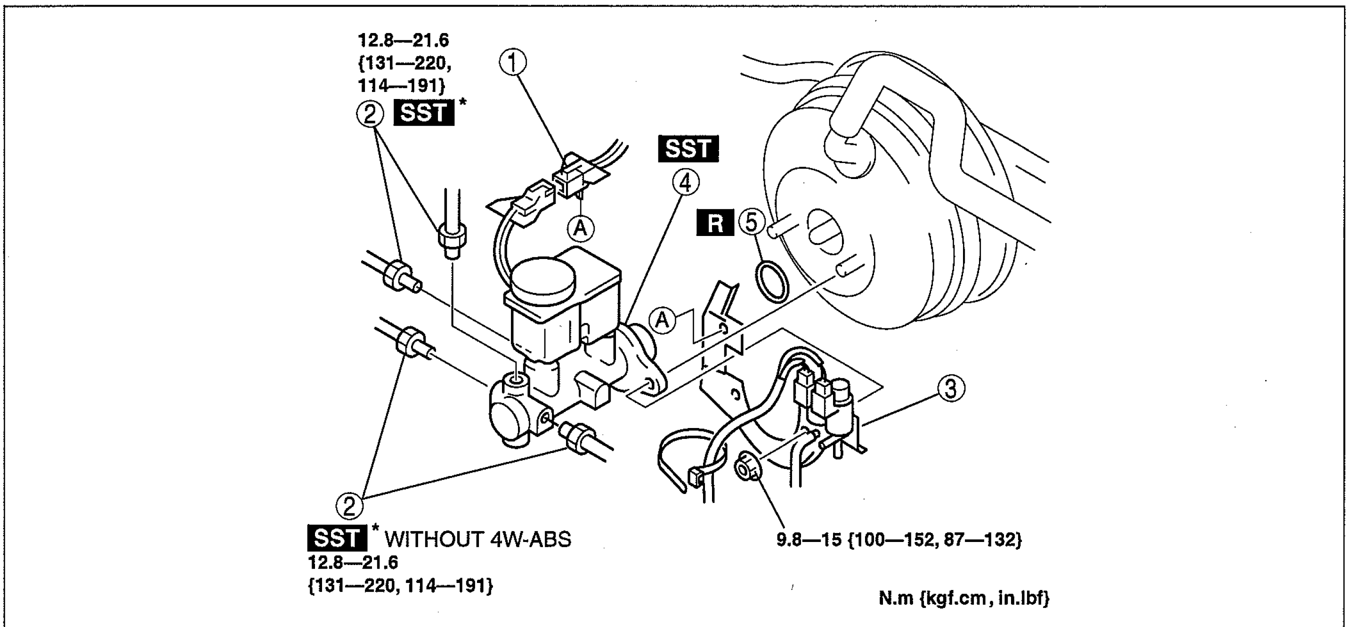


DBR411ZWB009

MASTER CYLINDER REMOVAL/INSTALLATION

dcf041143400w01

1. Remove in the order indicated in the table.
2. Install in the reverse order of removal.



arnffw00001966

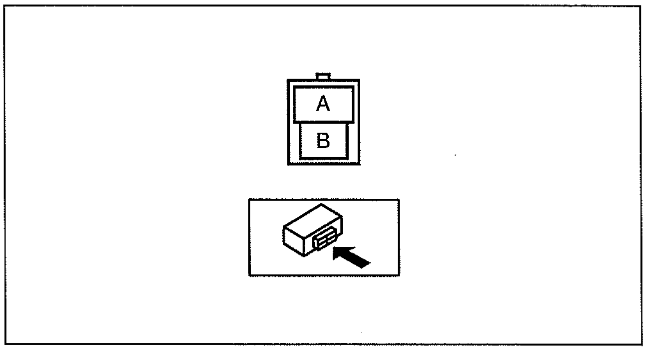
1	Brake fluid level sensor connector
2	Brake pipe
3	Bracket

4	Master cylinder
5	O-ring (AT)

BRAKE FLUID LEVEL SENSOR INSPECTION

dcf041143400w02

1. Disconnect the brake fluid level sensor connector from the master cylinder.
2. Inspect for continuity according to fluid level between the brake fluid level sensor terminals.
    - If not as indicated in the table, replace the brake fluid level sensor. (See 04-11-6 MASTER CYLINDER REMOVAL/ INSTALLATION.)



DBR411ZWB017

○—○: Continuity

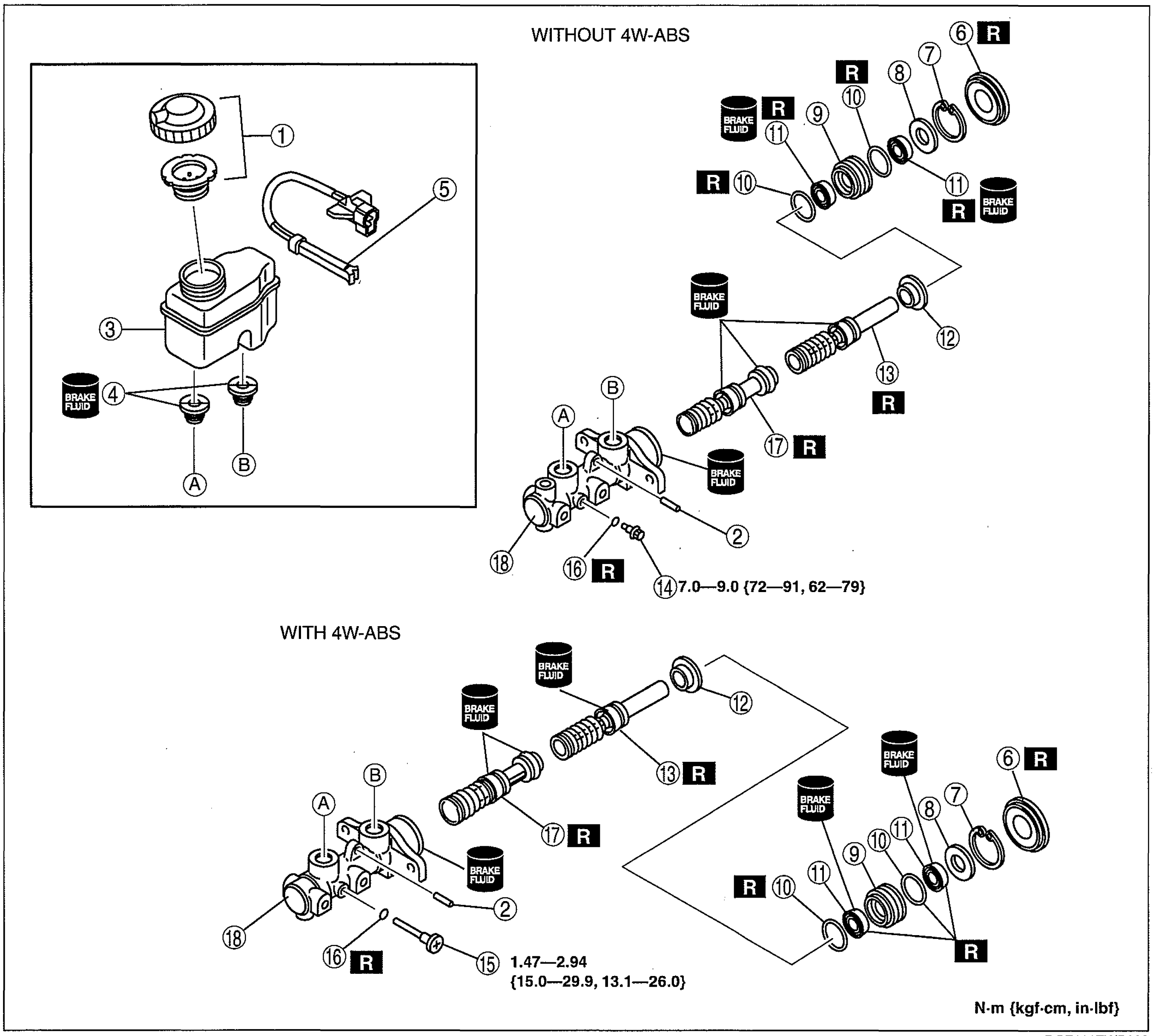
Condition	Terminal	
	A	B
Above MIN		
Below MIN	○—	—○

CHU0411W017

MASTER CYLINDER DISASSEMBLY/ASSEMBLY

dcf041143400w03

1. After removing the brake fluid, disassemble in the order indicated in the table.
2. Assemble in the reverse order of disassembly.



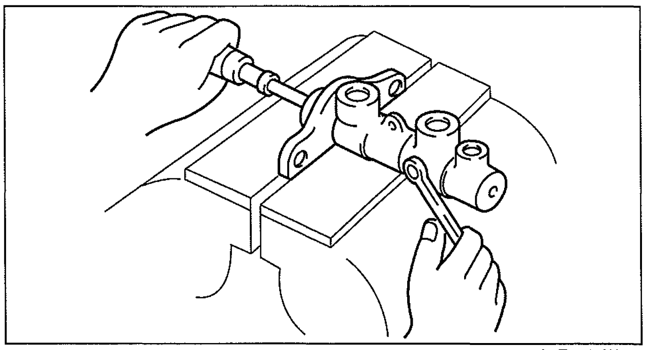
DCF411ZWB003

1	Cap set
2	Pin
3	Reservoir
4	Joint bushing
5	Brake fluid level sensor
6	Seal and plate assembly
7	Snap ring
8	Spacer
9	Piston guide
10	O-ring
11	Cup

12	Primary piston stopper
13	Primary piston
14	Stop screw (without 4W-ABS) (See 04-11-9 Stop Screw (without 4W-ABS) Assembly Note)
15	Stop screw (with 4W-ABS) (See 04-11-9 Stop Screw (with 4W-ABS) Assembly Note.)
16	Gasket
17	Secondary piston
18	Master cylinder body

Stop Screw (with 4W-ABS) Assembly Note

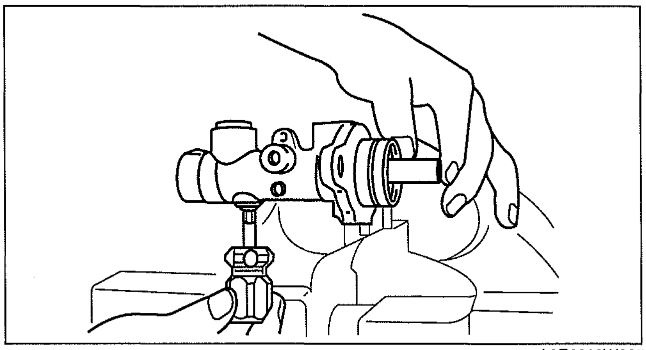
1. Install the secondary piston component with the piston hole facing the stop pin.
2. Install and tighten a new gasket and the stop pin.
3. Push and release the piston to verify that it is held by the stop pin.



A6E6912W043

Stop Screw (without 4W-ABS) Assembly Note

1. Push the primary piston component in fully.
2. Install and tighten a new gasket and the stop screw.
3. Push and release the piston to verify that it is held by the stop screw.



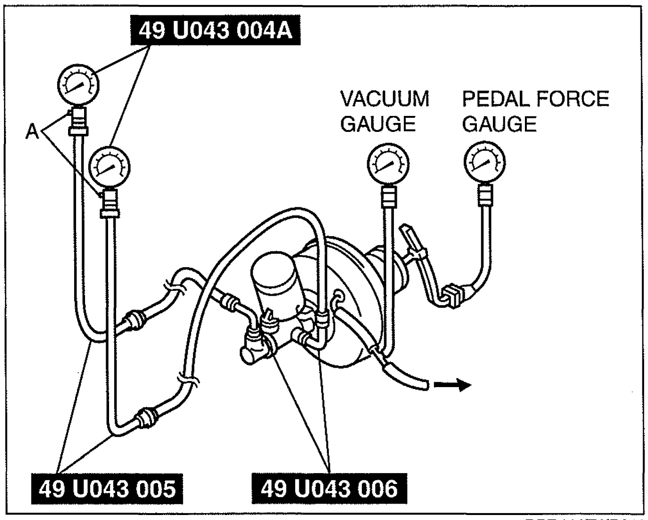
A6E6912W025

POWER BRAKE UNIT INSPECTION

dcf041143800w01

Power Brake Unit Function Inspection (Inspection using gauges)

1. Connect the **SST** gauges, a vacuum gauge, and a pedal depression gauge as shown in the figure. Bleed the air from the **SST** gauges before performing the following tests.



DBR411ZWB019

Checking for vacuum loss (unloaded condition)

1. Start the engine.
2. Stop the engine when the vacuum gauge indicates **67 kPa {500 mmHg, 20 inHg}**.
3. Observe the vacuum gauge for **15 s**.
   - If the gauge indicates **64—66 kPa {476—499 mmHg, 18.8—19.6 inHg}**, the unit is operating.

Checking for vacuum loss (loaded condition)

1. Start the engine.
2. Depress the brake pedal with a force of **200 N {20 kgf, 45 lbf}**.
3. With the brake pedal held depressed, stop the engine when the vacuum gauge indicates **67 kPa {500 mmHg, 20 inHg}**.
4. Observe the vacuum gauge for **15 s**.
   - If the gauge indicates **64—66 kPa {476—499 mmHg, 18.8—19.6 inHg}**, the unit is operating.

Checking for hydraulic pressure

1. If the engine is stopped (vacuum **0 kPa {0 mmHg, 0 inHg}**) and the fluid pressure is within the specification, the unit is operating.

   Power brake unit fluid pressure when pedal depressed at 200 N {20 kgf, 45 lbf}
   At 0 kPa {0 mmHg, 0 inHg}: 593 kPa {6.1 kgf/cm^2, 87 psi} or more

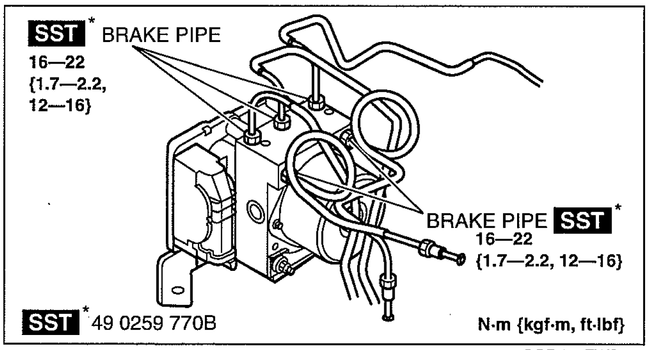
2. Start the engine. Depress the brake pedal when the vacuum reaches **67 kPa {500 mmHg, 20 inHg}**.
   - If the fluid pressure is within the specification, the unit is operating.
   - If the fluid pressure is not as specified, inspect for damage to the check valve or vacuum hose, and fluid leakage of the hydraulic line. Repair as necessary, and inspect again.

   Power brake unit fluid pressure when pedal depressed at 200 N {20 kgf, 45 lbf}
   At 67 kPa {500 mmHg, 20 inHg}: 6,950 kPa {71 kgf/cm^2, 1,008 psi} or more

POWER BRAKE UNIT REMOVAL/INSTALLATION

dcf041143800w02

1. Remove the master cylinder. (See 04-11-6 MASTER CYLINDER REMOVAL/INSTALLATION.)
2. For vehicles with 4W-ABS, disconnect all the ABS HU/CM brake pipes shown in the figure. (See 04-13B-3 ABS HU/CM REMOVAL/INSTALLATION [4W-ABS].)
3. Remove the brake pipe (ABS HU/CM—master cylinder).

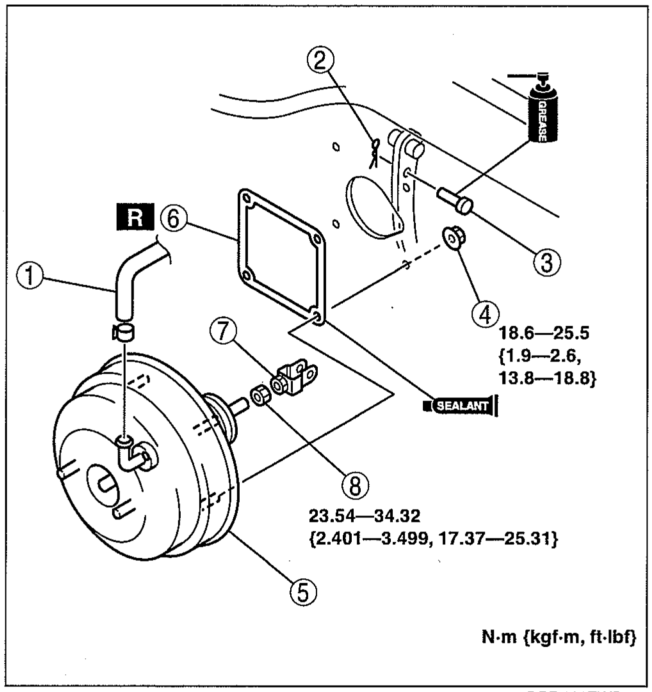


DBR411ZWB049

4. Remove in the order indicated in the table.

1	Vacuum hose
2	Snap pin
3	Clevis pin
4	Nut
5	Power brake unit
6	Gasket
7	Fork
8	Nut

5. Install in the reverse order of removal.



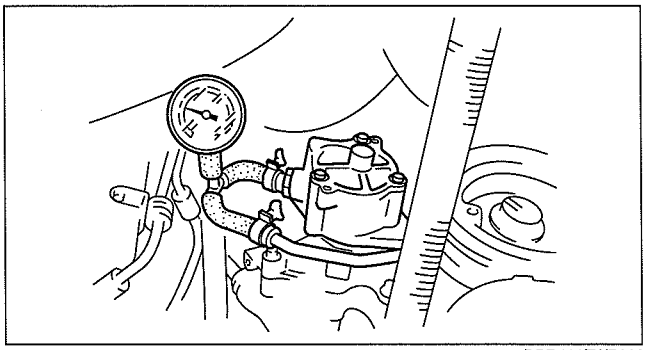
DBR411ZWB020

VACUUM PUMP INSPECTION

dcf041118777w01

1. Warn up the engine.
2. Disconnect the vacuum hose from the vacuum pump and connect a vacuum gauge as shown in the figure, then check the vacuum.
   - If the pressure is less than the specification, inspect for the following.
     — Malfunction of the vacuum pump
     — Shortage of the lubrication oil pressure

Vacuum specification (In 20 s)
Engine speed 1,270 rpm: 74 kPa {550 mmHg, 22 inHg} or more
Engine speed 2,450 rpm: 94 kPa {700 mmHg, 28 inHg} or more

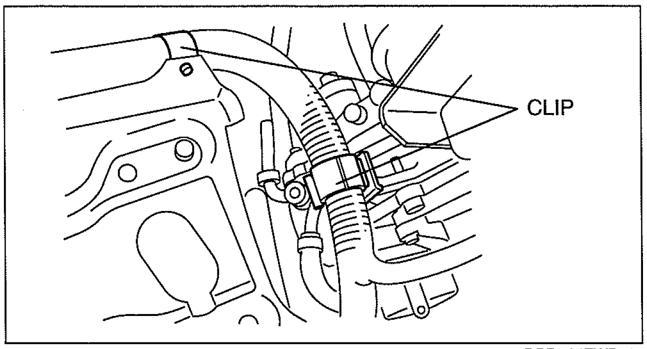


DBR411ZWB023

VACUUM PUMP REMOVAL/INSTALLATION

dcf041118777w02

1. Remove the battery and battery tray. (See 01-17-2 BATTERY REMOVAL/INSTALLATION.)
2. Remove the clip shown in the figure.

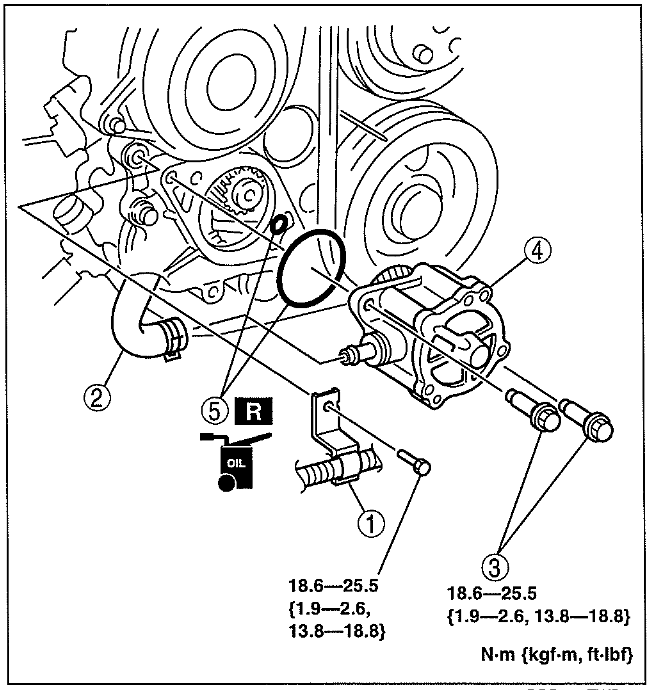


DBR614ZWB020

3. Remove in the order indicated in the table.

1	Bracket (WL-C, WE-C)
2	Vacuum hose
3	Bolt
4	Vacuum pump
5	O-ring

4. Install in the reverse order of removal.

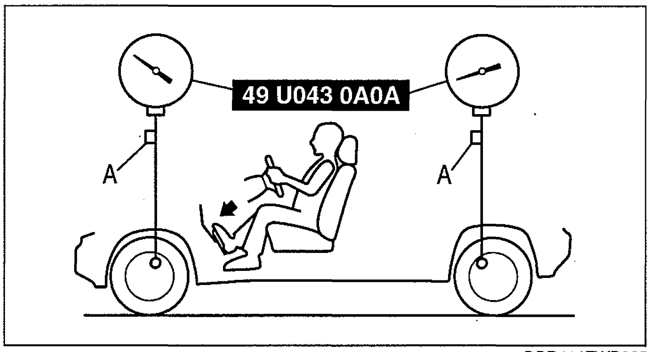


DBR411ZWB022

LOAD SENSING PROPORTIONING VALVE (LSPV) INSPECTION

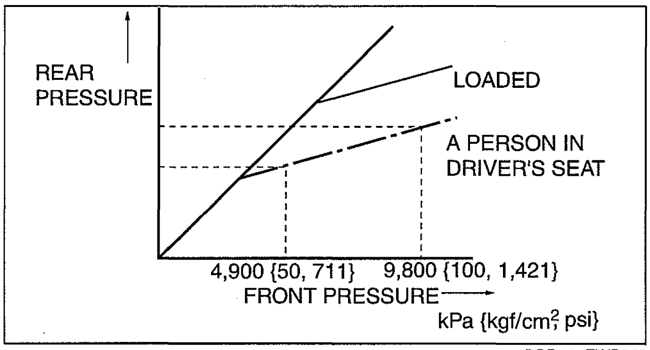
dcf041143903w01

1. Place the vehicle on level ground.
2. Check the tire inflations and set them to the recommended pressure if necessary.
    - If not as specified, replace or adjust.
3. Check the vehicle with a person in the driver's seat; include the standard vehicle tools, the spare wheel and tire, and a full tank of fuel.
4. Install the **SST** to the front wheel cylinder and rear wheel cylinder, then bleed the air in the brake line and LSPV using a air bleed valve A.



DBR411ZWB025

5. Inspect the fluid pressure of rear wheel cylinder while the fluid pressure of front wheel cylinder is **4,900 kPa {50 kgf/cm², 711 psi}** and **9,800 kPa {100 kgf/cm², 1,421 psi}**.
    - If not as specified, adjust the LSPV.



DBR411ZWB026

Load sensing proportioning valve (LSPV) fluid pressure

Type		Front wheel cylinder fluid pressure (kPa {kgf/cm², psi})	Rear wheel cylinder (kPa {kgf/cm², psi})
4x2 (except Hi-Rider)	Without 4W-ABS	4,900 {50, 711}	2,250—2,850 {23—29, 327—413}
		9,800 {100, 1,421}	3,130—3,930 {32—40, 454—569}
	With 4W-ABS	4,900 {50, 711}	2,910—3,710 {30—37, 423—538}
		9,800 {100, 1,421}	4,970—6,170 {51—62, 721—894}
Hi-Rider, 4x4	Without 4W-ABS	4,900 {50, 711}	2,250—2,850 {23—29, 327—413}
		9,800 {100, 1,421}	3,130—3,930 {32—40, 454—569}
	With 4W-ABS	4,900 {50, 711}	2,650—3,450 {28—35, 385—500}
		9,800 {100, 1,421}	4,700—5,900 {48—60, 682—855}

Note

- When applying the specified pressure, the brake pedal must not be "double pumped" or released.
- Read the rear pressure. After **approx. 2 s** elapse from setting the front wheel cylinder fluid pressure.

LOAD SENSING PROPORTIONING VALVE (LSPV) ADJUSTMENT

dcf041143903w02

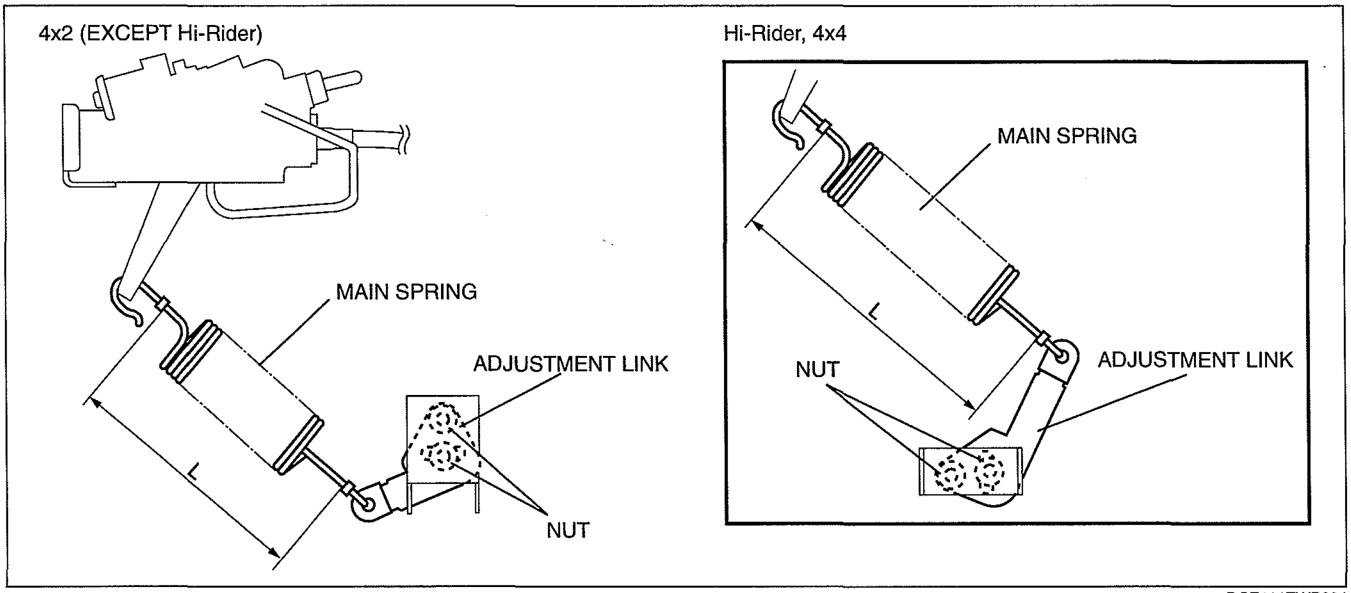
1. Place the unloaded vehicle on level ground. Unloaded: Fuel tank is full. Engine coolant and tire, jack and tools are in designated position.

Note

- A change of **5 mm {0.20 in}** in dimension L results in a change of the following.

Type		Change (kPa {kgf/cm^2, psi})
4x2 (except Hi-Rider)	without 4W-ABS	850 {8.7, 123}
	with 4W-ABS	1,020 {10, 148}
Hi-Rider, 4x4	without 4W-ABS	1,060 {11, 154}
	with 4W-ABS	1,080 {11, 157}

2. Adjust main spring dimension L between the LSPV and the adjustment link loosening and repositioning the LSPV.



DCF411ZWB004

- Decrease dimension L if the fluid pressure is low.
- Increase dimension L if the fluid pressure is high.

Specification
Dimension L

Type		Dimension L (mm {in})
4x2 (except Hi-Rider)	without 4W-ABS	147.5—154.5 {5.808—6.082}
	with 4W-ABS	
Hi-Rider, 4x4	without 4W-ABS	175.5—182.5 {6.910—7.185}
	with 4W-ABS	

3. Tightening the nut.

Specification
18.6—25.4 N·m {1.90—2.59 kgf·m, 13.8—18.7 ft·lbf}

4. After adjustment, recheck the fluid pressure.
   - If not as specified, replace the LSPV as a component.

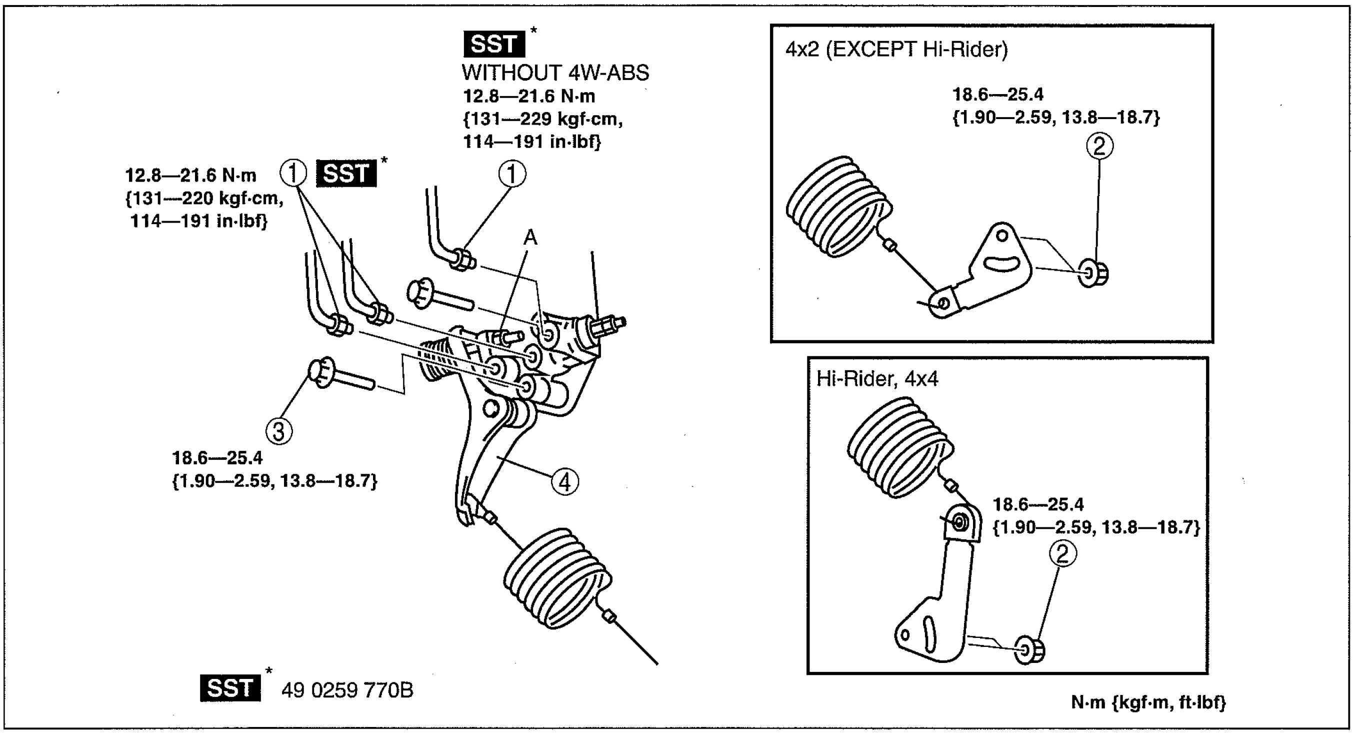
04

LOAD SENSING PROPORTIONING VALVE (LSPV) REMOVAL/INSTALLATION

dcf041143903w03

Caution
- **Do not disassemble the load sensing proportioning valve (LSPV) or move nut A, otherwise the LSPV may not function properly.**

1. Remove in the order indicated in the table.
2. Install in the reverse order of removal.
3. After installation, adjust the main spring dimension. (See 04-11-13 LOAD SENSING PROPORTIONING VALVE (LSPV) ADJUSTMENT.)

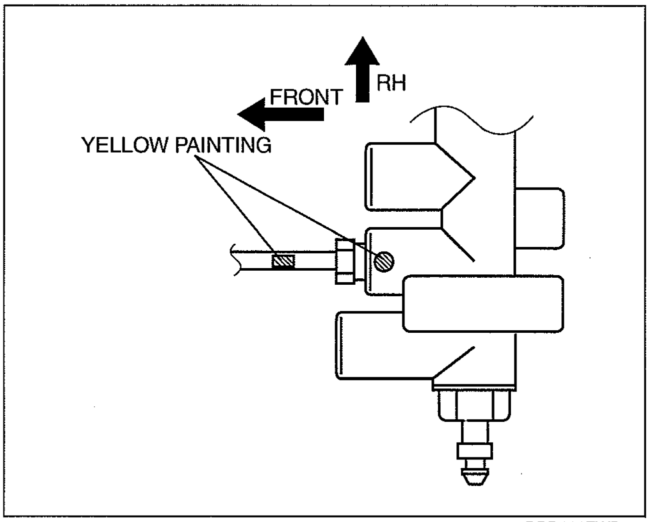


DCF411ZWB005

1	Brake pipe (See 04-11-14 Brake Pipe Installation Note)
2	Nut
3	Bolt
4	Load sensing proportioning valve (LSPV)

Brake Pipe Installation Note

1. Install the brake pipe to the LSPV and align yellow painting on the brake pipe and LSPV.



DBR411ZWB029

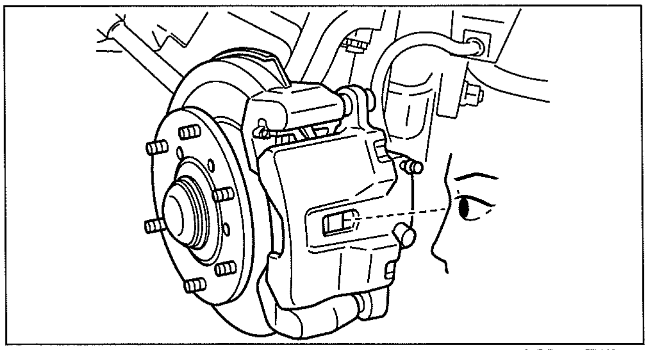
FRONT BRAKE (DISC) INSPECTION

dcf041133980w01

Disc Pad Thickness Inspection

1. Jack up the front of the vehicle on level ground and support it with safety stands.
2. Remove the wheels.
3. Look through the caliper inspection hole and verify the remaining thickness of the pad.

Minimum front disc pad thickness
2.0 mm {0.079 in} min.

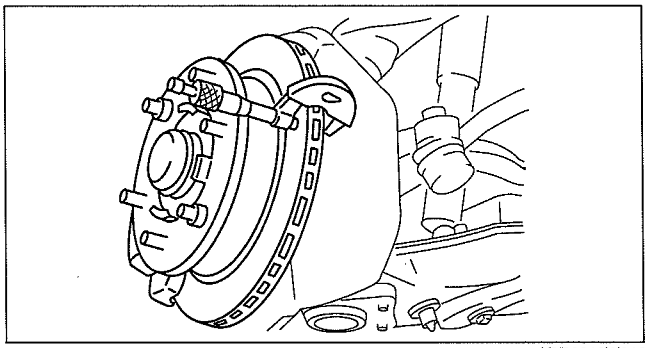


DBR411ZWB036

Disc Plate Thickness Inspection

1. Measure the thickness of the disc plate.
   - If the thickness is not within the specification, replace the disc plate.

Caution
- **When it is necessary to machine the disc plate, make sure the disc plate is not removed from the vehicle when machined, excessive runout may result. Machine the disc plate with it installed on the vehicle.**



ABR6912W010

Standard front disc plate thickness
4x2 (except Hi-Rider): 24.0 mm {0.945 in}
Hi-Rider, 4x4: 28.0 mm {1.11 in}

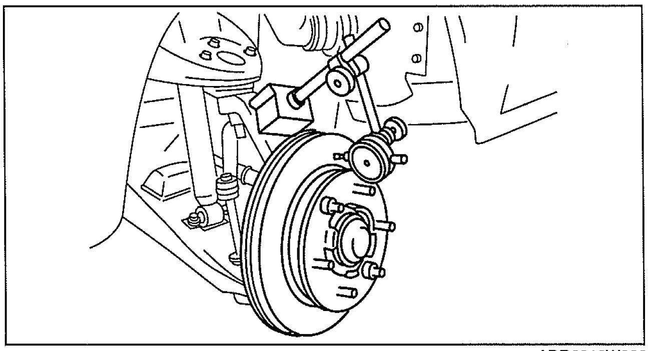
Minimum front disc plate thickness
4x2 (except Hi-Rider): 22.0 mm {0.867 in}
Hi-Rider, 4x4: 26.0 mm {1.03 in}

Minimum front disc plate thickness after machining using a brake lathe on-vehicle
4x2 (except Hi-Rider): 22.8 mm {0.898 in}
Hi-Rider, 4x4: 26.8 mm {1.06 in}

Disc Plate Runout Inspection

1. Tighten the disc plate to the wheel hub using two wheel nuts. When measuring runout, measure at the outer edge of the disc plate surface.

Front disc plate runout limit
0.05 mm {0.002 in} max.



ABR6912W009

04

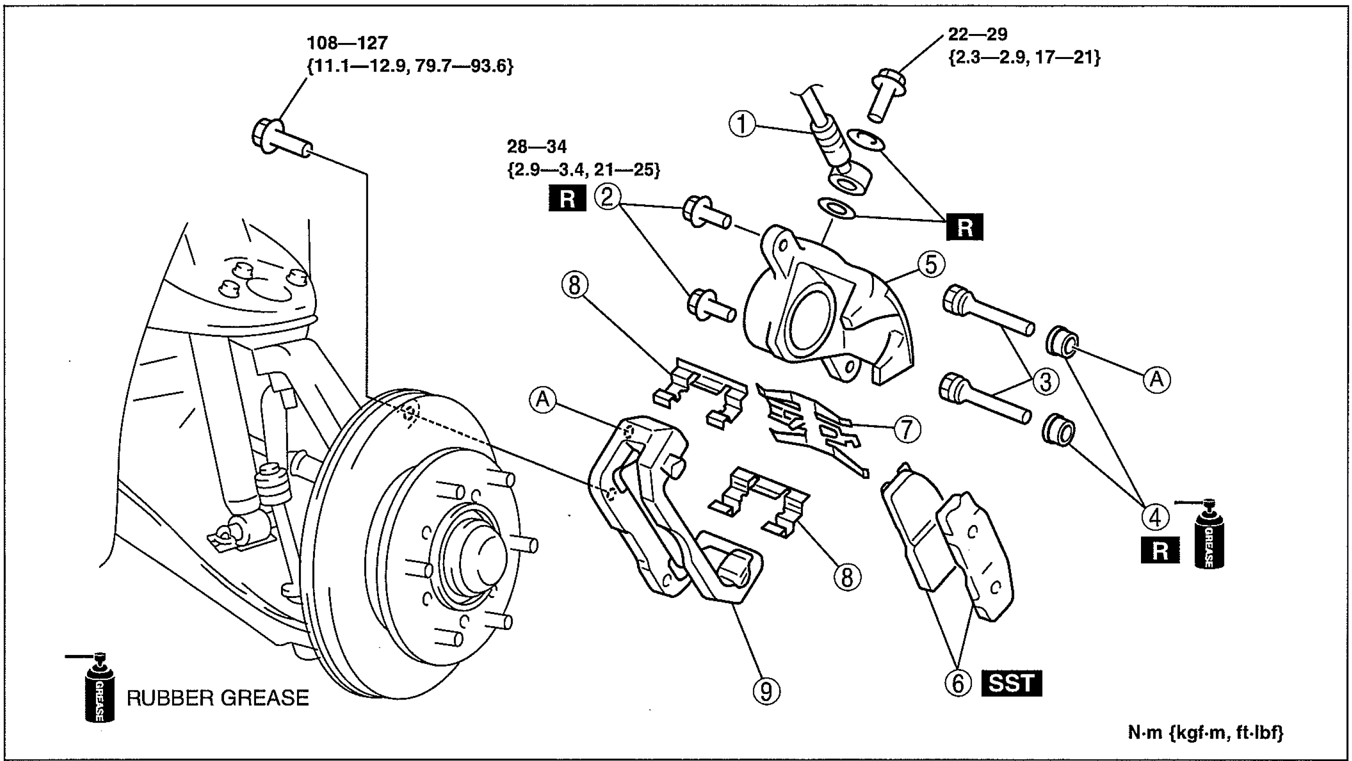
FRONT BRAKE (DISC) REMOVAL/INSTALLATION [4x2 (EXCEPT Hi-Rider)]

dcf041133980w02

Note

- Because the disc plate is installed to the front wheel hub, refer to WHEEL HUB, STEERING KNUCKLE REMOVAL/INSTALLATION to remove it. (See 03-11-4 WHEEL HUB, STEERING KNUCKLE REMOVAL/ INSTALLATION [4x2 (EXCEPT Hi-Rider)].)

1. Remove in the order indicated in the table.
2. Install in the reverse order of removal.
3. After installation, depress the pedal a few times, rotate the wheel by hand, and verify that the brake does not drag.



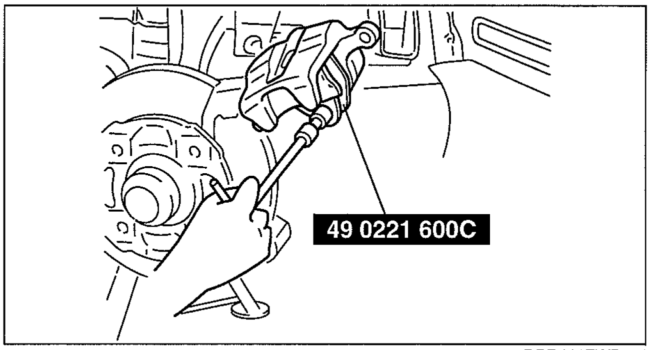
DBR411ZWB030

1	Brake hose
2	Bolt
3	Guide pin
4	Dust boot
5	Caliper

6	Disc pad (See 04-11-16 Disc Pad Installation Note)
7	Plate
8	Guide plate
9	Mounting support

Disc Pad Installation Note

1. Push the piston fully inward using the **SST**.
2. Install the disc pad.



DBR411ZWB031

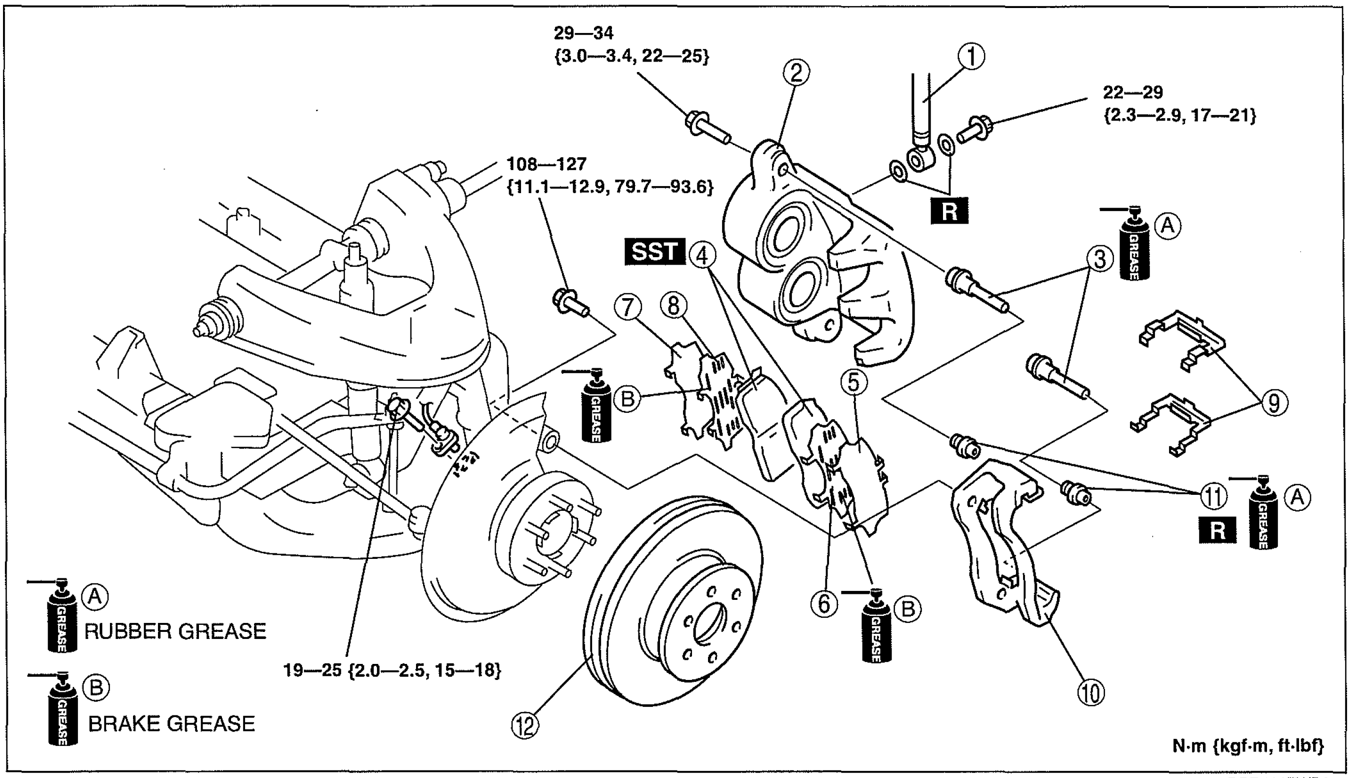
FRONT BRAKE (DISC) REMOVAL/INSTALLATION [Hi-Rider, 4x4]

dcf041133980w03

Caution

- **Performing the following procedures without first removing the ABS wheel-speed sensor may possibly cause an open circuit in the harness if it is pulled by mistake. Before performing the following procedures, remove the ABS wheel-speed sensor (axle side) and fix it to an appropriate place where the sensor will not be pulled by mistake while the vehicle is being serviced.**

1. Remove in the order indicated in the table.
2. Install in the reverse order of removal.
3. After installation, depress the pedal a few times, rotate the wheel by hand, and verify that the brake does not drag.



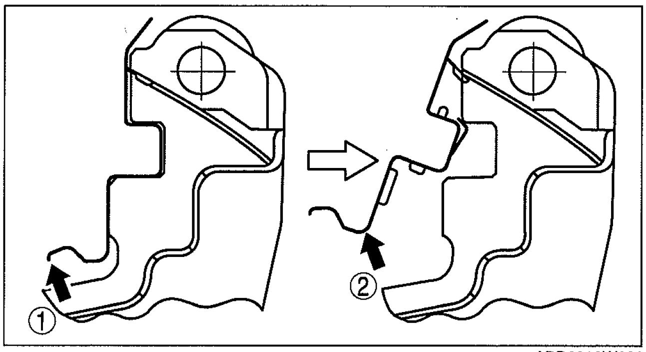
DBR411ZWB035

1	Brake hose
2	Caliper
3	Guide pin
4	Disc pad (See 04-11-19 Disc Pad Installation Note)
5	Outer shim (cover shim) (See 04-11-18 Outer Shim, Inner shim Installation Note)
6	Outer shim (rubber coated shim) (See 04-11-18 Outer Shim, Inner shim Installation Note)
7	Inner shim (cover shim) (See 04-11-18 Outer Shim, Inner shim Installation Note)
8	Inner shim (rubber coated shim) (See 04-11-18 Outer Shim, Inner shim Installation Note)
9	Guide plate (See 04-11-18 Guide Plate Removal Note)
10	Mounting support
11	Dust boot
12	Disc plate (See 04-11-18 Disc Plate Removal Note) (See 04-11-18 Disc Plate Installation Note)

04

Guide Plate Removal Note

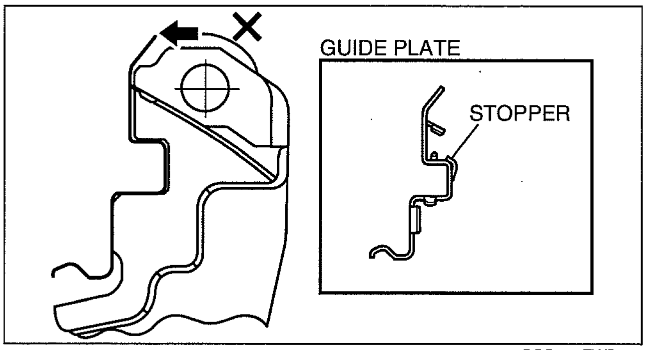
1. Pull up and at a slant on the supporting part of the guide plate in the direction indicated by the arrow (1) and (2) as shown in the figure, to remove the guide plate from the mounting support.



ABR6912W032

Caution

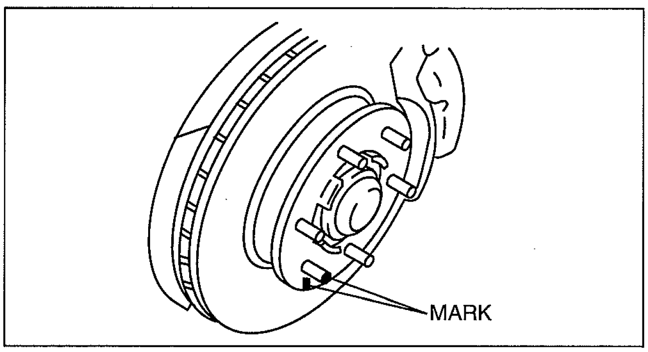
- **If the guide plate is removed by pulling in the direction indicated by the arrow as shown in the figure, the stopper may be deformed and not able to be reused. Do not remove in this manner.**



DBR411ZWB054

Disc Plate Removal Note

1. Mark the wheel hub bolt and disc plate before removal for reference during installation.



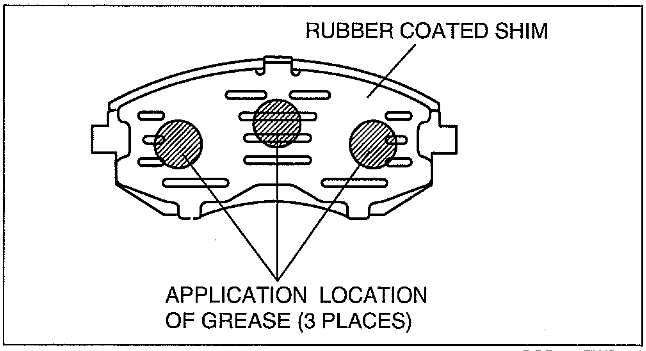
DBR411ZWB021

Disc Plate Installation Note

1. Remove any rust or grime on the contact face of the disc plate and wheel hub.
2. Install the disc plate and align the marks made before removal.

Outer Shim, Inner shim Installation Note

1. Assemble the rubber coated shim to the back plate of the disc pad.
2. Apply brake grease to the rubber coated shim as shown in the figure.

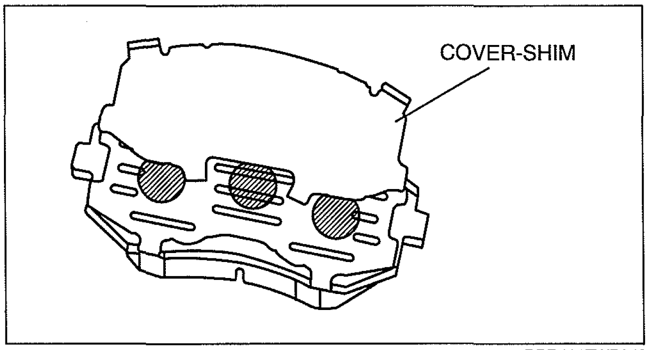


DBR411ZWB038

3. Assemble the cover-shim to the grease-applied rubber coated shim.

Caution

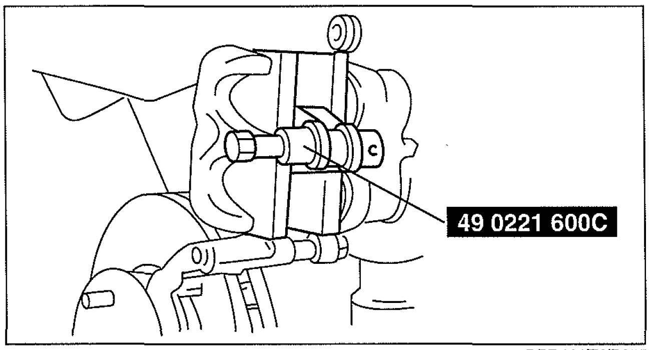
- **Be careful not to get any grease on the pad lining surface.**



DBR411ZWB048

Disc Pad Installation Note

1. Push the piston fully inward using the **SST**.
2. Install the disc pad.



DBR411ZWB055

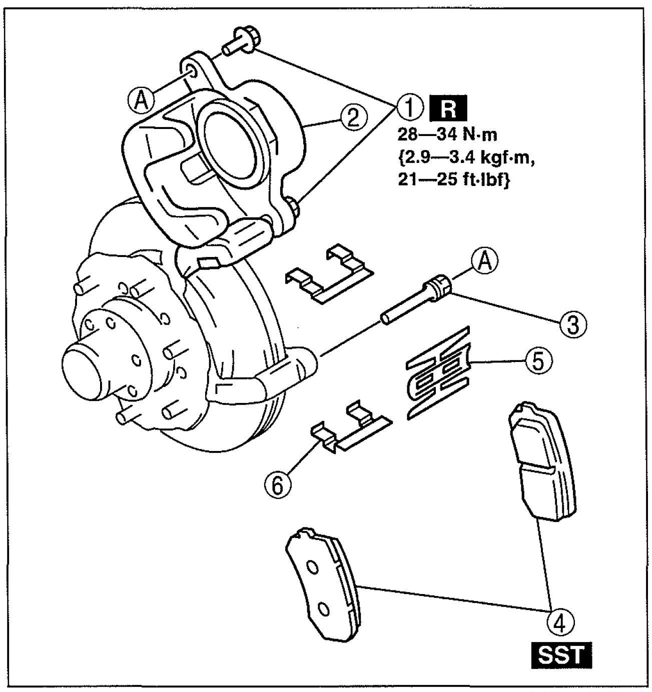
DISC PAD (FRONT) REPLACEMENT [4x2 (EXCEPT Hi-Rider)]

dcf041133630w01

1. Remove in the order indicated in the table.

1	Bolt
2	Caliper
3	Guide pin
4	Disc pad (See 04-11-16 Disc Pad Installation Note.)
5	Plate
6	Guide plate

2. Install in the reverse order of removal.



DBR411ZWB032

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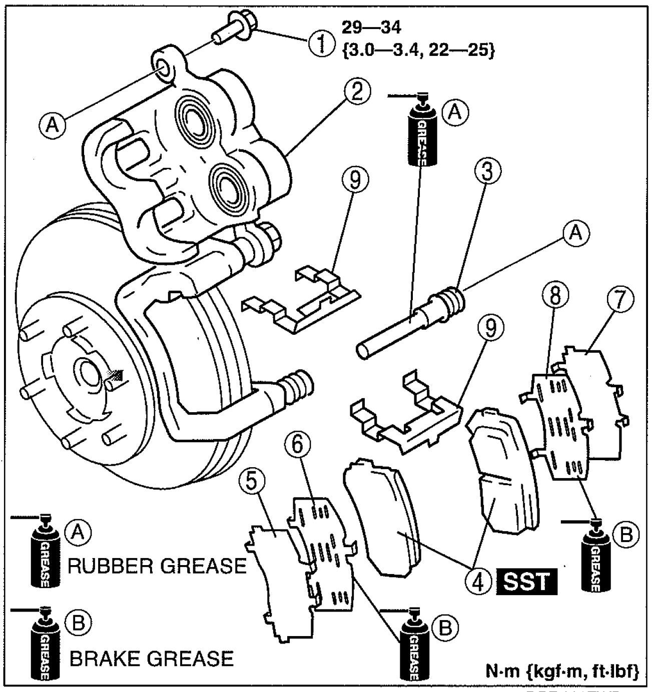
DISC PAD (FRONT) REPLACEMENT [Hi-Rider, 4x4]

dcf041133630w02

1. Remove in the order indicated in the table.

1	Bolt
2	Caliper
3	Guide pin
4	Disc pad (See 04-11-19 Disc Pad Installation Note)
5	Outer shim (cover shim) (See 04-11-19 Disc Pad Installation Note)
6	Outer shim (rubber coated shim) (See 04-11-19 Disc Pad Installation Note)
7	Inner shim (cover shim) (See 04-11-19 Disc Pad Installation Note)
8	Inner shim (rubber coated shim) (See 04-11-19 Disc Pad Installation Note)
9	Guide plate (See 04-11-18 Guide Plate Removal Note)

2. Install in the reverse order of removal.



DBR411ZWB037

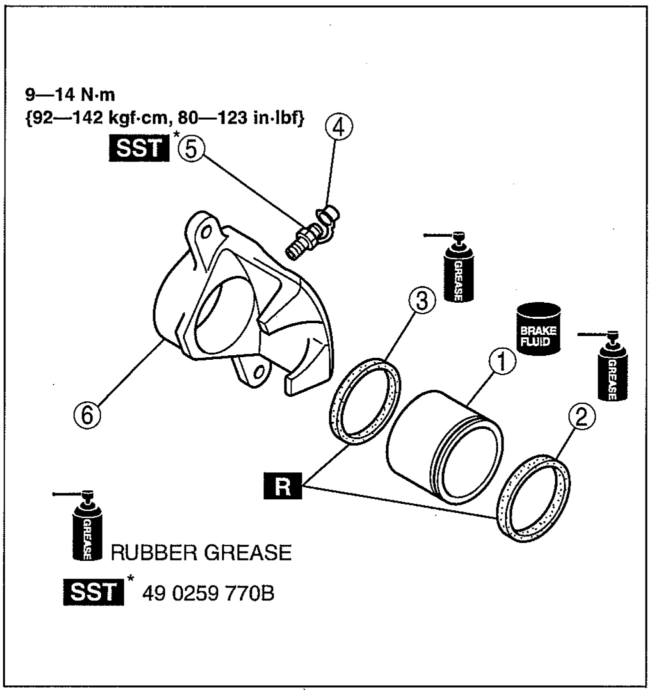
CALIPER (FRONT) DISASSEMBLY/ASSEMBLY [4x2 (EXCEPT Hi-Rider)]

dcf041133990w01

1. Disassemble in the order indicated in the table.

1	Piston (See 04-11-21 Piston Disassembly Note)
2	Dust seal
3	Piston seal
4	Bleeder cap
5	Bleeder screw
6	Caliper

2. Assemble in the reverse order of removal.



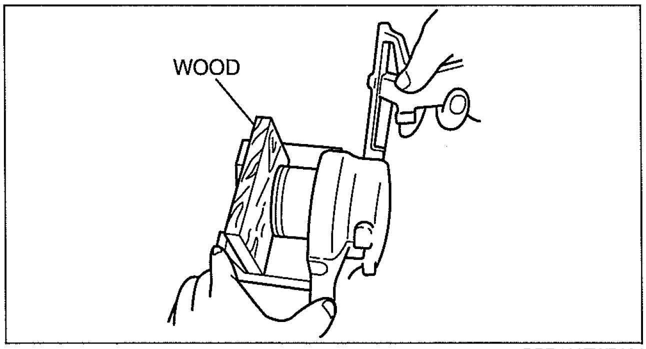
DBR411ZWB033

Piston Disassembly Note

Caution

- **The piston could be damaged if blown out with great force. Blow the compressed air slowly to prevent the piston from suddenly popping out.**

1. Place the piece of wood in the caliper, then blow compressed air through the hole to force the piston out of the caliper.



DBR411ZWB034

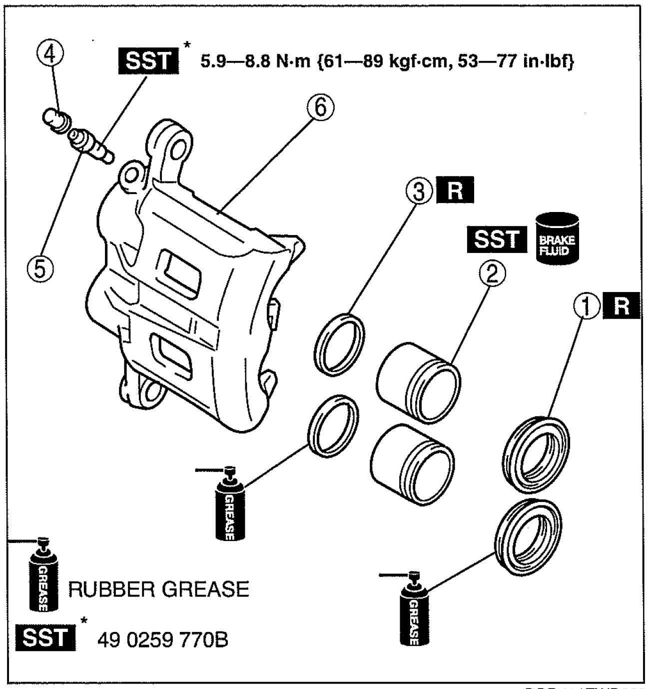
CALIPER (FRONT) DISASSEMBLY/ASSEMBLY [Hi-Rider, 4x4]

dcf041133990w02

1. Disassemble in the order indicated in the table.

1	Dust seal
2	Piston (See 04-11-21 Piston Disassembly Note)
3	Piston seal
4	Bleeder cap
5	Bleeder screw
6	Caliper

2. Assemble in the reverse order of removal.



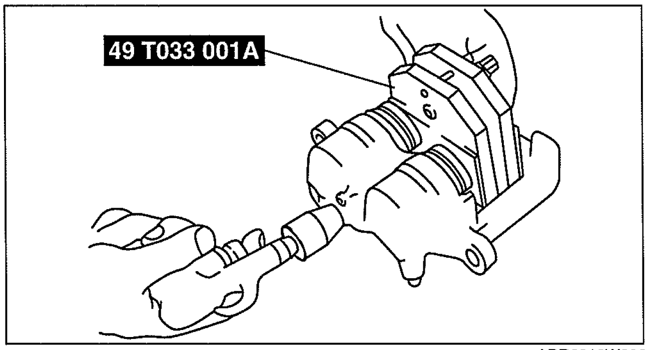
DBR411ZWB039

Piston Disassembly Note

Caution

- **The piston could be damaged if blown out with great force. Blow the compressed air slowly to prevent the piston from suddenly popping out.**

1. Place the **SST** in the caliper, then blow compressed air through the hole to force the piston out of the caliper.



ABR6912W006

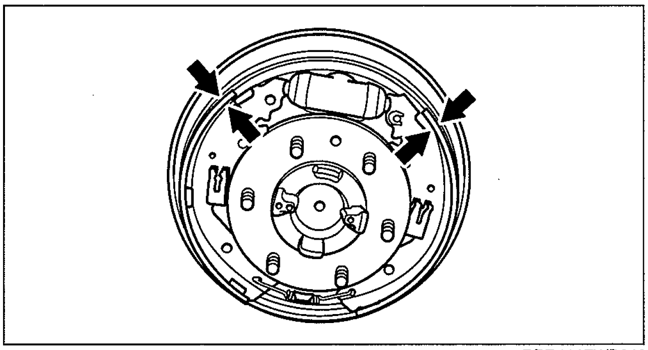
REAR BRAKE (DRUM) INSPECTION

dcf041126250w01

Brake Lining Thickness Inspection

1. Remove the brake drum. (See 04-11-23 REAR BRAKE (DRUM) REMOVAL/INSTALLATION.)
2. Inspect the brake lining thickness.
   - If it less than the minimum specification, replace the brake shoe. (See 04-11-23 REAR BRAKE (DRUM) REMOVAL/INSTALLATION.)

   Minimum rear brake lining thickness
   1.0 mm {0.04 in}

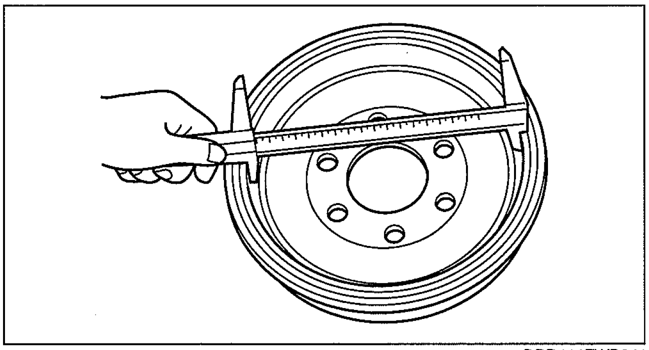


DBR411ZWB042

Brake Drum Inspection

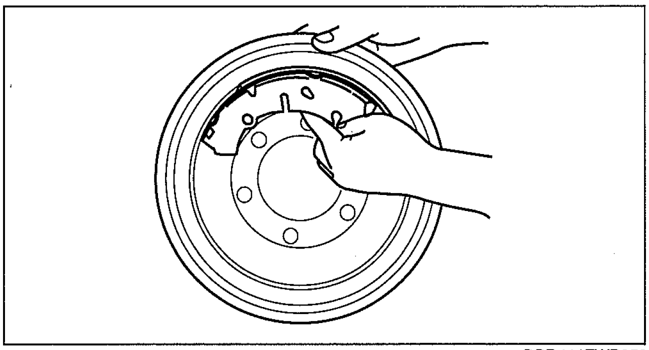
1. Remove the brake drum. (See 04-11-23 REAR BRAKE (DRUM) REMOVAL/INSTALLATION.)
2. Inspect the inner diameter of the brake drum.
   - If t exceeds the maximum specification, replace the brake drum. (See 04-11-23 REAR BRAKE (DRUM) REMOVAL/INSTALLATION.)

   Maximum rear brake drum diameter
   4x2 (except Hi-Rider): 271.5 mm {10.68 in}
   Hi-Rider, 4x4: 296.5 mm {11.67 in}



DBR411ZWB041

3. Apply chalk to the inside of the brake drum, rub it against the brake shoe, and then inspect for scratches and uneven or abnormal wear inside the drum.
   - If there is significant uneven wear of the brake drum, repair it within the maximum brake drum inner diameter or replace. (See 04-11-23 REAR BRAKE (DRUM) REMOVAL/INSTALLATION.)
4. After inspection, wipe the chalk off.



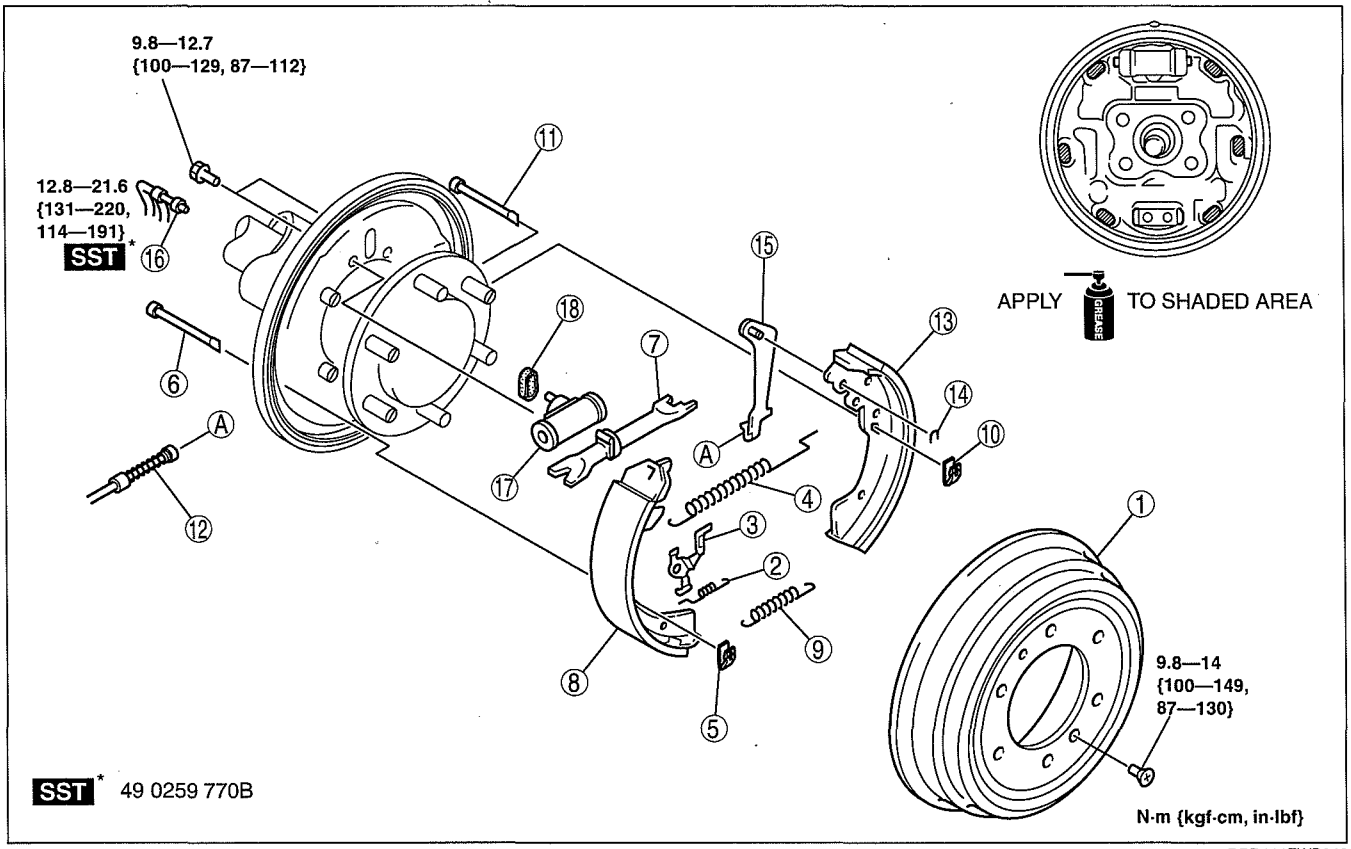
DBR411ZWB050

REAR BRAKE (DRUM) REMOVAL/INSTALLATION

dcf041126250w02

Warning
- **When removing/installing the brake drum component parts, the spring could fly off and cause injury. Remove/install the spring being careful not to allow the spring to fly off. Wear protective equipment such as safety glasses if necessary.**

1. Remove in the order indicated in the table.
2. Install in the reverse order of removal.
3. After installation, pump the brake pedal a few times and inspect the following:
   — Parking brake lever stroke
   — Brake drag



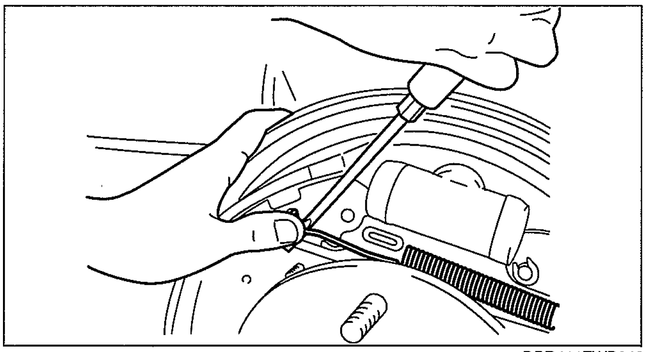
DBR411ZWB040

1	Brake drum (See 04-11-24 Brake Drum Installation Note.)
2	Pawl spring
3	Pawl lever
4	Upper return spring (See 04-11-24 Upper Return Spring Removal Note.) (See 04-11-24 Upper Return Spring Installation Note.)
5	Hold spring (leading shoe side)
6	Hold pin (leading shoe side)
7	Adjust strut
8	Brake shoe (leading shoe)

9	Lower return spring
10	Hold spring (trailing shoe side)
11	Hold pin (trailing shoe side)
12	Parking brake cable
13	Brake shoe (trailing shoe)
14	U-ring
15	Operating lever
16	Brake pipe
17	Wheel cylinder
18	Gasket

Upper Return Spring Removal Note

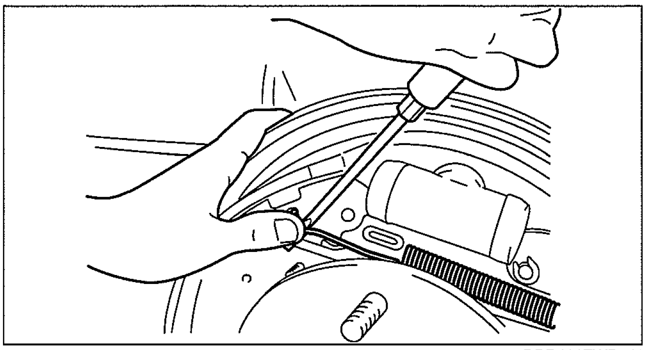
1. Remove the return spring using a flathead screwdriver while supporting it with the hand, as shown in the figure, to prevent it from flying off.



DBR411ZWB046

Upper Return Spring Installation Note

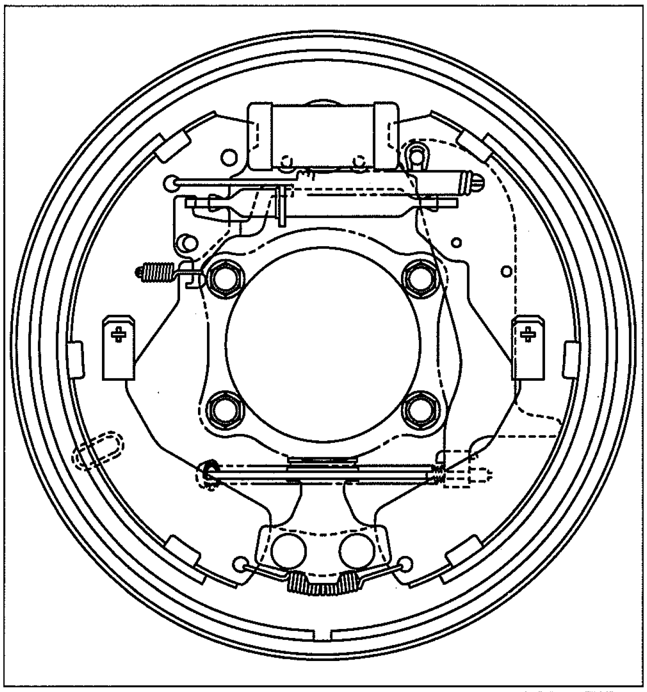
1. Install the return spring using a flathead screwdriver while supporting it with the hand, as shown in the figure, to prevent it from flying off.



DBR411ZWB046

Brake Drum Installation Note

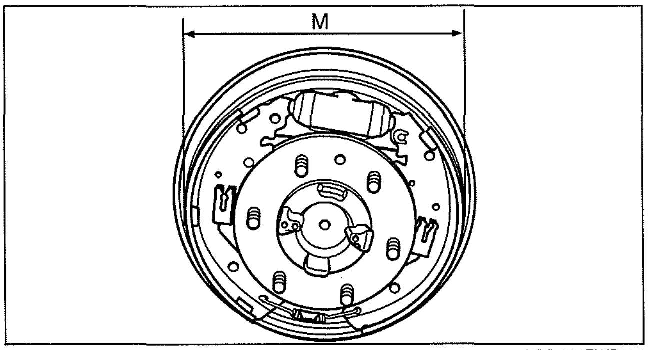
1. Verify that each component part of the rear brake (drum) is properly installed as shown in the figure.



DBR411ZWB056

2. Turn the adjuster on the adjustment strut to adjust the brake shoe until the outer diameter (M) of the brake shoe is as specified below.

Brake shoe adjustment value
- **4x2 (except Hi-Rider): 269.3—269.7 mm {10.603—10.618 in}**
- **Hi-Rider, 4x4: 294.3—294.7 mm {11.59—10.60 in}**



DBR411ZWB051

3. Install the brake drum.
4. Depress the brake pedal five times and operate the auto adjuster.
5. Verify there is no brake drag.
   - If there is any brake drag, readjust the brake adjustment

WHEEL CYLINDER DISASSEMBLY/ASSEMBLY

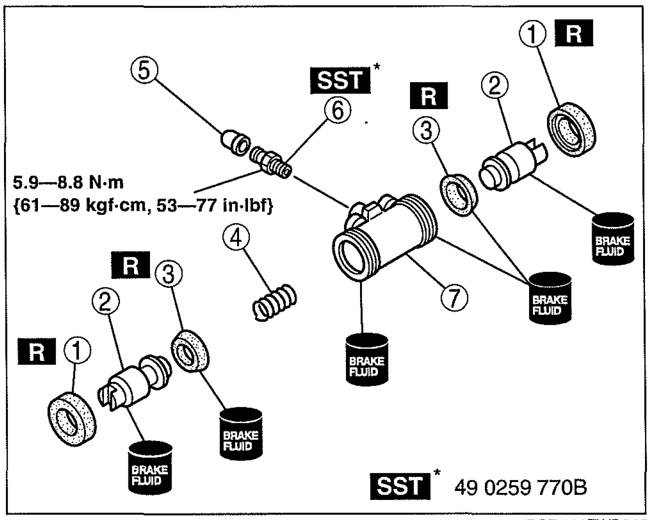
dcf041126610w01

Caution
- **Replace the wheel cylinder component if a problem is found.**

1. Disassemble in the order indicated in the table.

1	Dust boot
2	Wheel cylinder piston
3	Piston cup
4	Wheel cylinder spring
5	Bleeder cap
6	Bleeder screw
7	Wheel cylinder body

2. Assemble in the reverse order of disassembly.



DBR411ZWB045

04

04-12 PARKING BRAKE SYSTEM

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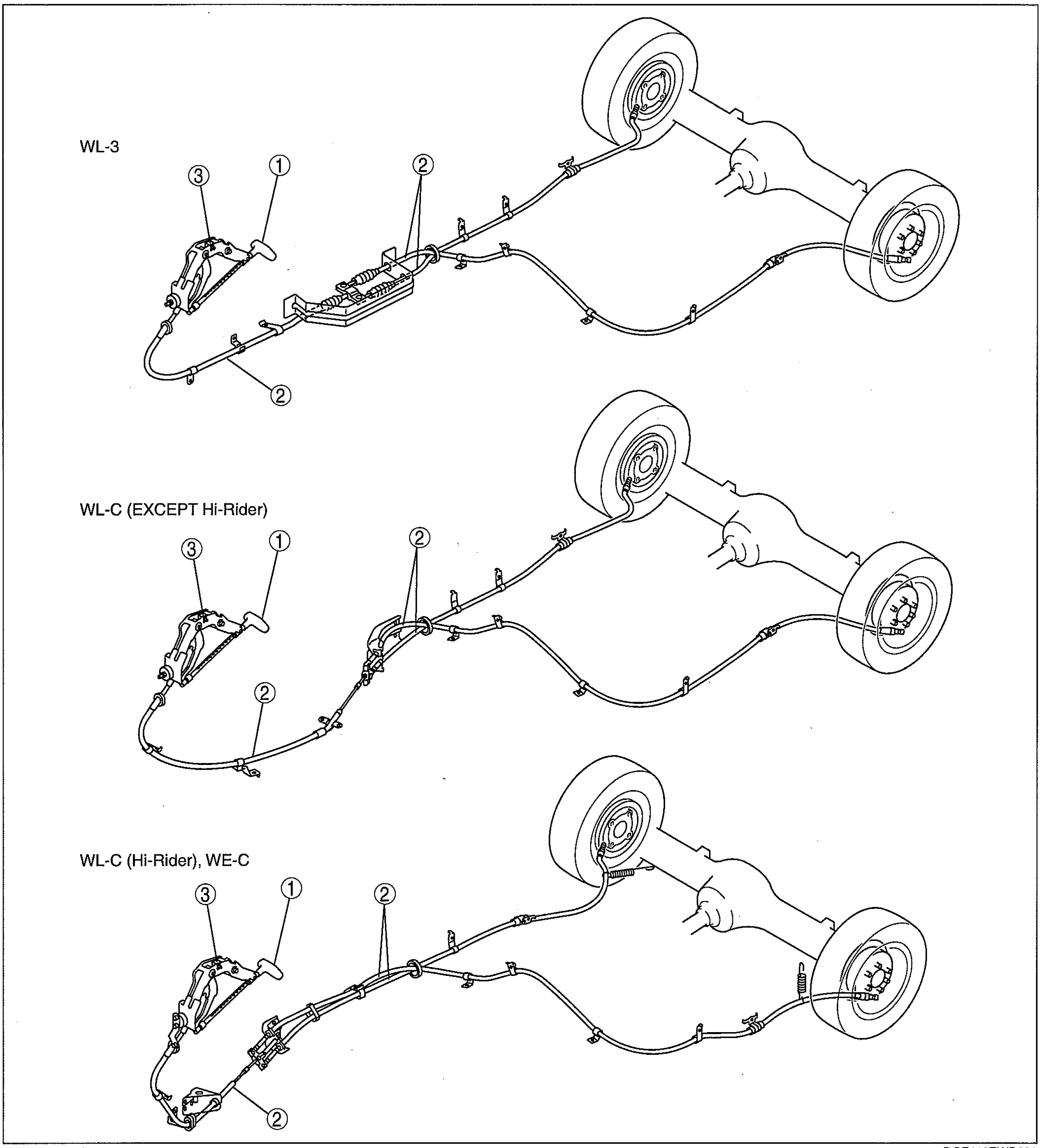
PARKING BRAKE SWITCH INSPECTION 04-12–8

PARKING BRAKE SYSTEM LOCATION INDEX

dcf041200000w01



DCF412ZWB001

1	Parking brake lever (See 04-12-3 PARKING BRAKE LEVER INSPECTION.) (See 04-12-3 PARKING BRAKE LEVER ADJUSTMENT.) (See 04-12-4 PARKING BRAKE LEVER REMOVAL/INSTALLATION.)

2	Parking brake cable (See 04-12-5 PARKING BRAKE CABLE REMOVAL/INSTALLATION [WL-3].) (See 04-12-6 PARKING BRAKE CABLE REMOVAL/INSTALLATION [WL-C (EXCEPT Hi-Rider)].) (See 04-12-7 PARKING BRAKE CABLE REMOVAL/INSTALLATION [WL-C (Hi-Rider), WE-C].)
3	Parking brake switch (See 04-12-8 PARKING BRAKE SWITCH INSPECTION.)

PARKING BRAKE LEVER INSPECTION

dcf041244300w01

Parking Brake Lever Stroke Inspection

1. Pump the brake pedal a few times.
2. Pull the parking brake lever two to three times.
3. Verify that the stroke is within the specification when the parking brake lever is pulled with a force of **98 N {10 kgf, 22 lbf}**.
   - If not within the specification, adjust the parking brake lever. (See 04-12-3 PARKING BRAKE LEVER ADJUSTMENT.)

Parking brake lever stroke when pulled at 98 N {10 kgf, 22 lbf}
1—7 notches

PARKING BRAKE LEVER ADJUSTMENT

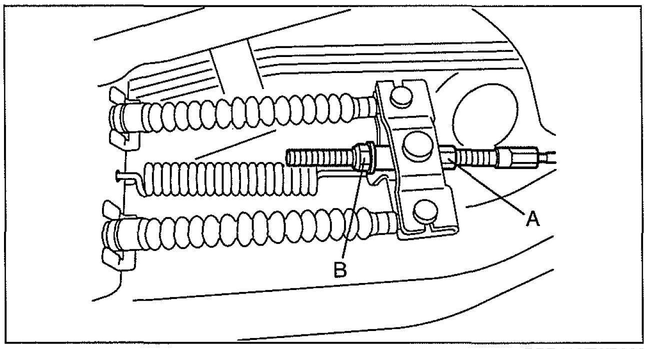
dcf041244300w02

Parking Brake Lever Stroke Adjustment

1. Before adjustment, depress the brake pedal several times while the vehicle is moving in reverse.
2. Loosen the locknut A and turn the adjustment nut B so that the stroke is within the above range.
3. After adjustment, tighten the locknut A.

Tightening torque
6.9—9.8 N·m {71—99 kgf·cm, 62—86 in·lbf}

4. After adjustment, pull the parking brake lever one notch and verify that the parking brake warning light illuminates.
5. Verify that the rear brakes do not drag.

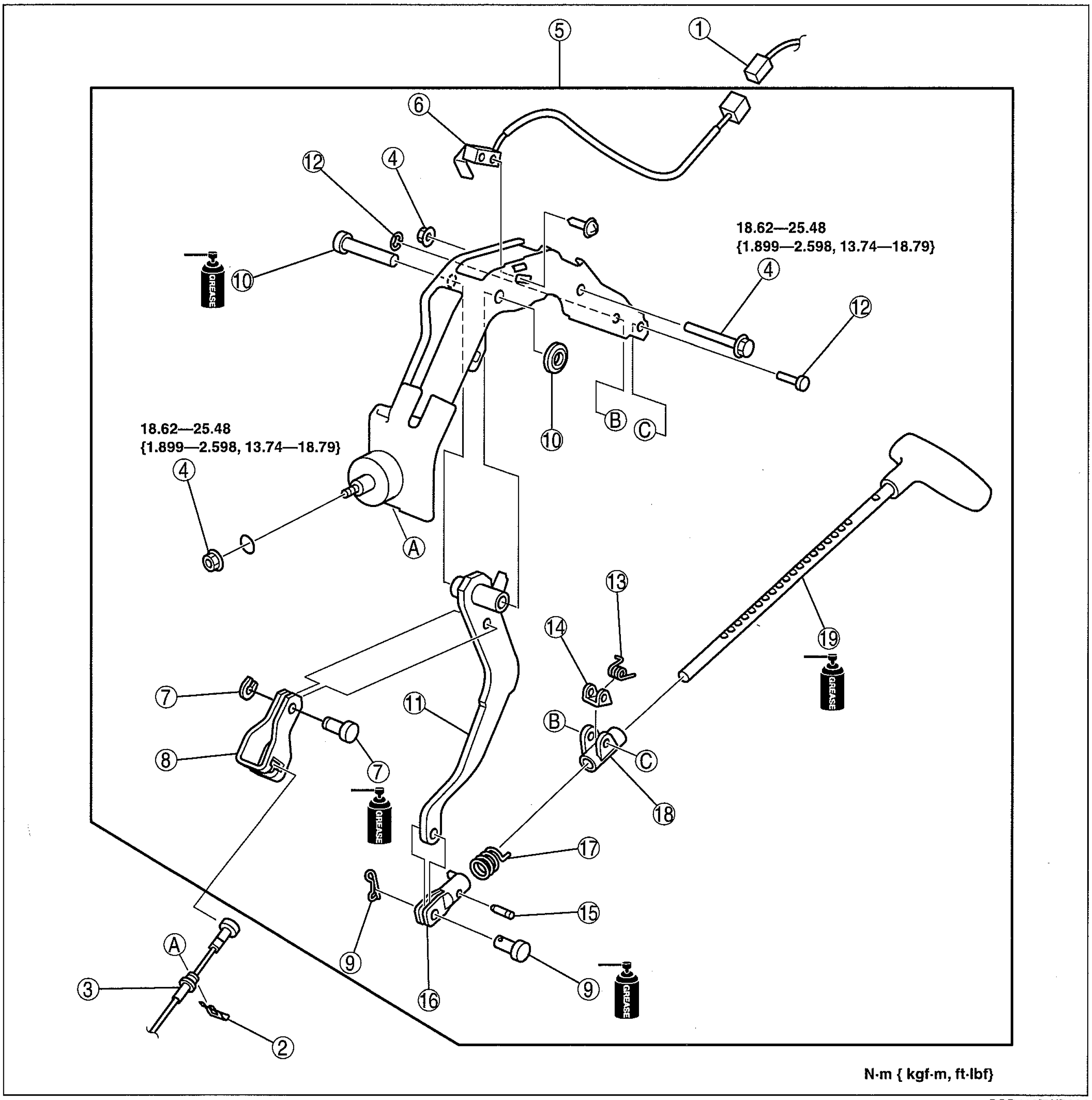


DBR412ZWB002

PARKING BRAKE LEVER REMOVAL/INSTALLATION

dcf041244300w03

1. Remove the lower panel. (See 09-17-10 LOWER PANEL REMOVAL/INSTALLATION.)
2. Remove in the order indicated in the table.
3. Install in the reverse order of removal.
4. After installation, inspect the parking brake lever stroke. (See 04-12-3 PARKING BRAKE LEVER INSPECTION.)



DBR412ZWB004

1	Parking brake switch connector
2	Clip
3	Front parking brake cable (See 04-12-5 PARKING BRAKE CABLE REMOVAL/ INSTALLATION [WL-3].) (See 04-12-6 PARKING BRAKE CABLE REMOVAL/ INSTALLATION [WL-C (EXCEPT Hi-Rider)].) (See 04-12-7 PARKING BRAKE CABLE REMOVAL/ INSTALLATION [WL-C (Hi-Rider), WE-C].)
4	Bolt and nut
5	Parking brake lever component

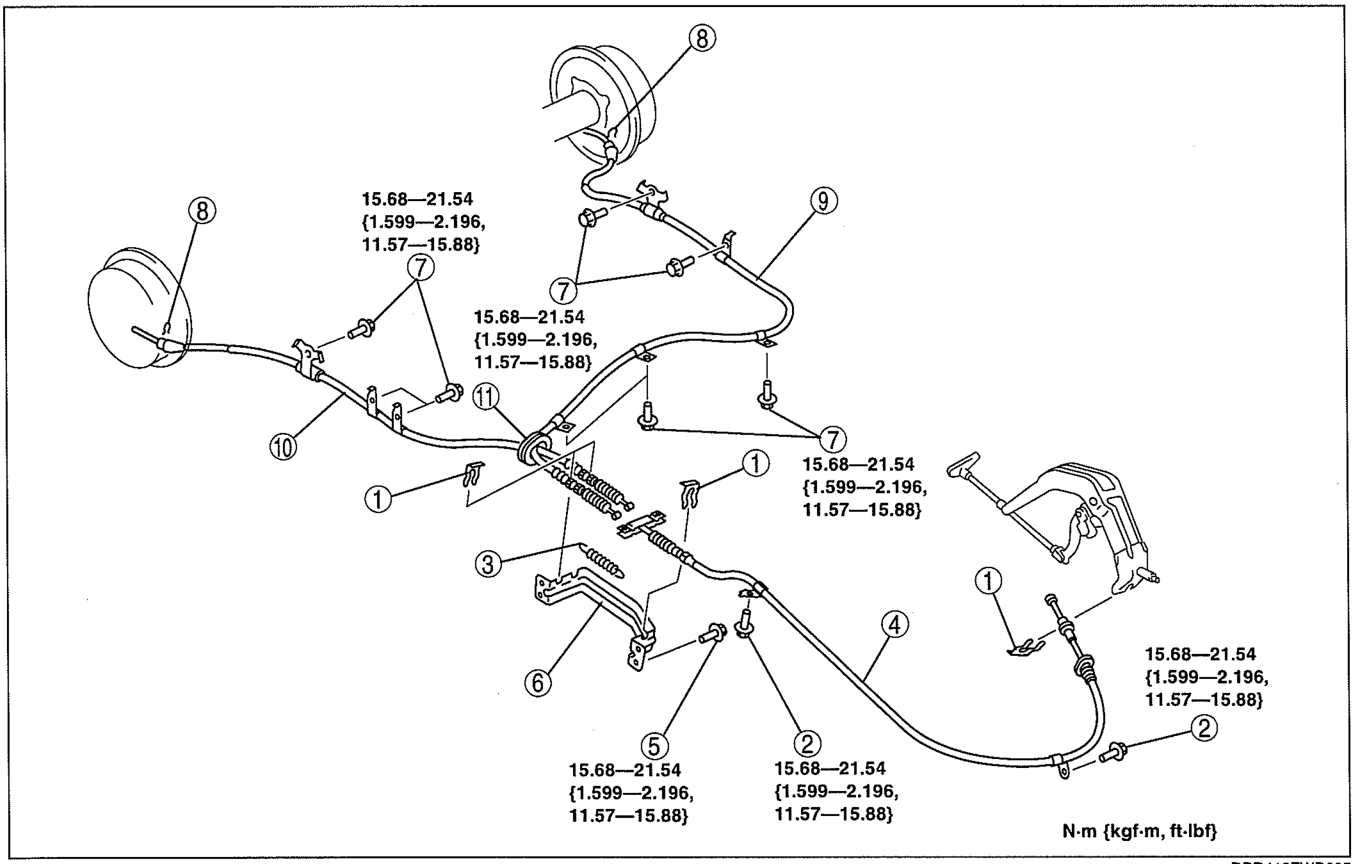
6	Parking brake switch
7	Clip and joint pin
8	Cable connector
9	Clip and joint pin
10	Clip and joint pin
11	Lever
12	Clip and joint pin
13	Spring
14	Ratchet pawl

15	Stopper
16	Fork joint
17	Spring
18	Guide
19	Rod

PARKING BRAKE CABLE REMOVAL/INSTALLATION [WL-3]

dcf041244410w01

1. Remove the fuel tank cover. (See 01-14A-5 FUEL TANK REMOVAL/INSTALLATION [WL-3].)
2. Remove the brake shoe (trailing shoe). (See 04-11-23 REAR BRAKE (DRUM) REMOVAL/INSTALLATION.)
3. Remove in the order indicated in the table.
4. Install in the reverse order of removal.
5. After installation, inspect the parking brake lever stroke. (See 04-12-3 PARKING BRAKE LEVER INSPECTION.)

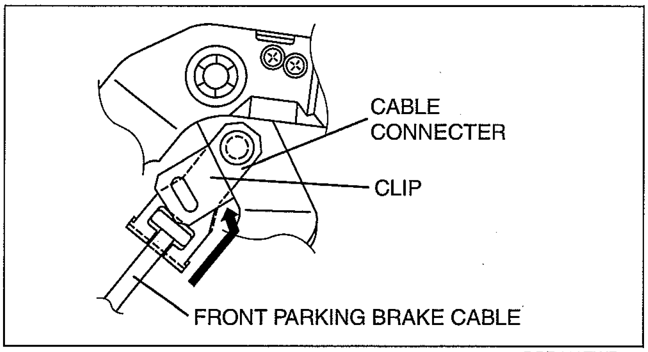


DBR412ZWB005

1	Clip
2	Bolt
3	Spring
4	Front parking brake cable (See 04-12-6 Front Parking Brake Cable Removal Note.)
5	Bolt
6	Bracket
7	Bolt
8	Clip
9	Rear parking brake cable (left)
10	Rear parking brake cable (right)
11	Grommet

Front Parking Brake Cable Removal Note

1. Slide the clip installed to the cable connector in the direction shown by the arrow.
2. Remove the end of the front parking brake cable from the cable connector.

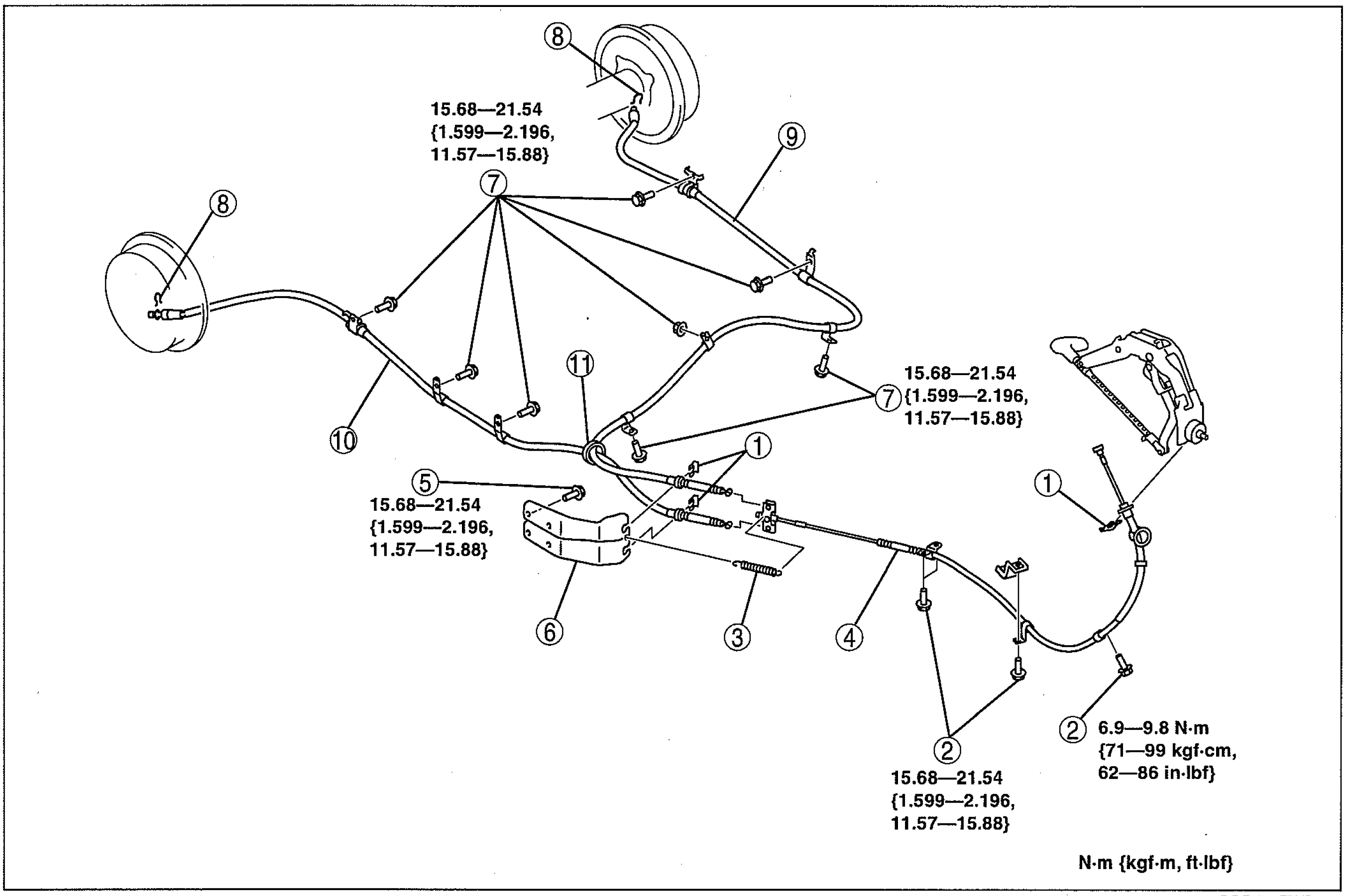


DBR412ZWB008

PARKING BRAKE CABLE REMOVAL/INSTALLATION [WL-C 4x2 (EXCEPT Hi-Rider)]

dcf041244410w02

1. Remove the fuel tank cover. (See 01-14B-4 FUEL TANK REMOVAL/INSTALLATION [WL-C, WE-C].)
2. Remove the brake shoe (trailing shoe). (See 04-11-23 REAR BRAKE (DRUM) REMOVAL/INSTALLATION.)
3. Remove in the order indicated in the table.
4. Install in the reverse order of removal.
5. After installation, inspect the parking brake lever stroke. (See 04-12-3 PARKING BRAKE LEVER INSPECTION.)



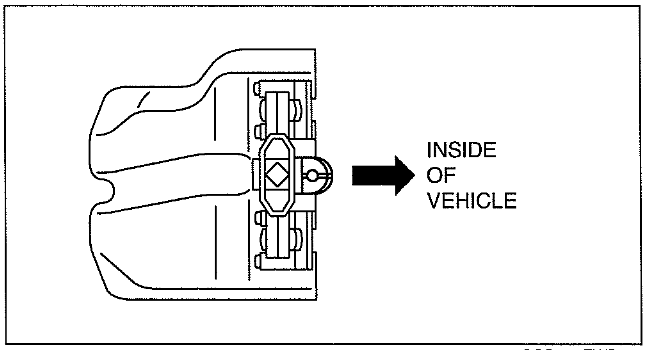
DBR412ZWB006

1	Nut
2	Bolt
3	Spring
4	Front parking brake cable (See 04-12-6 Front Parking Brake Cable Removal Note.) (See 04-12-7 Front Parking Brake Cable Installation Note.)
5	Bolt
6	Bracket

7	Bolt, nut
8	Clip
9	Rear parking brake cable (left)
10	Rear parking brake cable (right)
11	Grommet

Front Parking Brake Cable Installation Note

1. Install the front parking brake cable with the spring installation area of the equalizer pointed towards the inside of the vehicle.

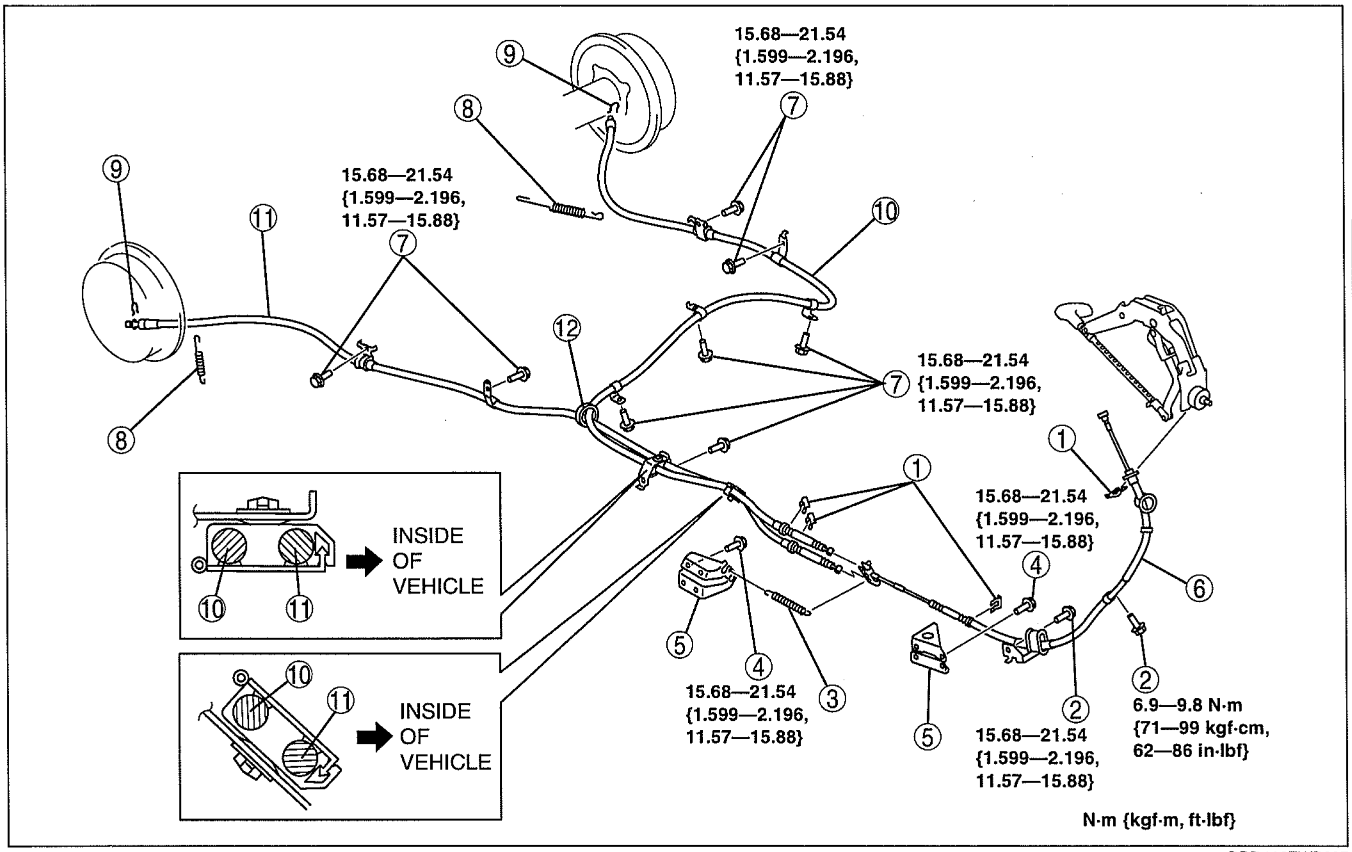


DBR412ZWB009

PARKING BRAKE CABLE REMOVAL/INSTALLATION [WL-C (Hi-Rider, 4x4), WE-C]

dcf041244410w03

1. Remove the fuel tank cover. (See 01-14B-4 FUEL TANK REMOVAL/INSTALLATION [WL-C, WE-C].)
2. Remove the brake shoe (trailing shoe). (See 04-11-23 REAR BRAKE (DRUM) REMOVAL/INSTALLATION.)
3. Remove in the order indicated in the table.
4. Install in the reverse order of removal.
5. After installation, inspect the parking brake lever stroke. (See 04-12-3 PARKING BRAKE LEVER INSPECTION.)



DBR412ZWB007

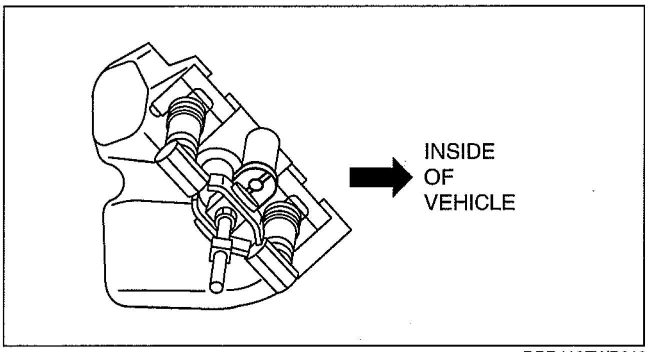
1	Clip
2	Bolt
3	Spring
4	Bolt
5	Bracket
6	Front parking brake cable (See 04-12-6 Front Parking Brake Cable Removal Note.) (See 04-12-8 Front Parking Brake Cable Installation Note.)

7	Bolt
8	Spring
9	Clip
10	Rear parking brake cable (left)
11	Rear parking brake cable (right)
12	Grommet

04

Front Parking Brake Cable Installation Note

1. Install the front parking brake cable with the spring installation area of the equalizer pointed towards the inside of the vehicle.

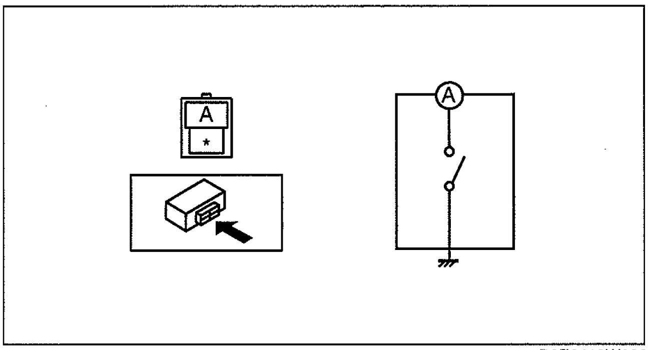


DBR412ZWB010

PARKING BRAKE SWITCH INSPECTION

dcf041266450w01

1. Disconnect the parking brake switch connector.
2. Verify that the continuity is as indicated in the table.
    - If not as indicated in the table, replace the parking brake switch.



B3E0412W006

O—O: Continuity

Condition	Terminal	
	A	Body ground
Parking brake lever pulled	O——	——O
Parking brake lever released		

CHU0412W003

04-13B ANTILOCK BRAKE SYSTEM [4-WHEEL ANTILOCK BRAKE SYSTEM (4W-ABS)]

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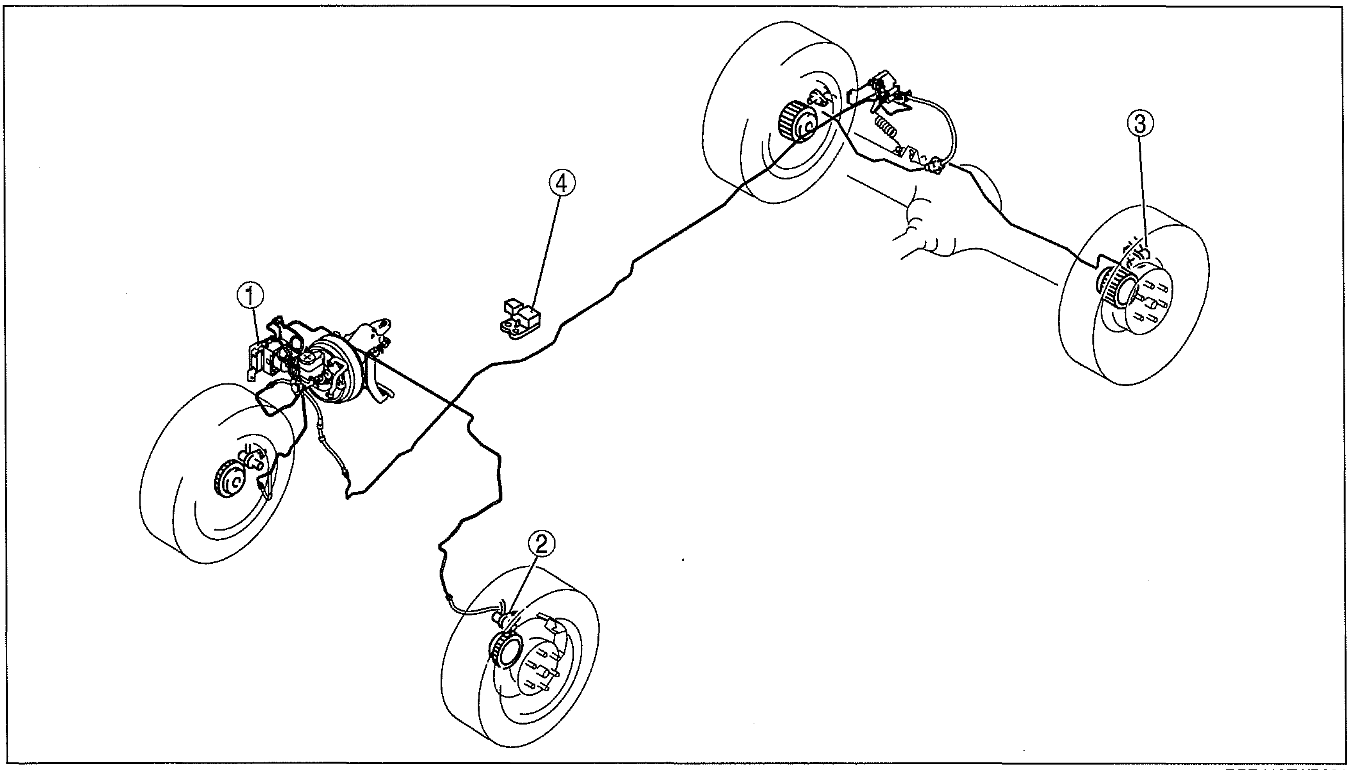
dcf041300000w02



DBR413ZWB001

1	ABS HU/CM (See 04-13B-2 ABS HU/CM SYSTEM INSPECTION [4W-ABS]) (See 04-13B-3 ABS HU/CM REMOVAL/INSTALLATION [4W-ABS]) (See 04-13B-5 ABS HU/CM INSPECTION [4W-ABS])
2	Front ABS wheel-speed sensor (See 04-13B-7 FRONT ABS WHEEL-SPEED SENSOR REMOVAL/INSTALLATION [4W-ABS (4x2 (EXCEPT Hi-Rider))]) (See 04-13B-8 FRONT ABS WHEEL-SPEED SENSOR REMOVAL/INSTALLATION [4W-ABS (Hi-Rider, 4x4)]) (See 04-13B-9 FRONT ABS WHEEL-SPEED SENSOR INSPECTION)
3	Rear ABS wheel-speed sensor (See 04-13B-10 REAR ABS WHEEL-SPEED SENSOR REMOVAL/INSTALLATION [4W-ABS (4x2 (EXCEPT Hi-Rider))]) (See 04-13B-11 REAR ABS WHEEL-SPEED SENSOR REMOVAL/INSTALLATION [4W-ABS (Hi-Rider, 4x4)]) (See 04-13B-11 REAR ABS WHEEL-SPEED SENSOR INSPECTION [4W-ABS])
4	G sensor (Hi-Rider, 4x4) (See 04-13B-12 G SENSOR REMOVAL/INSTALLATION [4W-ABS]) (See 04-13B-13 G SENSOR INSPECTION [4W-ABS])

ABS HU/CM SYSTEM INSPECTION [4W-ABS]

dcf041300000w03

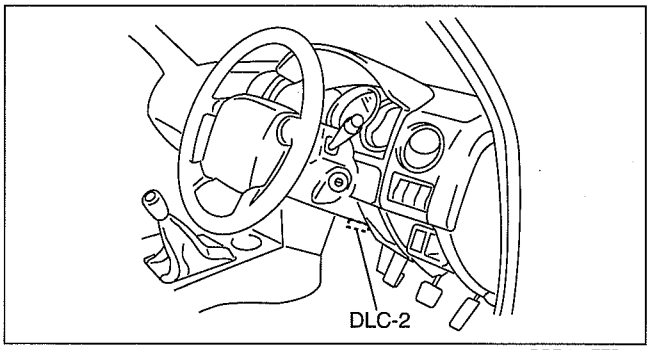
ABS Hydraulic Unit On-vehicle Inspection

Preparation

1. Verify that the battery is fully charged.
2. Turn the ignition switch to the ON position and verify that the ABS warning light goes out after **approx. 3.0 s.**
3. Turn the ignition switch off.
4. Jack up the vehicle and support it evenly on safety stands.
5. Shift to neutral.
6. Release the parking brake.
7. Verify that all four wheels rotate.
8. Rotate the inspected wheels by hand and verify there is no brake drag.
   - If there is any brake drag, perform regular brake inspection.
   - If there is no brake drag, perform ABS HU/CM operation inspection.

Operation inspection

1. Perform "Preparation".
2. Connect the current diagnostic tool to the DLC-2 connector.
3. Set up an active command mode inspection according to the combination of commands below.



DBR402ZTB003

Operation condition	Command name			Command transmission type
	PMP_MOTOR	V_RF_OTL	V_RF_INL	
Brake pressure retention	OFF	OFF	ON	Manual
Brake pressure reduction	ON	ON	ON	

The chart above shows an example of a right front wheel inspection.

Caution

- **When operating the solenoid valve and pump motor using the active command mode, make sure to keep the operation time within 10 s to prevent damaging the ABS HU/CM.**

Note

- When working with two people, one should press on the brake pedal, the other should attempt to rotate the wheel being inspected.

4. Send the command while depressing on the brake pedal and attempting to rotate the wheel being inspected.
5. Performing the inspection above determines the following:
   - The ABS HU/CM brake lines are normal.
   - The ABS HU/CM hydraulic system is not significantly abnormal (including inside ABS HU/CM).
   - The ABS HU/CM internal electrical parts (solenoid, motor and other parts) are normal.
   - The ABS HU/CM output system wiring harnesses (solenoid valve, relay system) are normal.
   — However, the following items cannot be verified.
     - Malfunction of ABS HU/CM input system wiring harnesses and parts
     - Extremely small leakage in the ABS HU/CM internal hydraulic system
     - Intermittent malfunction of the above items

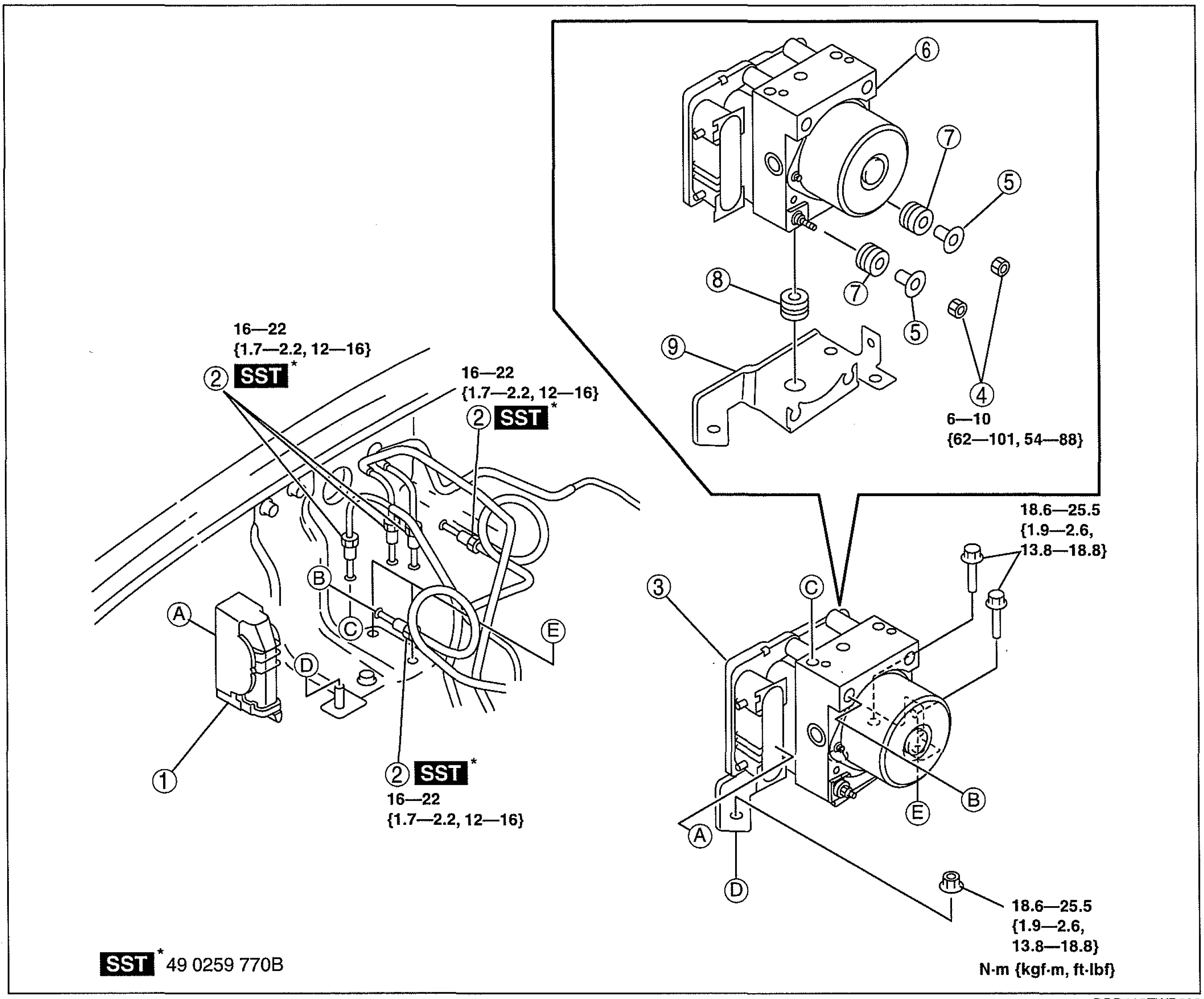
ABS HU/CM REMOVAL/INSTALLATION [4W-ABS]

dcf041343750w05

Caution

- **The internal parts of the ABS HU/CM could be damaged if dropped. Be careful not to drop the ABS HU/CM. Replace the ABS HU/CM if it is subjected to an impact.**

1. Remove in the order indicated in the table.
2. Install in the reverse order of removal.



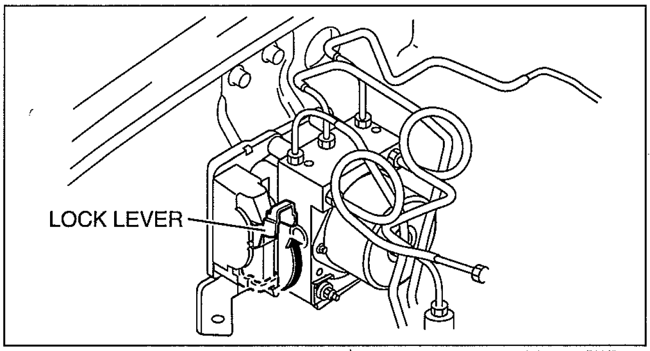
DBR413ZWB002

1	Connector (See 04-13B-4 Connector Removal Note.) (See 04-13B-4 Connector Installation Note.)
2	Brake pipe (See 04-13B-4 Brake Pipe Removal Note.) (See 04-13B-4 Brake Pipe Installation Note)
3	ABS HU/CM component

4	Nut
5	washer
6	ABS HU/CM
7	Bushing (ABS HU/CM side)
8	Bushing (bracket side)
9	Bracket

Connector Removal Note

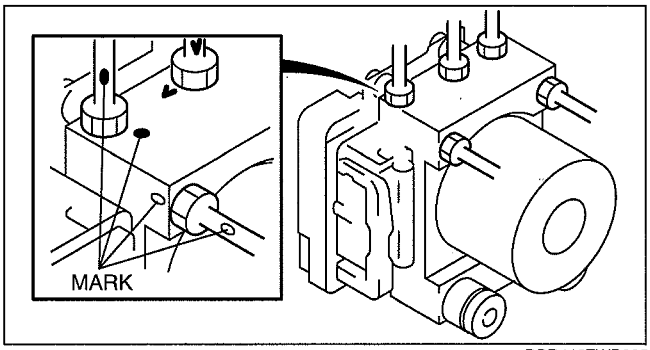
1. Rotate the lock lever in the direction of the arrow, and remove the ABS HU/CM connector.



DBR413ZWB003

Brake Pipe Removal Note

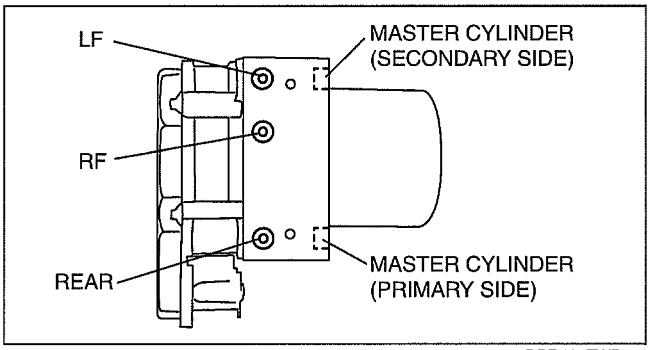
1. Place an alignment mark on the brake pipe and ABS HU/CM.
2. Apply protective tape to the connector to prevent brake fluid from entering.
3. Remove the brake pipe.



DBR413ZWB005

Brake Pipe Installation Note

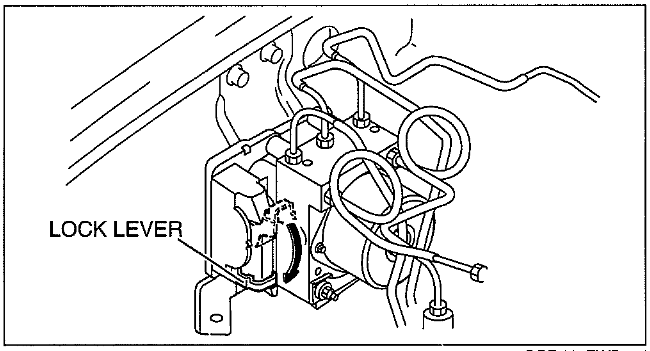
1. When install the brake pipe, align the marks made before removal with the ABS HU/CM as shown in the figure.
2. Tighten the brake pipe to the specified torque using the **SST** (49 0259 770B).



DBR413ZWB006

Connector Installation Note

1. After connecting the connector, rotate the lock lever in the direction of the arrow to install the ABS HU/CM connector.



DBR413ZWB004

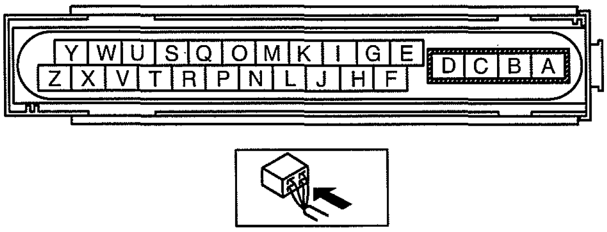
ABS HU/CM INSPECTION [4W-ABS]

dcf041343750w06

1. Disconnect the ABS HU/CM connector.
2. Connect the negative battery cable.
3. Attach the tester lead to the ABS HU/CM harness side connector, then inspect voltage, continuity or resistance according to the standard (reference value) on the table.

Standard (Reference Value)

ABS HU/CM WIRING HARNESS-SIDE CONNECTOR



DBR413ZWB017

Terminal	Signal name	Connected to	Measured item	Measured terminal (measured condition)	Standard	Inspection item(s)
A	Ground (ABS motor)	Ground point	Continuity	A—ground point	Continuity detected	• Wiring harness (A—ground point)
B	Power supply (ABS motor operation)	Battery	Voltage	Under any condition	B+	• Wiring harness (B—battery)
C	Power supply (solenoid)	Battery	Voltage	Under any condition	B+	• Wiring harness (C—battery)
D	Ground (ABS system)	Ground point	Continuity	D—ground point	Continuity detected	• Wiring harness (D—ground point)
E	LF wheel-speed sensor (single)	LF ABS wheel-speed sensor	Continuity	E—LF ABS wheel-speed sensor connector terminal A	Continuity detected	• Wiring harness (E—LF ABS wheel-speed sensor connector terminal A)
F	LF wheel-speed sensor (ground)	LF ABS wheel-speed sensor	Continuity	F—LF ABS wheel-speed sensor connector terminal B	Continuity detected	• Wiring harness (F—LF ABS wheel-speed sensor connector terminal B)
G	LR wheel-speed sensor (ground)	LR ABS wheel-speed sensor	Continuity	G—LR ABS wheel-speed sensor connector terminal B	Continuity detected	• Wiring harness (G—LR ABS wheel-speed sensor connector terminal B)
H	—	—	—	—	—	—
I	LR wheel-speed sensor (signal)	LR ABS wheel-speed sensor	Continuity	I—LR ABS wheel-speed sensor connector terminal A	Continuity detected	• Wiring harness (I—LR ABS wheel-speed sensor connector terminal A)
J	Power supply (system)	Ignition switch	Voltage	Ignition switch at ON	B+	• Wiring harness (J—ignition switch)
				Ignition switch is off.	1 V or less	—
K	RR wheel-speed sensor (ground)	RR ABS wheel-speed sensor	Continuity	K—RR ABS wheel-speed sensor connector terminal B	Continuity detected	• Wiring harness (K—RR ABS wheel-speed sensor connector terminal B)
L	RR wheel-speed sensor (signal)	RR ABS wheel-speed sensor	Continuity	L—RR ABS wheel-speed sensor connector terminal A	Continuity detected	• Wiring harness (L—RR ABS wheel-speed sensor connector terminal A)
M	RF wheel-speed sensor (ground)	RF ABS wheel-speed sensor	Continuity	M—RF ABS wheel-speed sensor connector terminal B	Continuity detected	• Wiring harness (M—RF ABS wheel-speed sensor connector terminal B)

04

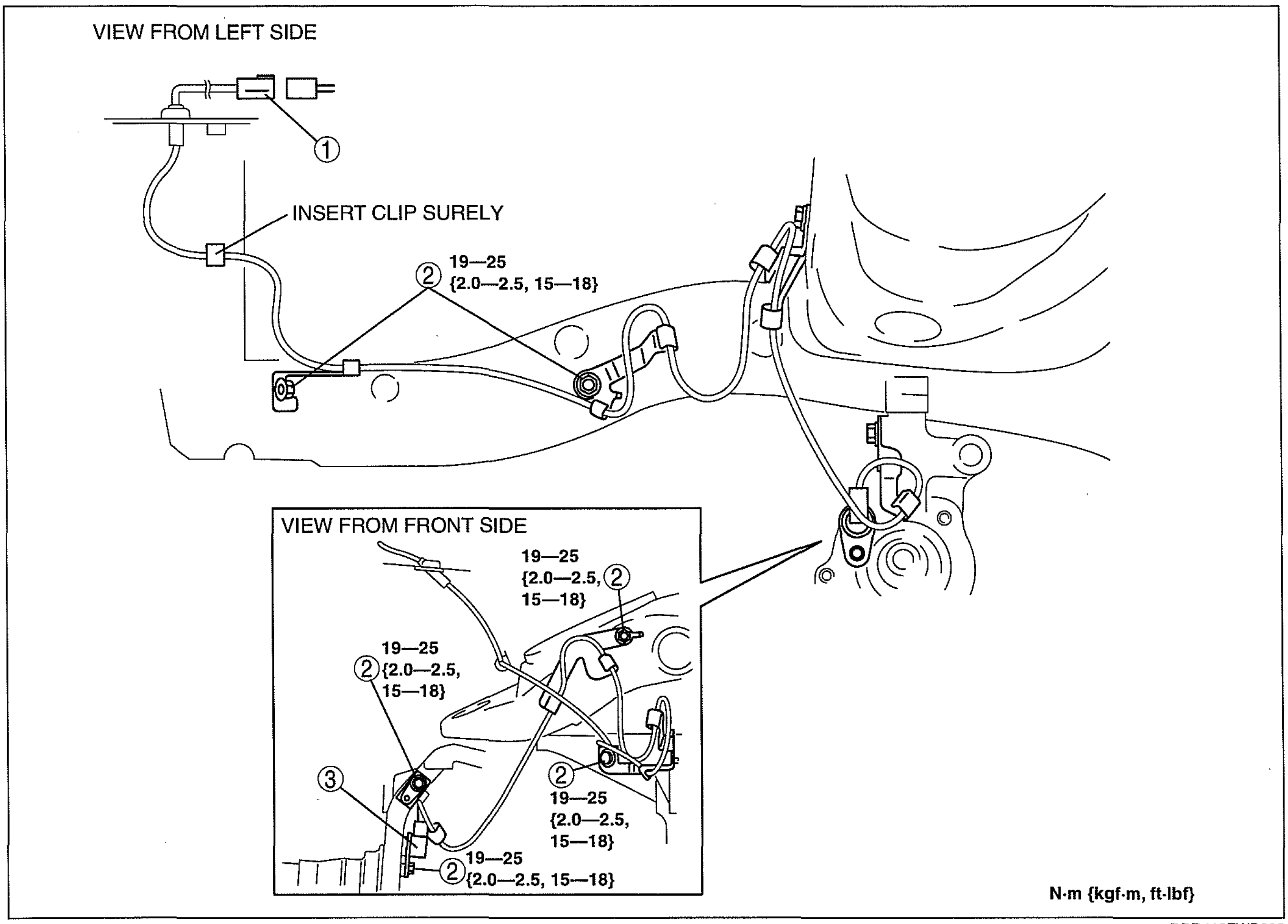
Terminal	Signal name	Connected to	Measured item	Measured terminal (measured condition)	Standard	Inspection item(s)
N	Brake switch	Brake switch	Voltage	N—brake switch (Brake pedal depressed)	B+	• Wiring harness (N—brake switch) • Brake switch
				Y—brake switch (Brake pedal not depressed)	1 V or less	
O	RF wheel-speed sensor (signal)	RF ABS wheel-speed sensor	Continuity	O—RF ABS wheel-speed sensor connector terminal A	Continuity detected	• Wiring harness (O—RF ABS wheel-speed sensor connector terminal A)
P*	G sensor (signal)	G sensor	Continuity	P—G sensor connector terminal B	Continuity detected	• Wiring harness (P—G sensor connector terminal B)
Q	On-board diagnostic	KLN terminal of DLC-2	Continuity	Q—DLC-2 connector terminal KLN	Continuity detected	• Wiring harness (Q—DLC-2 connector terminal KLN)
R	—	—	—	—	—	—
S	—	—	—	—	—	—
T	—	—	—	—	—	—
U	—	—	—	—	—	—
V*	G sensor (power supply	G sensor	Continuity	V—G sensor connector terminal A	Continuity detected	• Wiring harness (V—G sensor connector terminal A)
W*	G sensor (ground)	G sensor	Continuity	W—G sensor connector terminal C	—	• Wiring harness (W—G sensor connector terminal C)
X	—	—	—	—	—	—
Y	—	—	—	—	—	—
Z	—	—	—	—	—	—

* : Hi-Rider, 4x4

FRONT ABS WHEEL-SPEED SENSOR REMOVAL/INSTALLATION [4W-ABS (4x2 (EXCEPT Hi-Rider))]

dcf041343720w01

1. When removing the ABS wheel-speed sensor on the right side, remove the washer tank installation bolts and the connector, then move the washer tank to a place out of the way. (See 09-19-5 WINDSHIELD WASHER TANK REMOVAL/INSTALLATION.)
2. Remove in the order indicated in the table.
3. Install in the reverse order of removal.



DBR413ZWB008

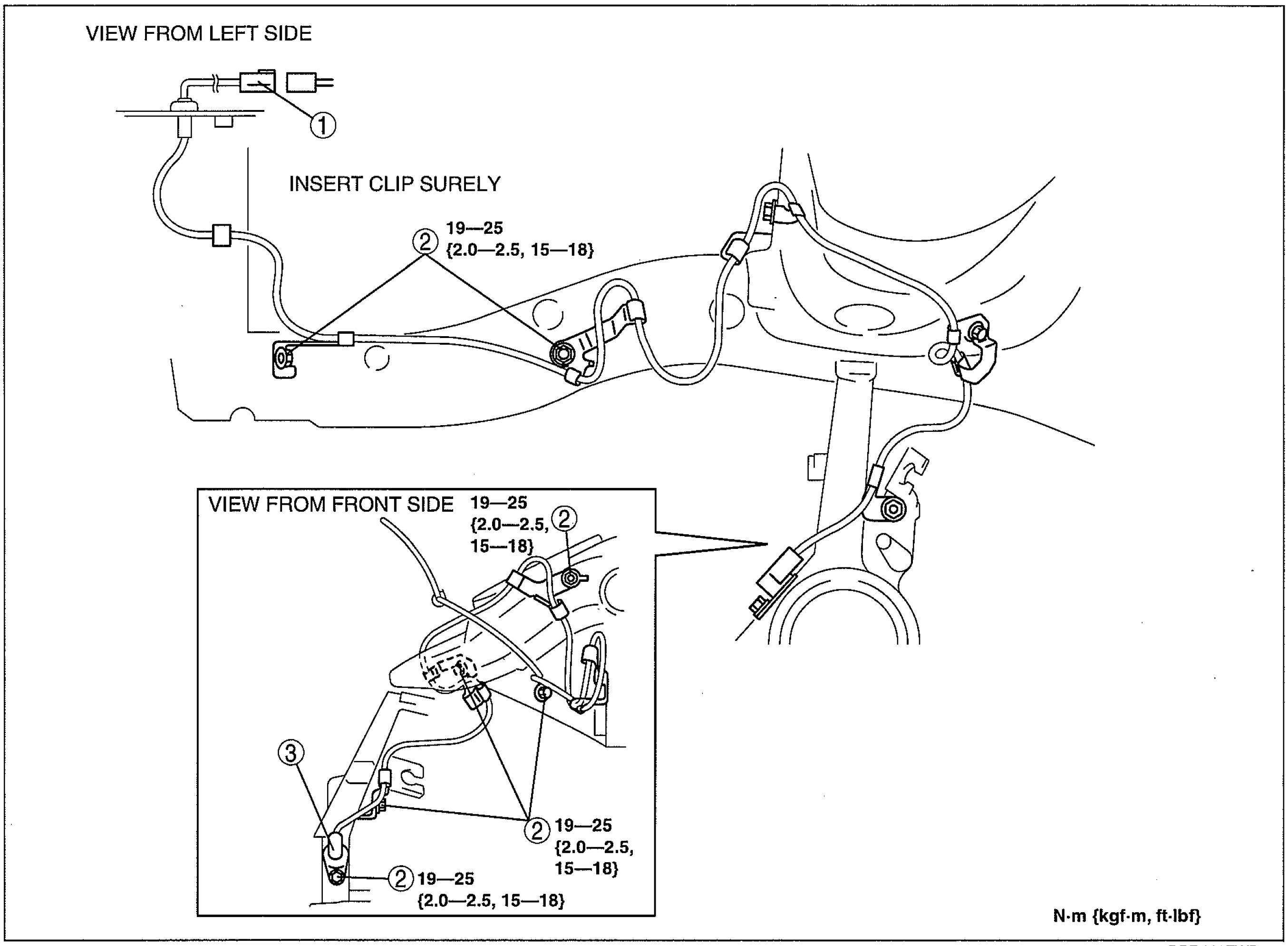
1	Connector
2	Bolt

3	Front ABS wheel-speed sensor

FRONT ABS WHEEL-SPEED SENSOR REMOVAL/INSTALLATION [4W-ABS (Hi-Rider, 4x4)]

dcf041343720w02

1. When removing the ABS wheel-speed sensor on the right side, remove the washer tank installation bolts and the connector, then move the washer tank to a place out of the way. (See 09-19-5 WINDSHIELD WASHER TANK REMOVAL/INSTALLATION.)
2. Remove in the order indicated in the table.
3. Install in the reverse order of removal.



DBR413ZWB009

1	Connector
2	Bolt
3	Front ABS wheel-speed sensor

FRONT ABS WHEEL-SPEED SENSOR INSPECTION

dcf041343720w03

Visual Inspection

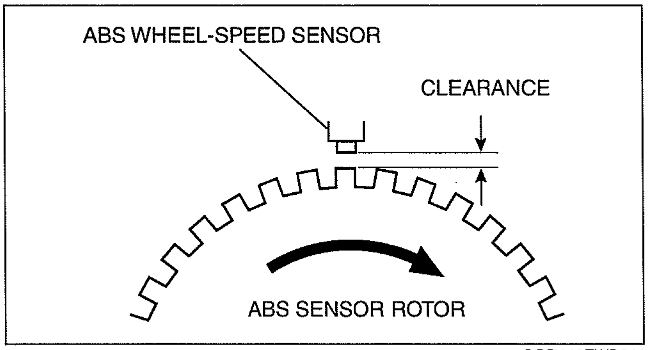
1. Remove the wheel and tire, and inspect the sensor for looseness and damage. Replace the sensor if necessary.

Clearance Inspection

1. Inspect the clearance between the wheel-speed sensor and the sensor rotor.

Clearance
0.3—1.1 mm {0.012—0.043 in}

- If not as specified, replace the defective part(s). (ABS wheel-speed sensor, ABS sensor rotor and steering knuckle etc.)



DBR413ZWB010

Resistance Inspection

1. Disconnect the ABS wheel-speed sensor connector.
2. Inspect the resistance at the ABS wheel-speed sensor.
    - If not as specified, replace the ABS wheel-speed sensor.

Resistance
1.2—1.6 kilohm

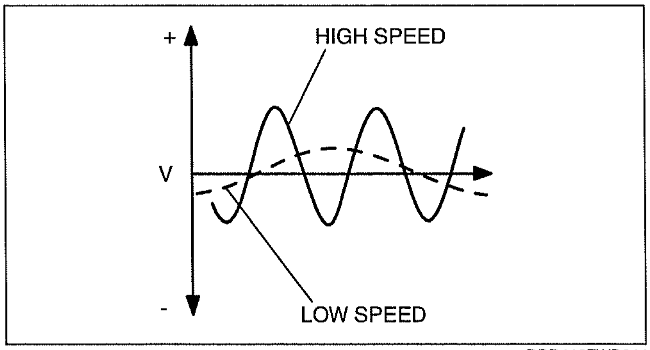
Voltage Inspection

1. Jack up the vehicle on level ground and support it evenly on safety stands.
2. Disconnect the ABS wheel-speed sensor connector.
3. Inspect each sensor by rotating each wheel one revolution per second.
    - If not as specified, replace the ABS wheel-speed sensor.

Voltage
0.15—1.2 V (AC)

Voltage Pattern Inspection

1. Jack up the vehicle on level ground and support it evenly on safety stands.
2. Disconnect the ABS wheel-speed sensor connector.
3. Using an oscilloscope, inspect voltage pattern for distortion and noise by rotating each wheel.
    - If there is distortion or noise, inspect the ABS sensor rotor.



DBR413ZWB011

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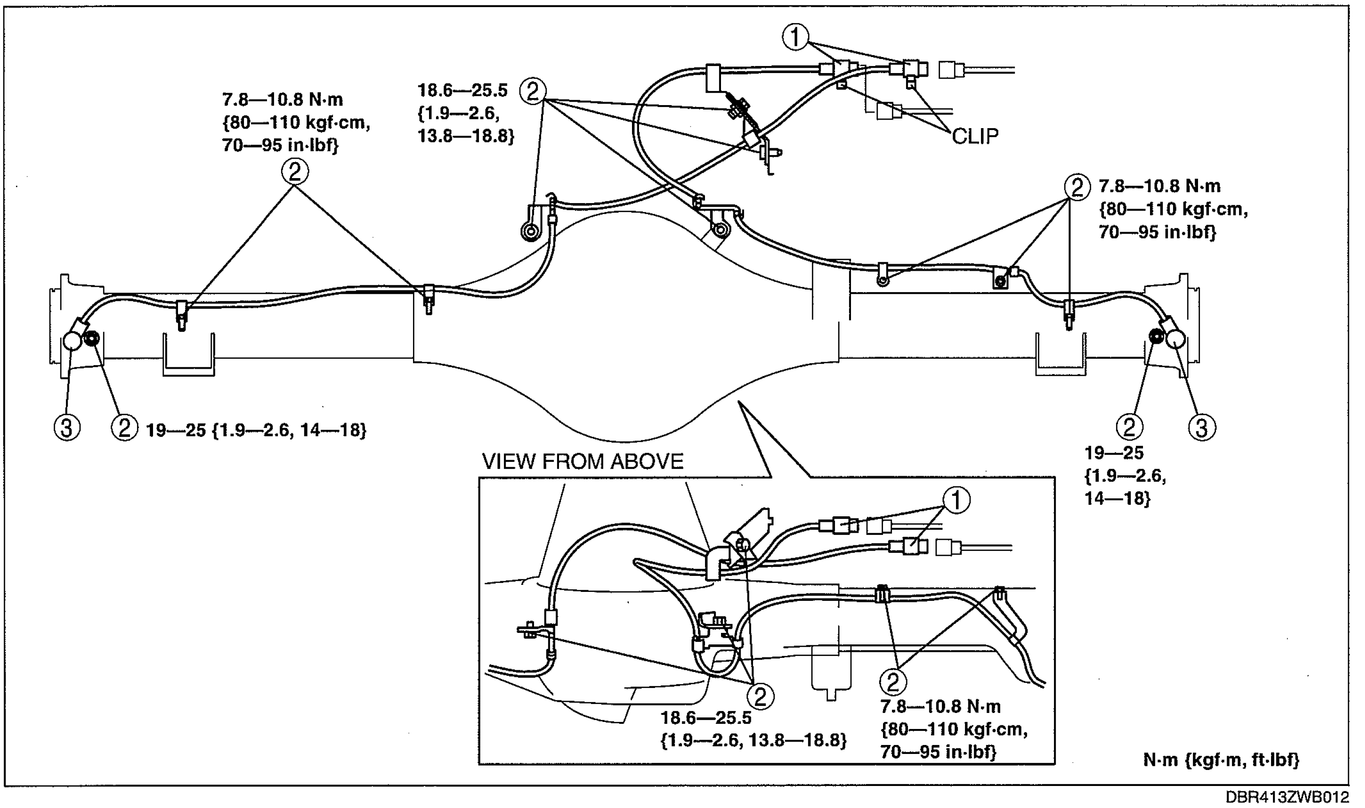
REAR ABS WHEEL-SPEED SENSOR REMOVAL/INSTALLATION [4W-ABS (4x2 (EXCEPT Hi-Rider))]

dcf041343710w03

1. Remove in the order indicated in the table.
2. Install in the reverse order of removal.



DBR413ZWB012

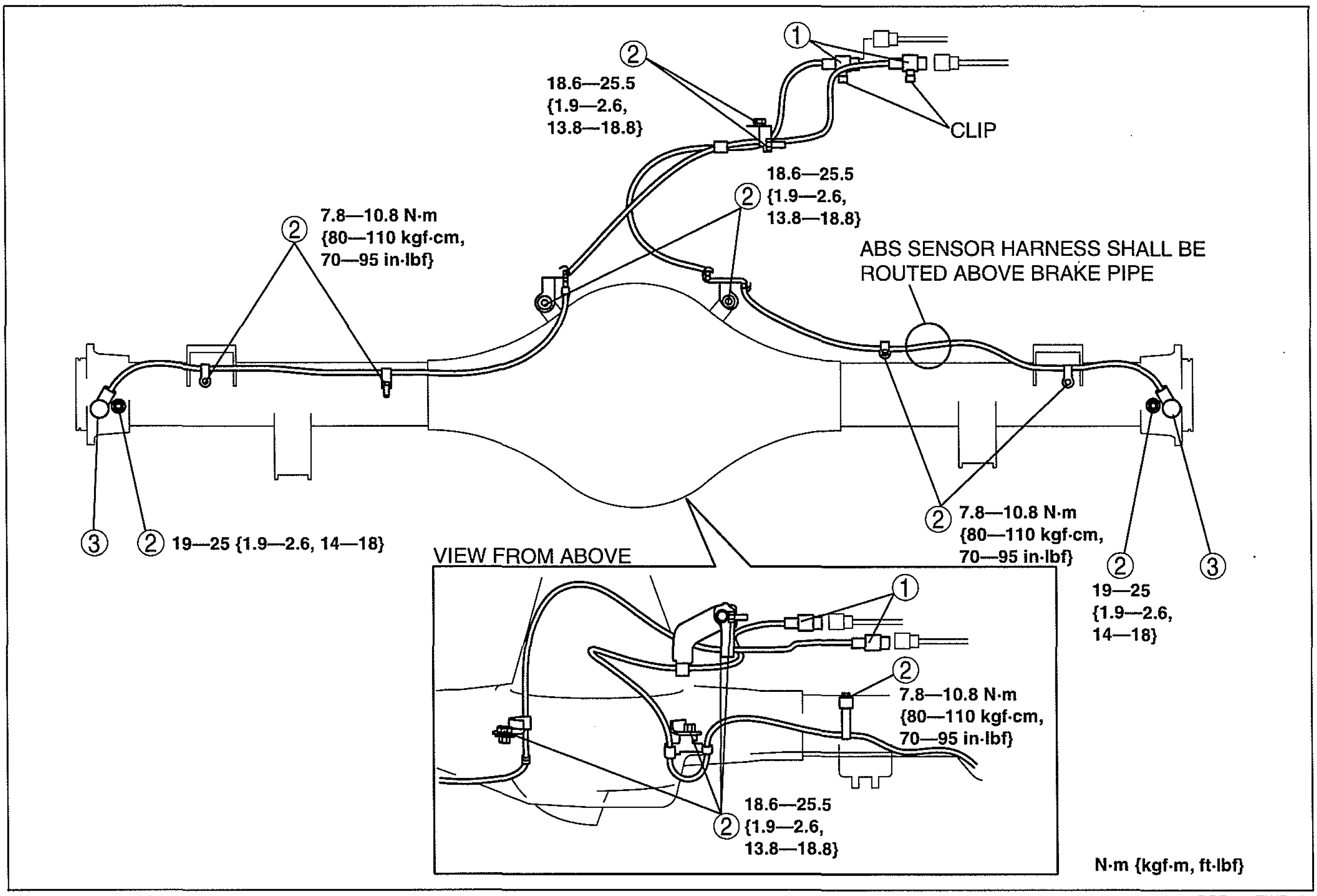
1	Connector
2	Bolt

3	Rear ABS wheel-speed sensor

REAR ABS WHEEL-SPEED SENSOR REMOVAL/INSTALLATION [4W-ABS (Hi-Rider, 4x4)]

dcf041343710w04

1. Remove in the order indicated in the table.
2. Install in the reverse order of removal.



DBR413ZWB013

1	Connector
2	Bolt
3	Rear ABS wheel-speed sensor

REAR ABS WHEEL-SPEED SENSOR INSPECTION [4W-ABS]

dcf041343710w05

Visual Inspection

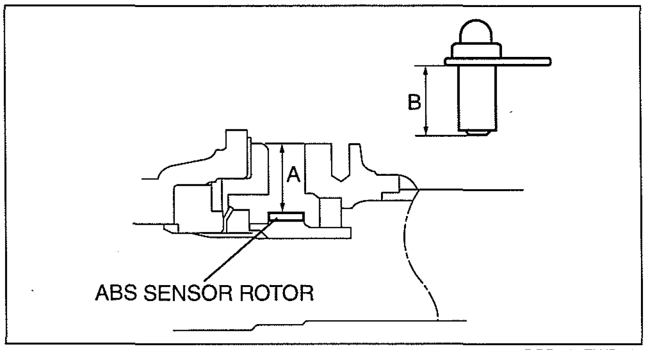
1. Inspect the sensor for looseness and damage. Replace the sensor if necessary.

Clearance Inspection

1. Remove the ABS wheel-speed sensor.
2. Measure the length A and B shown in the figure using vernier calipers.
3. Subtract B from A then verify the clearance between the wheel-speed sensor and the sensor rotor.

Clearance
0.3—1.1 mm {0.012—0.043 in}

- If not as specified, replace the defective part(s). (ABS wheel-speed sensor, ABS sensor rotor, rear axle shaft and rear axle housing etc.)



DBR413ZWB014

Resistance Inspection

1. Disconnect the ABS wheel-speed sensor connector.
2. Inspect the resistance at the ABS wheel-speed sensor.
    - If not as specified, replace the ABS wheel-speed sensor.

Resistance
1.2—1.6 kilohm

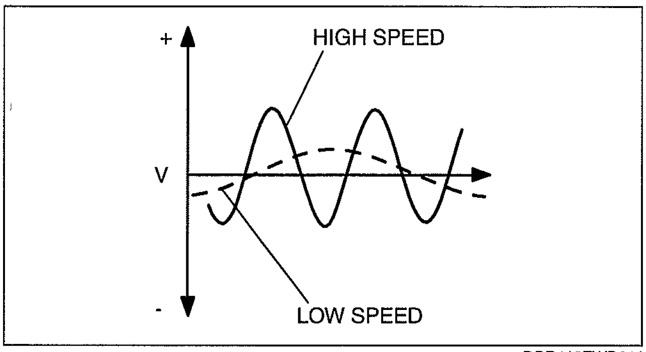
Voltage Inspection

1. On level ground, jack up the vehicle and support it evenly on safety stands.
2. Disconnect the ABS wheel-speed sensor connector.
3. Inspect each sensor by rotating each wheel one revolution per second.
    - If not as specified, replace the ABS wheel-speed sensor.

Voltage
0.15—1.2 V (AC)

Voltage Pattern Inspection

1. Jack up the vehicle on level ground and support it evenly on safety stands.
2. Disconnect the ABS wheel-speed sensor connector.
3. Using an oscilloscope, inspect voltage pattern for distortion and noise by rotating each wheel.
    - If there is distortion or noise, inspect the ABS sensor rotor.



DBR413ZWB011

G SENSOR REMOVAL/INSTALLATION [4W-ABS]

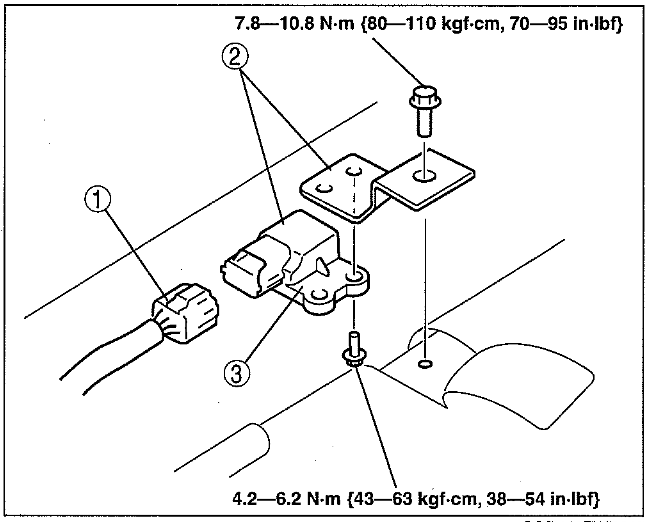
dcf041343770w01

Caution
- **Be careful and do not allow the combined sensor to fall. If by chance it is subjected to strong impact, replace it.**

1. Slide the RH front seat fully backward.
2. Remove in the order indicated in the table.

1	Connector
2	G sensor and bracket
3	G sensor

3. Install in the reverse order of removal.



DBR413ZWB015

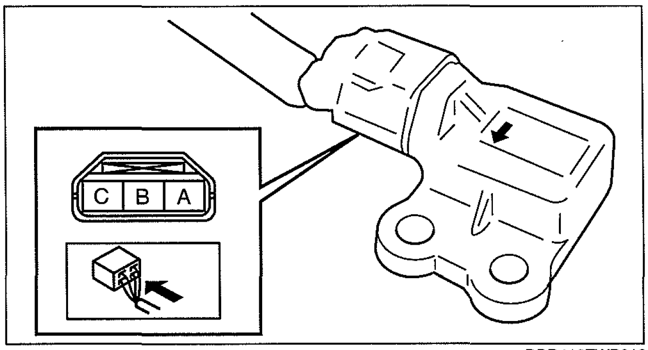
G SENSOR INSPECTION [4W-ABS]

dcf041343770w02

1. Connect the connector.
2. Turn ignition switch (engine switch) on, and verify the voltage between terminals A and B under the following conditions.
   - If not within the specification, replace the G sensor.

   (1) Horizontal

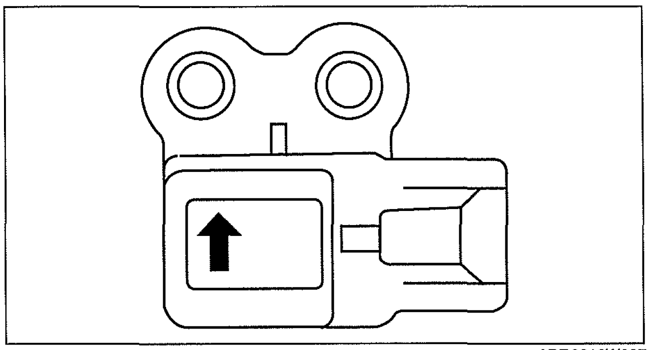
   Voltage
   2.4—2.6 V



DBR413ZWB016

   (2) Facing up (inclined **90**° from horizontal)

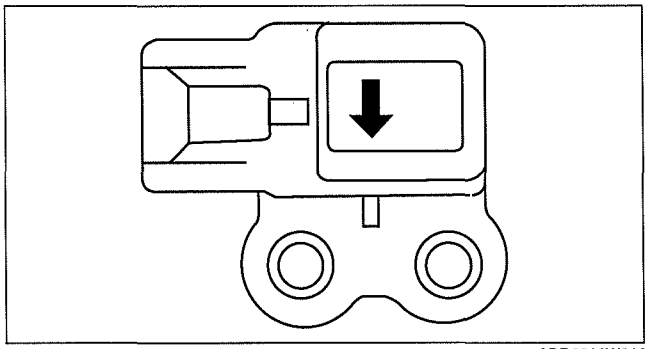
   Voltage
   1.3—1.7 V



ABR6916W007

   (3) Facing down (declined **90**° from horizontal)

   Voltage
   3.3—3.7 V



ABR6916W010

04-50 TECHNICAL DATA

BRAKE TECHNICAL DATA 04-50–1

BRAKE TECHNICAL DATA

dcf045000000w01

Item	Specification
Brake fluid type	SAE J1703, FMVSS 116 DOT-3
Brake pedal height	214—219 mm {8.43—8.62 in}
Standard pedal play	3.0—8.0 mm {0.12—0.31 in}
Standard pedal-to-floor clearance	105 mm {4.14 in} or more
Power brake unit fluid pressure when pedal depressed at 200 N {20 kgf, 45 lbf}	At 0 kPa {0 mmHg, 0 inHg}: 593 kPa {6.1 kgf/cm^2, 87 psi} or more
Power brake unit fluid pressure when pedal depressed at 200 N {20 kgf, 45 lbf}	At 67 kPa {500 mmHg, 20 inHg}: 6,950 kPa {71 kgf/cm^2, 1,008 psi} or more
Minimum front disc pad thickness	2.0 mm {0.079 in} min.
Minimum front disc plate thickness	4x2 (except Hi-Rider): 22.0 mm {0.867 in} Hi-Rider, 4x4: 26.0 mm {1.03 in}
Minimum front disc plate thickness after machining using a brake lathe on-vehicle	4x2 (except Hi-Rider): 22.8 mm {0.898 in} Hi-Rider, 4x4: 26.8 mm {1.06 in}
Front disc plate runout limit	0.05 mm {0.002 in} max.
Minimum rear brake lining thickness	1.0 mm {0.04 in}
Maximum rear brake drum diameter	4x2 (except Hi-Rider): 271.5 mm {10.68 in} Hi-Rider, 4x4: 296.5 mm {11.67 in}
Parking brake lever stroke when pulled at 98 N {10 kgf, 22 lbf}	1—7 notches

04

Load sensing proportioning valve (LSPV) fluid pressure

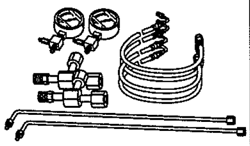
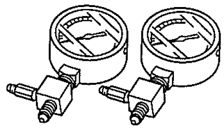
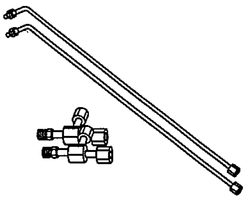
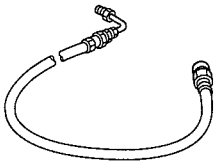
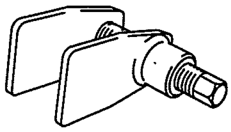
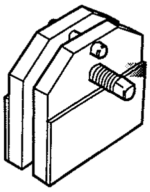
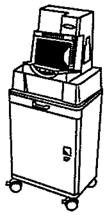
Type		Front wheel cylinder fluid pressure (kPa {kgf/cm^2, psi})	Rear wheel cylinder (kPa {kgf/cm^2, psi})
4x2 (except Hi-Rider)	Without 4W-ABS	4,900 {50, 711}	2,250—2,850 {23—29, 327—413}
		9,800 {100, 1,421}	3,130—3,930 {32—40, 454—569}
	With 4W-ABS	4,900 {50, 711}	2,910—3,710 {30—37, 423—538}
		9,800 {100, 1,421}	4,970—6,170 {51—62, 721—894}
Hi-Rider, 4x4	Without 4W-ABS	4,900 {50, 711}	2,250—2,850 {23—29, 327—413}
		9,800 {100, 1,421}	3,130—3,930 {32—40, 454—569}
	With 4W-ABS	4,900 {50, 711}	2,650—3,450 {28—35, 385—500}
		9,800 {100, 1,421}	4,700—5,900 {48—60, 682—855}

04-60 SERVICE TOOLS

BRAKES SST 04-60–1

BRAKES SST

dcf046000000w01

49 0259 770B Flare nut wrench 	49 U043 0A0A Oil pressure gauge set 	49 U043 004A Oil pressure gauge (Part of 49 U043 0A0A) 
49 U043 005 Joint (Part of 49 U043 0A0A) 	49 U043 006 Hose (Part of 49 U043 0A0A) 	49 0221 600C Disc brake expand tool 
49 T033 001A Disc brake piston stopper 	WDS 	—